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In this scientific journal, in the form of scientific articles, the results of scientific research conducted by professors and teachers of the Republic of Uzbekistan and international higher educational institutions, independent researchers, doctoral students, undergraduates were published. In addition to higher educational institutions, the journal also includes scientific articles by employees working in other research institutes, production organizations and enterprises of our region and republic.

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THE ROLE OF INDEPENDENT WORK IN THE CREDIT-MODULE TRAINING SYSTEM

U. T. Usarov¹ , G.Yu. Nodirov²

professor, department of «Social and natural sciences», Samarkand state university of
architecture and civil engineering, Samarkand, Uzbekistan¹

teacher, department of «Social and natural sciences», Samarkand state university of architecture
and civil engineering, Samarkand, Uzbekistan²

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Abstract. *The main task of higher education is to form the creative personality of a specialist capable of self-development, self-education, and innovative activity. Independent work of students plays a key role in the conditions of the credit-module education system (CMSE). The work examines the main aspects of the role of (CMSE) in the training of specialists.*

Key words: *Self-development, self-education, credit-module system, Moodle, Google Classroom.*

Introduction

In order to determine priority areas for systemic reform of higher education, raising the process of training independently thinking highly qualified personnel with modern knowledge and high spiritual and moral qualities to a qualitatively new level, modernizing higher education, development of the social sphere and economic sectors based on advanced educational technologies, the concept of development of the higher education system in the republic was approved. The transition in the republic to a credit-modular education system requires a more responsible attitude towards new pedagogical technologies in the education system. Independent work of students plays a major role in KMSE. The system of independent work of students is focused on the development of competencies, independence and responsibility of students. The paper examines the main aspects of the role of students' independent work in KMSE.

Discussion

In the credit-module system, the emphasis of responsibility for learning shifts from the teacher to the student. Independent work becomes the main tool for achieving educational goals. This encourages students to take a conscious approach to learning, develop time management and self-organization. Flexibility in learning. Independent work allows students to independently choose the pace and volume of studying the material. They can work in depth on those aspects of a subject that are most interesting or challenging to them. Development of key competencies. Independent work develops skills such as:

- Analysis and synthesis of information.
- Problem solving and research activities.
- Critical thinking.
- Skills in working with scientific sources and preparing scientific papers.

The role of the teacher as a mentor. In KMSE, the teacher plays the role of a consultant and mentor, guiding the learning process. Instead of presenting the material directly, the teacher provides resources, sets tasks, and corrects students' independent efforts.

Assessment of independent work. Independent work is assessed through a credit system, which makes its effectiveness measurable. An important role is played by:

- Constant monitoring of task completion (tests, controls).
 - Projects and practical assignments.
 - Self-assessment and reflection.

Interaction with modern technologies. KMSO actively uses electronic resources, such as:

- Distance learning platforms (LMS).
- Scientific literature databases.
- Interactive training materials.

Organizing independent work for students requires an integrated approach, which includes planning, motivation, provision of resources and control. Basic steps:

1. Defining goals and objectives. It is necessary to clearly formulate what students need to learn. Divide tasks into clear steps. Relate objectives to real-life examples or professional activities to highlight their significance.

2. Planning and structure. Prepare a study plan that includes clear deadlines. Create step-by-step instructions or checklists for students. Break large tasks into small parts to reduce your stress levels.

3. Providing resources. Prepare educational materials: articles, video lectures, presentations, links to useful resources. Use educational platforms (for example, Moodle, Google Classroom) for posting assignments and materials. Provide access to libraries, databases or simulations as needed.

4. Motivation: Explain how completing assignments will help them in their studies and professional lives. Encourage the use of independent work to improve skills (critical thinking, analysis, information retrieval). Use gamification elements (points, achievements) or projects that are interesting to students.

5. Control and support. Regularly check the completion of tasks (interim reports, tests, meetings). Organize consultations or discussions to answer questions. Give detailed feedback, highlighting successes and areas for improvement.

6. Use of technology. Introduce interactive formats: online forums, webinars, interactive tests. Use task management software (eg Trello, Notion).

7. Development of independent work skills. Teach students how to plan time (time management). Conduct master classes on searching and analyzing information. Teach them to set goals and record performance results. Regular support and creating a collaborative atmosphere will help students cope with tasks and increase their interest in learning.

Let's consider the difference between students' independent work under the modular-credit system (MCS) and traditional teaching methods in a number of aspects:

1. Organization of the educational process

- ISS: Training is divided into modules, each of which has clearly structured goals, objectives and expected results. Student independent work (SWS) is integrated into each module and is assessed for credits.

- Traditional methods: Independent work is usually less structured, and the volume can be set at the discretion of the teacher, without necessarily linking it to the credit system.

2. The role of the teacher

- MKS: The teacher acts more like a tutor or consultant who guides and coordinates the work of students.

- Traditional methods: The teacher plays a key role in imparting knowledge, and independent work is perceived as a complement to lectures and practices.

3. Scope and planning of CDS

- MKS: Independent work takes up a significant part of the educational process (up to 50% of the time). Specific tasks are developed for it that must be completed to achieve the competencies laid down in the module.

- Traditional methods: The amount of SRS is often determined at the discretion of the teacher or students are expected to allocate their time independently without clear instructions.

4. Assessment

- MKS: Completion of SRS assignments contributes to the final grade for the module. Each module ends with a knowledge test (test, project, report), which encourages students to work systematically.

- Traditional methods: Independent work is assessed less frequently and, as a rule, indirectly (for example, through checking notes, completing homework).

5. Skills and autonomy

- ISS: ISS promotes the development of skills in independent search, analysis and application of information. Students learn to organize their time and work for the long term.

- Traditional methods: Students' independence develops to a lesser extent, since the main focus is on mastering the material provided by the teacher.

6. Use of technology.

- MKS: Often involves the active use of educational platforms, electronic resources and interactive forms of work (video lessons, online tests, databases).

- Traditional methods: The use of technology may be less intensive and optional.

7. Target orientation.

- MKS: Independent work is aimed at achieving specific competencies prescribed in the educational program.

- Traditional methods: Focus more on overall learning rather than on specific skills and results.

Thus, within the framework of the modular-credit system, students' independent work becomes more structured, focused and integrated into the educational process, which requires greater responsibility and independence from students.

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ANALYSIS OF AUTOMATIC GAS PRESSURE REGULATION IN GAS SUPPLY SYSTEMS

**Mahmudov Ruzmat Mahmudovich¹, Aymatov Rustam Ruzibayevich², Mavlonov
Khusniddin Zokir O‘g‘li³**

¹Samarkand state university of architecture and civil engineering, associate professor.
Samarkand, Uzbekistan

²Samarkand state university of architecture and civil engineering, senior lecturer. Samarkand,
Uzbekistan

³Samarkand state university of architecture and civil engineering, student

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Abstract. *The article analyzes some of the shortcomings that arise during the operation of gas networks and measures to eliminate these shortcomings. At present, saving natural gas and automatic regulation of gas pressure are among the main issues*

Keywords: *gas networks, gas pipelines, energy efficiency, gas consumers, reliability, equipment*

Introduction. In the process of fuel extraction their value increases as they are extracted deeper and deeper. Therefore, it is natural that the need to save fuel costs becomes the main problem of the development of the national economy of our country. The growing cost of fuel energy throughout the world shows that fuel is essentially a limited resource. Therefore, it is the duty of every conscious person to save fuel energy in our country [1-4].

Gas distribution networks are the energy arteries of populated areas; therefore, any technical and structural problems are of great importance.

Gas supply and gasification enterprises and organizations face enormous challenges in ensuring uninterrupted and reliable gas supply.

Experimental studies. Gas supply systems operate around the clock with variable modes depending on the nature of gas consumption. The greatest unevenness of gas consumption is inherent in small household consumers, however, fluctuations in gas consumption for household needs have a certain pattern.

Hourly fluctuations confirm the lack of complete repeatability of daily schedules even on ordinary days of the same week and a significant change in the nature of uneven gas consumption by season, on pre-holiday days. Usually, the daily consumption schedule is characterized by morning and evening peaks, and at night gas consumption decreases several dozen times. Uneven gas consumption is caused by a large number of factors, the main ones being: climatic conditions, the way of life of the population of a particular area, the working hours of enterprises and institutions, the state of the housing stock, the degree of gasification of different categories of consumers. Uneven gas consumption determines the pressure modes in urban gas networks. Continuous periodic deviations in gas consumption by hours of the day from the average daily value have a major impact on the operating modes of gas equipment and automatic control devices.

The nature of the average daily gas consumption of municipal and household consumers has been sufficiently studied and can be considered as a continuous periodic function with a period of 24 hours (1 day) in the annual context (Fig. 1.0), represented by a Fourier series. The sum of

the Fourier series taking into account the numerical values of the coefficient for the average daily gas consumption will have the form:

$$Q(t) = 4.2 + (-1.2)\cos\frac{2\pi t}{24} + (0.05)\cos2\frac{2\pi t}{24} + (0.08)\cos3\frac{2\pi t}{24} + (-0.08)\cos4\frac{2\pi t}{24} + 0.17\cos5\frac{2\pi t}{24} + (-2.02)\sin\frac{2\pi t}{24} + (-1.86)\sin2\frac{2\pi t}{24} + (-0.04)\sin3\frac{2\pi t}{24} + 0.09\sin4\frac{2\pi t}{24} + 0.04\sin\frac{2\pi t}{24}2 \quad (1.0)$$

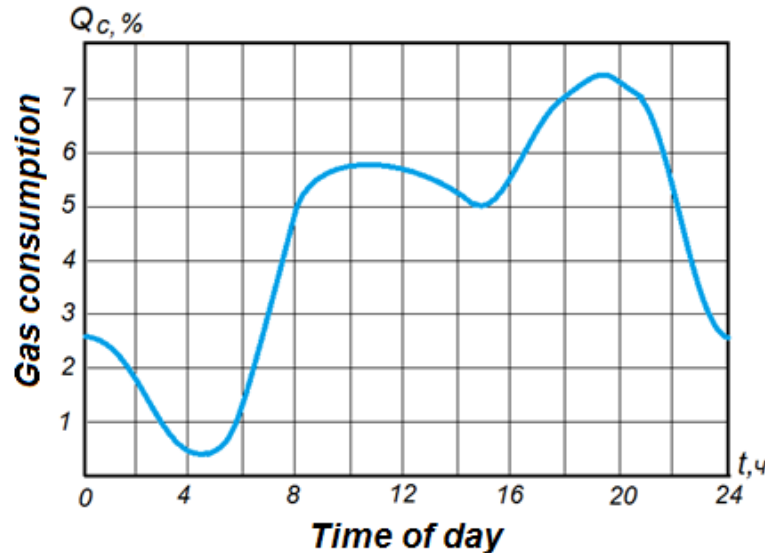


Fig. 1 – Graph of average daily gas consumption with household load.

All harmonic components of the Fourier series of the expression have a very definite effect on the main controlled pressure parameter, since these harmonic components approximate the average daily gas consumption very well. Higher harmonic components are intensively suppressed by the object and do not have a significant effect on the fluctuations of the controlled pressure. Expression (1.0) can be used to determine the upper limit of the frequency, the effect of which must be taken into account. This frequency corresponds to $\omega = 0.022 \text{ min}^{-1}$. The lower limit of the frequency can be taken equal to zero, as the limit, which corresponds to the stepwise effect of the disturbance on the system ($t = \infty$), i.e. It is possible to limit ourselves to the spectrum of frequencies acting on the system in the range $\omega = 0 \div 0.022 \text{ min}^{-1}$.

As is known, the most effective solution to the issues of pressure control in gas supply systems can be achieved by analyzing the main controlled process - the unsteady movement of gas in distribution gas pipelines. Since urban gas supply systems consist of several stages of gas distribution, the unsteady movement of gas in low-pressure distribution gas pipelines will be caused by hourly unevenness of gas consumption and the presence of variable pressures at the inlet of the gas distribution pipeline. This is explained by the fact that low-pressure gas pipelines supply mainly mass household consumers [5-13].

The main requirement for the gas supply system and at the same time one of the most difficult aspects to implement is maintaining the gas pressure of equipment and devices using gas at a certain permissible value with arbitrary changes in the flow rate in the network, in a wide range. When the gas pressure increases compared to the nominal, the operating modes of devices

and apparatus using gas are disrupted, and when the pressure decreases, their efficiency and productivity decrease.

The cost of designing, constructing and operating gas supply systems depends mainly on the population, the density and number of storeys of buildings, the structure of the fuel and energy balance, the number of gas consumers and energy capacities, as well as their geographic location.

As is known, gas is supplied to gasified cities, towns or industrial facilities from main gas pipelines through gas distribution stations (GDS). GDS are the final objects of the main gas pipeline and perform the following tasks: gas purification from mechanical impurities; reducing the gas pressure to a given value and automatically maintaining this value; heating of gas before pressure reduction, preventing the release of solid crystal hydrates and freezing of pipelines and fittings; protection of pipelines from unacceptable increases in pressure; gas odorization; accounting of the flow rate and quantity of passing gas.

From the GDS, gas is transported through a medium or high pressure network to gas regulation stations (GRS), located in heated detached buildings, where the gas pressure is reduced and it is fed into distribution gas pipelines of different pressure categories. The most branched and, therefore, long and expensive are low-pressure distribution gas pipelines, which supply the mass consumer (residential buildings, small industrial and municipal household consumers). Gas pipelines are laid mainly underground, their diameters usually vary from 50 to 400 mm. This requires the installation of electrochemical protection, insulation of pipelines, gas wells, control and measuring points; therefore, gas pipelines are characterized by significant metal consumption.

The main requirement for the gas supply system and at the same time the most difficult to fulfill is maintaining the gas pressure in gas-using equipment and devices at a given optimal value with arbitrary changes in flow in the network over a wide range. when the gas pressure increases against the nominal value, the operating modes of gas-using devices and installations are disrupted, and when the pressure decreases, their efficiency and productivity decrease[2-9].

The difficulty of maintaining the nominal gas pressure with the required accuracy for consumers is due to the fact that the service radius of a separate gas distribution point often reaches 900-1500 m, which leads to a significant drop in gas pressure depending on the distance of consumers from the gas distribution point.

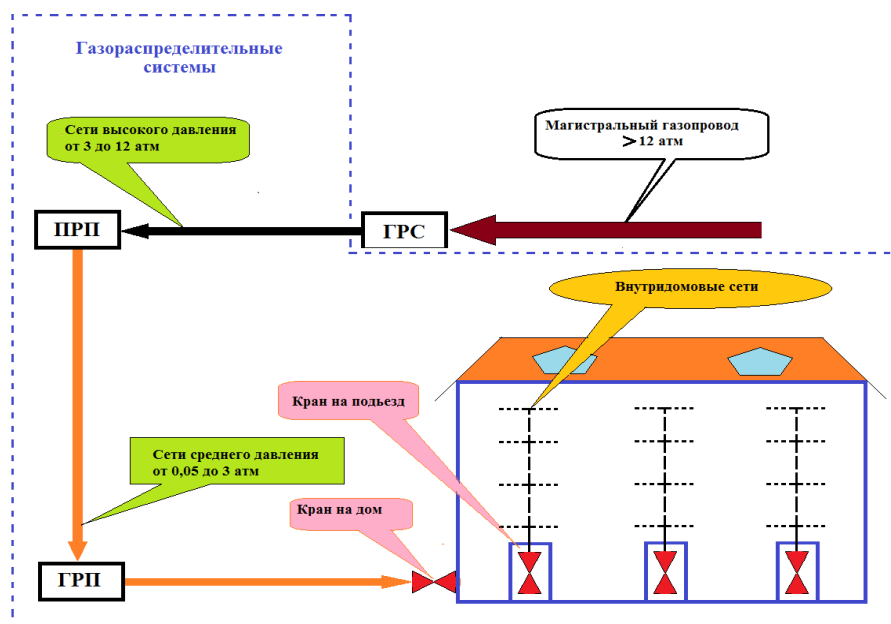


Fig. 2 - Diagram of gas network layout.

Reliable and stable operation of gas supply systems is impossible without reliable operation of control and safety shut-off valves and equipment. The first and main condition for stable and safe operation of the gas supply system is to ensure constant pressure; the second condition is protection against a possible increase or decrease in gas pressure at a controlled point of the gas pipeline or in front of a gas-using installation, unit or consumer device in excess of permissible values.

In accordance with these conditions, the GRP, GRU or combined pressure regulator includes the following elements:

- 1) pressure regulator, reducing gas pressure and automatically maintaining it at a given level regardless of changes in flow rate and input pressure;
- 2) safety shut-off valve, stopping gas supply in case of emergency increase or decrease in gas pressure after the regulator 50 above the set limits;
- 3) safety relief device, preventing an increase in gas pressure after the regulator to exclude false operation of the safety shut-off valve. This is usually observed in the system during transient modes or no gas consumption and during gas leaks through a closed pressure regulator valve;
- 4) filter for cleaning gas from mechanical impurities.

Currently, the economic factor is becoming increasingly important. Thus, the use of polyethylene pipes in the construction of gas pipelines reduces the costs of construction work and operation.

When designing or reconstructing gas supply systems, the choice of gas pressure in gas pipelines is of great importance. The higher it is accepted, the smaller the diameter of the gas pipeline is required. Often in populated areas there is a need to install regulating devices directly at gas consumers [2-5, 7-14].

When reconstructing worn-out gas pipelines, the most effective, from the point of view of the cost of construction work and subsequent operation, is to pull polyethylene pipes through them, while the cross-section of the gas pipeline is reduced and there is a need to increase the pressure in it, and therefore the need to install house regulators or cabinet regulating points.

Conclusions. It can be said that the closer the control device is to the gas consumer, the more accurately the pressure in front of it is maintained, that is, the gas equipment operates with good efficiency in the passport mode and less harmful emissions into the atmosphere.

The correct choice of the number, type and location of control devices determines not only the technical and economic indicators, but also the reliability of the entire gas distribution system.

With the introduction of a regional automated gas distribution control system, the level of technical and economic indicators will be even higher.

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THE ROLE OF SPEECH ACTIVITY IN THE DEVELOPMENT AND EVOLUTION OF LANGUAGE

Malikova Iroda

Samarkand State Institute of Foreign Languages

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Abstract. *Modern linguistics studies language as a complex system, which necessitates the analysis of paradigmatic and syntagmatic relationships. These relationships allow individual language units to interact with others across various levels of linguistic structure. Each language unit is connected and realized through paradigmatic and syntagmatic relationships, making them essential components for the functioning of the language system. All linguistic units participate in paradigms, while syntagmatic interactions transform these units into speech elements during speech activity.*

Keywords: *language, process, interpretation, external linguistic factors, smaller paradigms, Interdependent.*

Introduction

In his book "F. de Saussure's Linguistic Conception", I. Quchqortoyev writes: "The primary concern for a language user is the current state of the language (its synchronic state), as this is defined by the existing system of linguistic tools. From this perspective, only the synchronic state of language constitutes a system. However, this view, while significant for linguistics, cannot fully explain the nature of language. Language is an adaptive tool that evolves in response to communicative needs, and this adaptation occurs continuously during the speech process." [1:31-32]. According to Saussure, speech activity has two aspects: individual and social, which are inseparable. In addressing the question, "What is language?", he emphasizes that the concept of "language" is not equivalent to "speech activity." Instead, language is a component—though the most significant one—of speech activity. It is a social product and exists in the memory of individual speakers, allowing them to engage in speech activity. This connection between individual memory and speech activity aligns with cognitive linguistics, which is grounded in the theory of knowledge and emphasizes the human factor. Information about reality finds expression in human cognition and consciousness. Similarly, pragmalinguistics, with its focus on human-centric factors, directly intersects with cognitive linguistics. Sh. Safarov notes: "When discussing cognitive activity, it is important to consider not only reasoning and intellectual processes but also the outcomes of thought and associated non-mental phenomena." [3:10]. A. Kuchiboyev elaborates on the role of cognitive knowledge in speech activity, stating: "Cognitive knowledge facilitates the perception of implicit meaning through presuppositions and contextual interpretation, leading to the comprehension of speech." [4:15]. These views highlight the crucial role of the human factor in addressing linguistic problems from cognitive and pragmatic perspectives. E.A. Popova affirms this, stating: "It is difficult to imagine the progress of linguistics without considering the human factor. In earlier periods, linguistics was entirely devoid of human considerations." [5:69]

Popova seems to blur the lines between internal and external linguistic issues. While the human factor undeniably plays a critical role in language evolution, it is considered an external linguistic element. Despite the direct connection between language systems and human cognition,

language operates within its internal rules. This distinction validates Saussure's division of linguistics into internal and external domains.

However, recognizing the autonomy of internal linguistics does not diminish the importance of external linguistic factors. Saussure himself acknowledges: "Studying external linguistic factors—non-linguistic phenomena—is essential. However, the internal structure of language can still be studied independently of these factors, as language is a system governed by its internal order." [2:31]. This insight suggests that while external factors contribute to language application, internal linguistics is centered on the structural principles unique to the language system.

R. Saifullayeva and colleagues classify paradigms into larger and smaller systems, as well as external and internal paradigms. For example, consonants can be categorized as voiced and voiceless, representing smaller paradigms. The broader consonant system, in turn, constitutes a larger paradigm.

These paradigms highlight associative relationships within linguistic systems, often referred to as systemic or paradigmatic relations. Saussure also differentiates between the linguistics of language (*langue*) and speech (*parole*), advocating for the study of both while emphasizing their distinction. He writes: "For three thousand years, linguistics has focused on language. Now, it is time to study speech, which remains an underexplored field awaiting researchers." [2:29].

The very existence of the language system is tied to speech activity, as its nature and essence are revealed through communicative processes. Consequently, the semantic properties of linguistic units are realized and clarified in speech. In his work "Semantics", Sh. Safarov highlights the critical role of nomination (naming) in speech activity, stating: "The process of linguistic sign selection during naming—whether arbitrary or motivated—is a fundamental issue in theoretical linguistics." [8:20]. The interconnected nature of linguistic elements ensures that their paradigmatic and syntagmatic relationships become apparent during speech. Though differing in linguistic status, these relationships are interdependent and can be observed in the interaction of units across language levels.

A. Nurmonov, reflecting on speech activity, references V. Humboldt's idea that language is not a static *ergon* (product) but rather a dynamic *energeia* (energy). Based on this premise, he concludes that the primary object of linguistic study should be speech activity. He argues: "Speech activity integrates the social and individual dimensions of language, forming a complex process rooted in the dialectical relationship between language (*langue*) and speech (*parole*)." [9:17].

Paradigmatic relationships are defined by several key features:

1. When one member of a paradigm is recalled, other members of the same paradigm are also brought to mind.
2. In any specific speech situation, only one member of a paradigm is chosen.
3. Paradigm members share similarities while differing in at least one distinctive feature.
4. Members can replace or substitute one another in specific contexts.

These characteristics underline the essential role paradigmatic relationships play in shaping language as a system and enabling its use in speech.

Studying language as an integrated system aids in understanding speech activity. Modern linguistics distinguishes between language and speech, resulting in the scientific development of both linguistic domains. While language linguistics has been extensively studied over three millennia, speech linguistics remains an emerging field requiring further exploration. Sh. Safarov notes: "The system-structural paradigm resolved many of the shortcomings of earlier approaches,

which often analyzed linguistic phenomena in isolation. However, this paradigm introduced its own limitations, such as the separation of language from its user—humans. Efforts to address these shortcomings led to the development of pragmatic and cognitive linguistic paradigms." [7:38].

The hierarchical organization of linguistic elements involves selecting units from various levels—phonemes, morphemes, words, phrases, and sentences—each contributing to the construction of meaning. For example, phonemes combine to form sounds and letters, while morphemes contribute to lexemes, and words build sentences and texts.

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IMPORTANCE OF READING COMPREHENSION

Kurbanova Zarina Odiljonovna

Teacher of Samarkand State Institute of Foreign Languages

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Abstract. *Reading comprehension is a fundamental skill that plays a pivotal role in academic success and personal development. It goes beyond simply decoding words; it involves understanding and extracting meaning from text. In this article, we will explore the significance of reading comprehension and its far-reaching impact on various aspects of education.*

Key words: *reading skill, problem solving, global language, critical thinking, career advancement.*

Introduction

The value of mastering all four language skills—reading included—has increased as English becomes more and more important as a global language. Because English is widely used in so many fields, including science, technology, education, and business, reading comprehension is essential for good communication.

1. Academic Achievement:

Reading comprehension is at the core of learning across all subjects. Whether it's understanding a math problem, interpreting a historical event, or comprehending scientific concepts, the ability to grasp written information is crucial. Students with strong reading comprehension skills tend to perform better in exams and assessments.

2. Critical Thinking:

Proficient reading comprehension goes hand in hand with critical thinking. It allows individuals to analyze, evaluate, and synthesize information. Engaging with complex texts promotes the development of analytical skills, enabling students to think critically about the content they encounter.

3. Effective Communication:

A high level of reading comprehension correlates with improved communication skills. People who can understand written information more effectively can articulate their thoughts and ideas more clearly, both in writing and verbally. This skill is invaluable in various professional and personal contexts.

4. Lifelong Learning:

Reading comprehension is not only essential for formal education but also for continual self-directed learning. In an era where information is abundant, those who can comprehend and apply new knowledge are better equipped to adapt to evolving circumstances and pursue lifelong learning.

5. Problem Solving:

Many real-world challenges require the ability to comprehend and analyze complex information. Whether in the workplace or daily life, individuals who can navigate and understand written material are better equipped to solve problems and make informed decisions.

6. Cultural Awareness:

Reading exposes individuals to diverse perspectives, cultures, and experiences. Strong reading comprehension fosters cultural awareness and empathy, as readers gain insights into the lives and perspectives of others through literature and informational texts.

7. Career Advancement:

In the professional world, effective communication and critical thinking, both rooted in reading comprehension, are highly valued. Employees who can comprehend complex documents, analyze data, and communicate clearly often have a competitive edge, contributing to career advancement.

Teaching Reading rules: There are a number of rules that teachers can follow to help English language learners learn to read. Among these guidelines are:

Reading is an active ability: Reading ought to be an active activity in which students actively interact with the text, comprehend word meanings, analyse images, comprehend arguments, and develop opinions. There is a greater chance of forgetting and a surface-level grasp when reading passively. Pupils must be enthusiastic about their work: Instructors must to design reading assignments that actively involve pupils and encourage sincere curiosity and drive. Students that actively participate get the most out of their reading experiences. Students should react to a reading text's context in addition to its language:

Understanding the text's meaning and message is reading's primary objective. Students should react to a reading text's context in addition to its language:

One important aspect of reading is prediction: An important part of reading is making predictions. Before reading, teachers should help students develop predictions about the text by employing contextual signals including book covers, images, contents, and titles. This facilitates the activation of pupils' past knowledge and creates the environment for fulfilling reading experiences. Align the task with the topic: Instructors should make sure that reading assignments are in line with the text's purpose and topic. It is essential to create questions and activities that stimulate students' thinking and encourage them to interact with the book.

In conclusion, the importance of reading comprehension cannot be overstated. It forms the foundation for academic success, critical thinking, effective communication, and lifelong learning. As we recognize its significance, educators and learners alike should continue to prioritize and enhance reading comprehension skills to prepare for the challenges and opportunities of the future.

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NEGATIVE IMPACTS OF INTERFERENCE ON SECOND LANGUAGE LEARNING: AN ANALYSIS OF CONTRIBUTING FACTORS

Rasulova Istoda Abdulatifovna

Samarkand State University

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Abstract. *This article studies the nature and types of interference phenomena, their causes, and analyzes the opinions of various authors on this issue and ways to eliminate interferentive errors in the speech of bilinguals.*

Keywords: *Interlanguage interference, bilingualism, internal interference, external interference, phonetic and grammatical interference*

Introduction

The contemporary sociolinguistic landscape, characterized by increasing multilingualism, demands a reevaluation of foreign language teaching methodologies. President Shavkat Mirziyoyev's emphasis on educational reform underscores the urgency of this need. The development of effective strategies to prevent and remediate negative linguistic interference is crucial for preparing students for diverse and globally interconnected environments.

The best option depends on the specific context and intended audience. The original text is a bit rambling; these options aim for greater clarity and conciseness.

The increasing prevalence of multilingualism necessitates a deeper understanding of the factors influencing second language acquisition (SLA) in multilingual contexts. Uzbek primary school students, exposed to Uzbek, Russian, and a foreign language, provide a case study for investigating the impact of prior language learning on subsequent acquisition. The interplay of these languages affects linguistic development, resulting in both positive transfer (facilitation) and negative interference (inhibition). Linguistic interference, defined as the intrusion of features from one language into another, can impede SLA at various linguistic levels, from phonology to syntax and pragmatics. This phenomenon is the result of the dynamic interaction of language systems in the multilingual mind.

Choose the option that best suits your needs and intended audience. The original text contains some redundancy; these rephrased versions aim for greater clarity and conciseness.

In most of cases, interference is realized as the result of language contact, it is to say that, as the result of speaking two or more languages, bilinguals do not follow language norms and get out of them¹.

In order to study interference as a sociolinguistic phenomenon and to prevent the mistakes occurring because of interference a number of works have been completed by world linguists such as U.Weinreich, Ellis, Rod, E.D.Polivanov, Vishnevskaya, Barannikova, L.I.Aliev; and by Uzbek

¹ Вайнрайх У. Однoязычие и многoязычие. В "Новое в лингвистике" / Под ред. В.Ю. Розенцвейга. / Вып. 6. М.: Изд. Прогресс, 1972. -С. 25-60.

linguists like U.K.Yusupov², D.D.Djalolov³, M.Djusupov⁴, G.T.Makhkamova⁵, K.Nabieva⁶, J.Buronov⁷.

When the representatives of two different languages or dialects communicate or write, there occurs inter-lingual communication. In other words, a contact that happens between two or more language owners is a inter-lingual communication. For instance, when the Uzbeks and the Tajiks communicate with each other, i.e. enter into a language contact, there happens Uzbek-Tajik language contact. In this case, if they communicate in Uzbek, Uzbek becomes language of communication. If they speak Tajik, Tajik becomes language of communication. However, they may also communicate with each other in Russian. In this case, Russian is the third language that is used as a mean of communication, and it is called “a lingua franca”. As a result of the language contact, these speakers may get detached or closer. The former situation is called “convergence”, and the latter – “divergence”. For the language contact to happen, one or both sides of the communication must know the native language of the opposite speaker or the lingua franca. Speaking two languages is called “bilingualism”, and the person able to speak two languages is called “bilingual”. A bilingual may transfer the rules of one language into another in a wrong way. This phenomenon is defined as “interference”. Interference usually happens in transferring the rules of the native language into another language.

It is known that teaching a foreign language provides to clarify a subject matter from all perspectives. Based on the existing experiments it is possible to identify and describe the main reasons of interference phenomenon in the range of the system of the following four components:

- linguistic;
- psychological;
- sociolinguistic;
- Stylistic inappropriateness.

To the linguistic factors that indicate appearance and development of interference, primarily belong inappropriateness and similarity of the languages of contact.

According to U.Weinreich, “mixing is always connected to the structure of the language, as if it is a particular answer of the language structure to a strange element that entered into it [9, p. 1]⁸. It is necessary to note that identifying the reason of interference only by similarity or on the contrary by the difference between the structure of the languages is likely to be simplified. It is possible to suppose about the closeness of these concrete structures that they create the most comfortable conditions for the appearance of mixing of the languages that mutually influence each other. The founder of Saratov’s linguistic school L.I.Barannikova said: “From one hand linguistic

² У.К.Юсупов. Проблемы сопоставительной лингвистики. Ташкент: Фан, 1987

³ Жалолов Ж.Ж. Тил таълимида методик компрессия ва турли интерференциялар хақида // Хорижий тилларни ўқитиш методикаси ва замонавий тилшуносликнинг долзарб муаммолари. – Андижон: АнДУ, 2017.

⁴ Звуковые системы русского и казахского языков. Слог. Интерференция. Обучение произношению / М. Джусупов; М-во нар. образования УзССР, Респ. пед. ин-т рус. яз. и лит. - Ташкент : Фан, 1991. - 239 с.;

⁵ Формирование межкультурной компетенции студентов филологического вуза: (английский язык): автореферат диссертации на соискание ученой степени доктора педагогических наук: 13.00.05 / Махкамova Гульнара Турдахуновна

⁶ Набиева, Карима Методика преодоления межъязыковой лексической интерференции при обучении письменному высказыванию в национальных группах языкового вуза : диссертация ... кандидата педагогических наук

⁷ Ж.Бўронов “Инглиз ва ўзбек тилларининг қиёсий грамматикаси”–Тошкент “Ўқитувчи”, 1973.

⁸ Вайнрайх, У. Языковые контакты: состояние и проблемы исследования / пер. с англ. и коммент. Ю. А. Жлуктенко / У. Вайнрайх. – К., 1979

closeness facilitates the identifies transfer from one language into another, the development of bilingualism and the mixing of languages ... but the closeness of the languages is very limited in it itself. It identifies particular boundaries of language mixing that is connected with numerous similar examples in the structures, because the second appears on the basis of similarity and difference” [2, p. 16]⁹. In this way, it would be honest to denote the reasons of interference as similarities and differences in language structures that mutually influence each other.

Here, without a doubt, the level of formation of the speech skills in the learning language is considered as an additional factor: an imperfect language acquisition form comfortable conditions for the originating interference and contrarily, developing the skills diminishes the level of interference. While analyzing the linguistic reasons of interference, we consider that the criteria like the conditions of the communicative actions, more clearly, the field of contact and the features of its participants is not of little importance. Talking about daily communication, the participants of the contact may ignore to follow the concreteness of pronunciation and some grammatical norms, what about in academic environment, a formal speech provides meaning of speech, concreteness and clearance of its form, and in the result there will be formed a chance to control possible interference. Also, it is purposeful to take into consideration whether a speaker taking part in a communication is a monolingual or bilingual. Bilingualism of the speaker and his awareness of the chance to understand the transfer from language to language or the mixing forms help to appear mixing mistakes in his speech. But monolingualism of the speaker serves to control the correctness of the speech and prevent a potential mixing¹⁰.

As U.Weinreich pointed out that while studying interference it is necessary to take into account the linguistic and extra-linguistic factors [9, p. 1]¹¹. Continuing to define the reasons of interference, we highlight the psychological group and among them we analyze the factor of language ability.

A popular specialist in the fields of psychology and linguistics A.A.Leontyev defines linguistic ability as following: “A language ability is a complex of psychological and physiological conditions that provide the language signs to be assimilated, formed, improved and perceived in a required level by a linguistic community” [6, p. 54]¹². Based on this definition, it is possible to suppose that a weak development of linguistic abilities is one of the main psychological reasons of speech disorder, because in this case the level of accepting and perceiving the rules of one language diminishes for transferring them into another language.

As one of the main features we can also show the connection between the emotional state of the speaker (as well as his tiredness level) and the range of interference. It is reasonable to address the words of U.Weinreich as a proof of our opinion: “To overcome interference demands such actions that are not always easy” [3, p. 54]¹³. Here we must talk about psychological actions and emotional stress which are important to control over the confusions between native and learning languages causing numerous interferentive mistakes.

⁹ Баранникова, Л. И. Вопросы методики преподавания иностранных языков в связи с проблемой языковой интерференции / Л. И. Баранникова. – Саратов, 1966.

¹⁰ Вайнрайх, У. Одноязычие и многоязычие / У. Вайнрайх // Новое в лингвистике. Вып. VI. – М., 1972.

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¹² Леонтьев, А. А. *Язык, речь, речевая деятельность* / А. А. Леонтьев. – М., 2010.

¹³ Вайнрайх, У. *Языковые контакты: состояние и проблемы исследования* / пер. с англ. и коммент. Ю. А. Жлуктенко / У. Вайнрайх. – К., 1979.

The existence of psychological and sociolinguistic factors in bilingual's conscience has a syncretic character. In the 50's of the XX century famous American psychologist Ch.Osgood and sociolinguist S.M.Ervin divided bilingualism into two types: "compound" and "coordinate". In order to study this bilingualism phenomenon language acquisition is defined as following: in the case of learning languages in a "mixed" environment (exercising often to move from one language into another, to explain similar meanings), the "compound" type is developed. Turning from language into another when this environment or action changes, is denoted by "coordinate" bilingualism [6, p. 241-253]¹⁴. In the former situation, languages form connected parts of the united system in speaker's conscience. What about the latter type bilingualism, in it the languages form separate existing systems that own important functional independence. Based on this fact, we consider that compound bilingualism can motivate interference to origin and develop more than coordinate bilingualism can.

The sociolinguistic factors that give rise to interference phenomena in a speaker's speech should include, first and foremost, the socio-historical context in which the language is associated.

We support L.I.Barannikova's opinion, she writes as following: "Among the factors influencing the level of development of interference are, first of all, the existence of general socio-historical conditions, the speakers of the languages with which they communicate" [2, p. 12]¹⁵. Except the factors mentioned by the linguist, in our opinion, social status of the language can be expressed by the fact whether the language is given state status, which directly affects its authority and, consequently, its attitude to the purity of language and its compliance with norms.

From this arises the next social relation which we have pointed out. According to Weinreich, "in some societies the transition from language to language is considered normal, while in others it is viewed negatively. If interference is allowed in one language, it is condemned in another". As an example the author shows English-French bilingualism in Canada: the attitude to the state of interference may be in the nature of individual and social phenomena. In Canada, making pronunciation errors (accents) when speaking English are less socially beneficial than in the United States, for example, when a French cleaner speaks English with a French accent, on the contrary, it may be useful in terms of his social background and position [3, p. 53]¹⁶.

Now we analyze methodological factors by identifying such reasons of interference phenomena as linguistic, psychological, sociolinguistic features (as well as, the crossroad of these subfields), firstly, we approach to them and address the teaching method that is shown by this approach. A practical and comparative study, the most wide-spread method among the group of the methods based on the conscious approach, gives a little a chance for interference. For the influence of one language on another to be less, it is purposeful to study similarities and differences between the structures of native and foreign languages. Perfect knowledge of one's native language serves as an important factor in learning a foreign language.

Thereby, interference is produced mainly in presence of four aspects – language norms, bilingual's character, relationships between native and foreign languages, social conditions. Based on these aspects, there was created a four-dimensional typology of interference:

Based on these aspects, they created the following four-dimensional typology of interference:

¹⁴ Weinreich, U. Languages in contact / U. Weinreich. – N. Y., 1953. – P. 1

¹⁵ Баранникова, Л. И. Вопросы методики преподавания иностранных языков в связи с проблемой языковой интерференции / Л. И. Баранникова. – Саратов, 1966.

¹⁶ Вайнрайх, У. Языковые контакты: состояние и проблемы исследования / пер. с англ. и коммент. Ю. А. Жлуктенко / У. Вайнрайх. – К., 1979.

- 1) due to deviation from the language norms (linguistic aspect);
- 2) according to the specifics of the speech activity of the bilingual (psychological aspect);
- 3) according to the relationship between the native language and the languages studied (inter-linguistic aspect);
- 4) according to the social situation (socio-ethnolinguistic aspect)¹⁷.

Interference is a phenomenon that is not up to the will of the speaker. Its amount corresponds to the increase of person's acquisition of language and it totally disappears when the language is fully mastered. The structural reason of interference depends on the differences at each level of two languages, while the nonstructural reasons are equivalent to the psychological laws of inter-lingual equality. Of all the events that occur as a result of language interactions (interference, transference, intercalation, transcalation), only interference can be predicted, i.e., a linguist can target an interference phenomenon as the result identifying similarities and differences by comparing two languages. Other phenomena are unpredictable as their formation depends on the influence of many extra-linguistic aspects.

Regarding interference G.M.Vishnevskaya argues that as a result and process of the influence of the language system on bilingual's speech, one language system becomes dominant, and this effectively affects the system of the second learned language.¹⁸

For example, in Uzbek word "tulki" (fox) there is no need to put a soft sign after the letter "l", though it is pronounced softly. But if the Russians want to write this word, they write and pronounce it like this: "тульки". Such kind of misunderstandings cause difficulties in learning language.

It is important to study the didactic and psychological aspects of not only the theoretical problems of phonetic interference (reasons of origin, types), but also preventing and overcoming potential mistakes. In particular, Methodist K.Kadyrova in teaching Arabic to Uzbek students to explain the reasons of phonetic interference, addresses O.A.Yamshchikova's following opinion: "Phonetic interference is the result of an effective influence of particular skills system as a leader on the skills acquired later. One of the main features of phonetic interference as a psychological phenomenon in the methodology of teaching another language is the violation of auditory-pronunciation skills as a result of interactions and the opposite perception of the phonetic system of the studied language.

Methodist scholar V.S.Dmitreyeva quotes N.S.Trubetskoy's opinion, who paid special attention to the problems of bilingualism in the field of linguistics: "In the process of listening and analyzing speech, we usually use our phonological "filters", which are primarily considered in our native language. The fact that our "filters" do not correspond to that of another language (which, of course), leads to a number of errors and misunderstandings. The sounds of other language accept the wrong phonological interference of our language.¹⁹ So, an accent in a foreign language is not an inability to pronounce this or that sound, but it is connected to "the wrong

¹⁷ Карлинский А.Е. Экспериментальное изучение лексической интерференции в прикладных целях // -Алма-Ата, 1989, с.60.

¹⁸ Вишневская, Г.М. Билингвизм и его аспекты [Текст] / Г.М. Вишневская. – Иваново, 1997. – 100 с. 5,с24

¹⁹ Дмитриева, Ю.В. Проблема межъязыковой интерференции в преподавании иностранных языков в вузе Электронный ресурс. / Ю.В. Дмитриева. Сайт Саратовского государственного университета имени 49. Н.Г. Чернышевского. Режим доступа: <http://www.sgu.rti/files/nodes/13579/Dmitrieva.doc>. - Загл. с экрана

conclusion about the sounds” based on differences between the phonological systems of native and foreign languages.

In the early stages of language learning, the sounds of foreign language are learned side by side with those of native language. A relatively common type of phonetic interference is similarity-based interference, and it is usually connected to a direct transition of native language pronunciation skills to similar foreign sounds while acquiring the sound system of studied language.

Methodists divide skills into three groups:

- 1) Transferring speech skills of native language into the data of a new language;
- 2) Transferring previously acquired skills into the data of a new language;
- 3) Newly acquired skills.

According to S.G.Nikolaev²⁰, it is possible to make a conclusion that there are three approaches in teaching foreign languages taking into account the features of native language:

- 1) To compare the facts and similarities of the two languages, and to acquire by means of the similarities;
- 2) To compare the facts of the two languages and to set the similarities, and to acquire by means of differences;
- 3) To compare the facts of the two languages and to set the incompatible ones, and to acquire by means of contrast.

We can use all aforementioned three approaches, if we take Uzbek as a native language, and Turkish as a foreign language. Both languages belong to the same language family, and therefore, they have common features. But Uzbek and Russian belong to different language families, thus there should be paid more attention to many different features in these languages.

While teaching children to the second language, it is important to take into account the experience of teaching speech in native language. For example, acquiring separate sounds articulation is the most difficult task for a child. O.I.Solovyeva pointing out the main directions of developing the sound culture of speech, argues that the educator has the task to teach children to pronounce the sounds in words clearly, to pronounce words correctly in accordance with the rules of orthoepy of the language, to teach them to cultivate an expressive speech.²¹

Thus, it is significant to pay attention, firstly, to hearing correctly and pronouncing clearly. Only if we pronounce the sounds purely and correctly, the process of learning a language becomes easy.

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PRIORITY AREAS FOR THE DEVELOPMENT OF QUALIFIED PERSONNEL COMPETENCIES FOR SUSTAINABLE ECONOMIC GROWTH AND EFFECTIVE MANAGEMENT

Koshkalda Iryna¹, Dombrovska Olena², Kovalov Oleksandr³, Abdukhamidov Abilkosim⁴

¹Doctor of Economic Sciences, Professor, Head of the Department of Land Administration and Cadastre, State Biotechnological University, Kharkiv, Ukraine, irinavit1506@gmail.com

²PhD in Economics, Associate Professor of the Department of Land Administration and Cadastre, State Biotechnological University, Kharkiv, Ukraine, domolena73@gmail.com

³PhD student, State Biotechnological University, Kharkiv, Ukraine, kovsashaa@gmail.com

⁴Ph.D., associate professor, Samarkand State University of Architecture and Construction.

Uzbekistan, abilkosim.abduxamidov.@email.ru

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Abstract. *The rapidly changing modern economy puts forward new requirements for the personnel qualifications and competencies. The article discusses the main priority areas for the development of qualified personnel competencies to ensure sustainable economic growth and effective management. The key competencies are outlined, including leadership, communication, decision-making, innovation, teamwork and result orientation. A model of competencies covering three levels is proposed: basic (knowledge), operational (skills) and strategic (values and attitudes). The key methods for training and development of competencies are identified.*

Keywords: *management competencies, sustainable development, competency-based approach, leadership, communication, motivation, innovation, decision-making.*

Introduction

In today's world, characterized by rapid socio-economic, scientific and technological changes, the need to develop a personality capable of adaptation, creative thinking and innovation comes to the fore. Such a personality is an important element in ensuring efficient management and promoting sustainable economic growth, as it is the driving force behind transformations in the modern world.

Sustainable economic development is based on the development of competencies that allow not only responding to challenges but also creating new opportunities for growth. Effective management requires the ability to think strategically, innovate and continuously improve, which directly depends on the level of professional training of qualified personnel and their ability to implement changes. Increasing requirements for staff competence, professional qualification level and adaptability contribute to the formation of the basis for stable economic development and improved management efficiency at all levels of government agencies, organizations and enterprises.

Some researchers believe that "competence is a set of interrelated human qualities (knowledge, skills, abilities) that are necessary for effective work in a particular field, and competence is the ability to possess these competencies, including personal attitude to the object of activity"[1]. It should be added that competence is not only about possessing competencies, but also about the ability to practically apply them in everyday activities. The author proposes to consider competencies at three levels: understanding, activity and being [1]. The first level, "understanding," includes the basic set of knowledge necessary for professional activity; the second level, "activity," is the ability to apply this knowledge in practice; the third level, "being,"

covers the system of professional and human values, as well as the ability to coexist harmoniously in society. Thus, competence can be viewed as a characteristic of a personality that allows him or her to solve problems and make decisions in a particular field.

The competency-based approach to the development of skilled personnel is a modern methodology aimed at developing not only professional knowledge but also a wide range of competencies that allow employees to work effectively in a constantly changing environment. This approach emphasizes the importance of developing the abilities, skills and attitudes necessary for sustainable economic growth and effective management.

The analysis of competencies characteristics and their structure identifies three levels at which competence is formed (Fig. 1). Each level of employee competence formation affects the other. For example, the personal level affects the cognitive and activity levels, because the employee's attitude to the activity will determine the level of knowledge and the way they obtain it, which in turn will affect the dynamics of competence development [2].

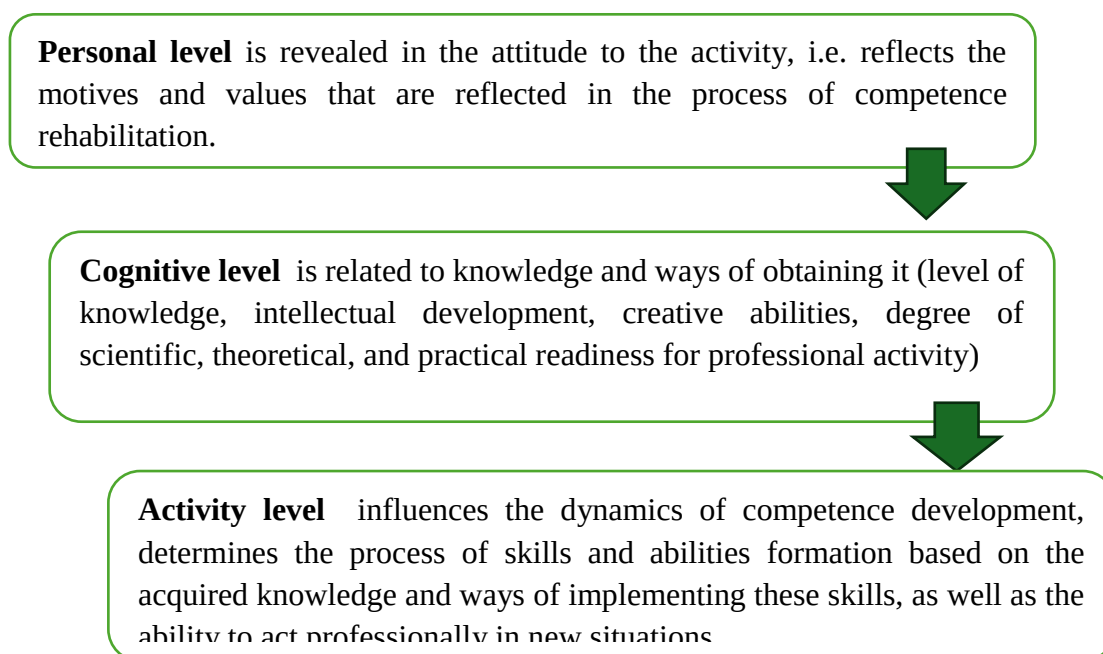


Fig.1. Levels of competence development [2].

In our opinion, the uniqueness of the competency-based approach to the development of skilled personnel for sustainable economic growth and effective management lies in its systemic approach focused on the development of not only professional knowledge but also critical practical skills. This approach differs from traditional methods in the following important aspects:

1. Focus on market needs:

- the competency-based approach ensures that skills are constantly updated according to changes in the labor market. It facilitates a rapid response to new economic challenges, including digitalization, globalization, environmental challenges and new business models.
- sustainable economic development requires not only preservation of existing knowledge, but also the ability to quickly adapt to new conditions, introduce innovations and work with the latest technologies.

2. Integration of professional skills: the unique approach is to combine technical competencies (hard skills) with social, communication, creative and leadership skills (soft skills). This harmonious integration helps employees not only to perform tasks efficiently, but also to cooperate effectively in teams, solve problems and initiate changes.

3. Focus on sustainable development and long-term goals: the competency-based approach considers the importance of sustainable economic growth by training personnel who understand long-term development strategies, environmental responsibility, efficient resource management, and the implementation of sustainable production principles. This allows to combine economic needs with social and environmental goals.

4. Continuous process of learning and self-improvement:

- this approach emphasizes continuous learning and development throughout life (life-long learning). This allows employees to be prepared for new challenges and take advantage of opportunities that arise in the labor market.

- continuous improvement of competencies allows to maintain a high level of competitiveness of both employees and enterprises in the global environment.

5. Comprehensive approach to management and leadership: the competency-based approach prepares personnel not only to perform specific professional tasks but also to make management decisions at the strategic level. It ensures the development of the ability to effectively manage organizations, which is essential for successful business in a challenging economic environment.

6. Implementation of innovations and technological changes: the modern labor market is actively changing due to technological innovations, and the competency-based approach ensures the training of personnel capable of not only responding to these changes, but also initiating them. This is especially important for ensuring effective management and sustainable growth in innovation-oriented economies.

Such a system promotes the integration of theoretical knowledge, practical skills and personal qualities, making future personnel competitive, flexible and ready to innovate.

To apply the competency-based approach, it is important to have a clearly defined classification of competencies [3]. D. McClelland, M. Dalziel, R. Page offer the following list of competency's types: 1) theoretical and professional knowledge; 2) job skills and universal skills (communication, management, etc.); 3) ability to master new activities; 4) value orientations and characteristics of motivation; 5) personal traits and psychophysiological characteristics [4].

Study [5] offers a model that provides a simple presentation of the direct and positive relationship between the identified managerial competencies and the main pillars of sustainable development. The authors found that there is a direct positive relationship between competencies in communication, leadership, development orientation, achievement orientation, motivation, teamwork, innovation and decision-making, economic viability, social inclusion and environmental protection.

In our opinion, important areas for developing the competencies of qualified personnel for sustainable economic growth and effective management are the following:

1. Professional (technical) competencies: this is a basic element of successful professional activity. Without proper knowledge and skills in a particular field, an employee will not be able to perform tasks at a high level. Professional competencies help to increase production

efficiency, improve the quality of products and services, which directly affects the competitiveness and sustainable economic growth of the organization and the state.

2. Managerial competencies: effective management of resources, people and processes is key to ensuring growth and sustainable development. Managers with these competencies are capable to plan strategies, optimize processes and make rational decisions. This helps to increase productivity, minimize costs and achieve long-term goals.

3. Digital competencies: given the rapid digitalization of the economy, employees with advanced digital skills can adapt to new working conditions and use the latest technologies to increase productivity. The introduction of digital tools allows automating processes, increasing efficiency, and providing access to global markets. This is critical for economic growth in the digital economy.

4. Innovative and creative competencies: innovation and creativity are the drivers of development. A workforce that is capable to generate new ideas and solutions helps businesses stay competitive, introduce new products and services, and optimize internal processes. This stimulates economic growth and the development of new markets.

5. Communication competencies: effective communication is the basis for successful interaction within the organization and with external partners. It helps to understand customer needs better, establish cooperation, resolve conflicts, and make quick decisions. This contributes to effective management and ensures sustainable development.

6. Social and ethical competencies: nowadays, when attention to social and environmental aspects is constantly growing, having ethical and social competencies helps organizations to adhere to high standards of responsibility. This strengthens reputation, increases trust from customers and society, which contributes to sustainable long-term development.

7. Analytical and strategic competencies: the ability to analyze data and strategic planning makes it possible to respond to market changes in a timely manner, predict risks and identify promising areas of development. This helps to reduce uncertainty and ensure sustainable development of the enterprise, which, in turn, contributes to stable economic growth.

8. Adaptability and resilience competencies: the modern labor market requires employees to be flexible and able to respond quickly to changes. In the face of global challenges and crises (such as pandemics or war), adaptability and resilience competencies allow organizations not only to survive but also to develop. This is the key to long-term stability and development in a changing environment.

These areas cover not only professional knowledge but are also crucial for sustainable economic growth and effective management. They allow organizations and institutions not only to carry out their activities effectively, but also to be prepared for the challenges of the modern world, to innovate, optimize processes and ensure long-term success.

Today, higher education institutions play a key role in shaping the competencies needed for sustainable economic growth and effective governance. They not only provide basic knowledge in various fields, but also contribute to the development of important skills such as critical thinking, leadership, innovation, teamwork, and decision-making. Through the integration of practical tasks, project activities, and research, universities develop students' ability to adapt to rapid changes in the business environment and successfully implement sustainable development strategies.

This is confirmed by the study of the authors [6], who point out that participants from different areas within the university, as well as external individuals and organizations, can come together to jointly address real-world sustainability challenges. Their Living Lab model is a dynamic network that connects the intellectual and other resources of an institution with practical sustainability challenges on or off campus. It effectively breaks down the boundaries between traditionally separated activities in education, research, external engagement, operations, and administration. It also allows the professional sustainability team on campus to enter into a closer dialogue with faculty and students to address real-world issues through experiential learning and teaching and/or research projects implemented both within the university and with external partners.

Competencies, such as knowledge, skills, abilities and attitudes, help to create a sustainable management system that is focused on long-term economic growth and efficiency. They allow not only to respond to changes but also to anticipate them, using innovative solutions and approaches that ensure sustainable development. We propose a model of competencies for achieving sustainable economic growth and effective management (Fig.2).

The development of these competencies will allow future professionals to effectively implement solutions aimed at sustainable economic development, while contributing to the improvement of management efficiency. A balanced combination of professional knowledge, innovation, leadership, social responsibility, and digital competencies will create the basis for the long-term competitiveness and sustainability of organizations and the economy as a whole.

To optimize the functioning of competencies, it is important to develop them comprehensively. This includes continuous training, development of leadership skills and analytical abilities, as well as the formation of ethical responsibility for decisions (Tab.1).

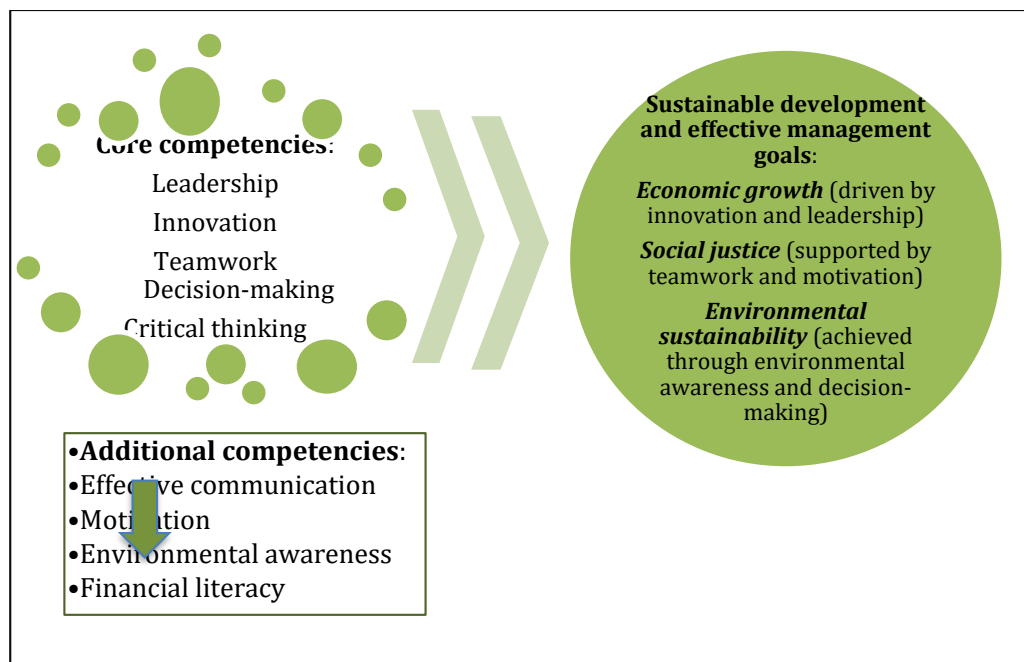


Fig. 2. Competency model for achieving sustainable economic growth and effective management.

Table 1. Key methods for training and competence development

	Specifications	Advantages	Illustration
1. Trainings and seminars	Practical classes with a focus on developing specific skills (communication, leadership, decision-making).	Ensure active participation, exchange of experience and feedback from experts.	Training in emotional intelligence, strategic planning or project management to improve management skills.
2. Coaching and mentoring	Individual approach, which provides support from an experienced specialist (mentor or coach) to develop competencies through consultations and recommendations.	Focus on personal and professional development through individual assignments, case studies and personalized learning strategies.	Coaching on leadership or decision-making, mentoring on implementing innovations in work processes.
3. Simulations and scenario modeling	Use of realistic models and simulations to develop competencies in conditions that simulate real-life situations.	It provides an opportunity to test and improve skills without risk, allowing employees to explore different solutions in difficult conditions.	Crisis management or strategic decision-making simulations.
4. Case-methods	Analyzing real or fictional business cases to develop analytical thinking, problem-oriented approaches and collective problem solving.	Allows to practice decision-making skills in real-life situations, analyze and solve problems that arise in professional activities.	Case studies on change management or innovation.
5. Team building and group work	Activities that promote teamwork, trust and interaction in groups.	Improves communication skills and the ability to work in a team, develops cooperation and joint problem solving.	Team building activities or working together on projects in interdisciplinary groups.

These methods can be adapted to the specifics of an organization or industry, and their combined use will ensure the comprehensive development of competencies necessary for sustainable economic growth and effective management.

Conclusions. The development of qualified personnel key competencies is an important factor in ensuring sustainable economic growth and effective management. Changes in

educational programs, emphasis on priority skills, and the introduction of innovative approaches will form the basis for building a human resource capacity which is able to successfully respond to the modern challenges. Ensuring sustainable economic growth requires professionals to have knowledge in the field of sustainable development, covering environmental, social and economic aspects. These competencies are becoming critical to preserving natural resources and ensuring social responsibility of business. In the context of global competition, there is a growing demand for innovative and entrepreneurial competencies. Educational programs should stimulate creative thinking and readiness to implement startups, new technologies and products that meet the requirements of the times.

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INTEGRATION OF DIGITAL TECHNOLOGIES IN THE TRAINING OF SPECIALISTS IN THE FIELD OF GEODESY, LAND MANAGEMENT AND URBAN PLANNING

Riasnianska Alona¹, Gurskienė Virginija², Pukite Vivita³, Abdukhamidov Abilkosim⁴

¹PhD in Economics, Associate Professor of the Department of Land Administration and Cadastre, State Biotechnological University, Kharkiv, Ukraine, alona.ryasnyanska@gmail.com;

²Dr. Sc., Associate Professor of the Department of Land Use Planning and Geomatics, Vytautas Magnus University, Kaunas, Lithuania, virginija.gurskiene@vdu.lt;

³Dr. Sc., Professor of the Department of Land Management and Geodesy, Latvia University of Life Sciences and Technologies, Jelgava, Latvia, vivita.pukite@llu.lv

⁴Ph.D., associate professor, Samarkand State University of Architecture and Construction. Uzbekistan, abilkosim.abduxamidov@email.ru

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Abstract. *The article investigates the impact of digital technologies on the specialists training in the field of geodesy, land management and urban planning. The methods of introducing geographic information systems (GIS), remote sensing and automated data processing into the educational process are considered. The practical skills that students acquire through the use of new technologies are analyzed, as well as the adaptation of curricula to the requirements of the modern labor market to increase the competitiveness of graduates.*

Keywords: *digital technologies, geographic information systems (GIS), remote sensing, data processing automation, training, geodesy, land management, urban planning, professional skills, innovative technologies.*

Introduction

The modern development of information technology is fundamentally changing approaches to education in various fields, including geodesy, land management and urban planning. The use of digital technologies is becoming an integral part of training, as they provide fast and accurate data processing, reduce time and resources, and increase the efficiency of spatial planning and land management processes. The integration of digital tools into the educational process allows not only to improve students' knowledge and skills, but also to adapt them to the real needs of the labor market, where more and more attention is paid to automation and digital transformation of work processes. Modern educational theories state that skills should be acquired through active research, in which teachers facilitate student learning [1].

In the context of globalization and rapid changes taking place in the engineering and geoinformation fields, training qualified specialists requires the introduction of innovative teaching methods. This includes the use of GIS (geographic information systems), BIM (building information modeling), drones for geodata collection, automated land management systems, and other technologies that contribute to an integrated approach to spatial object management. This approach not only ensures quality education but also creates conditions for the active participation of future professionals in the processes of sustainable development of cities and territories, infrastructure development, and support for environmentally sound resource use.

A geographic information system (GIS) is software that processes digital geospatial data, allowing users to store, visualize, and analyze spatial information [2]. Studies show a positive relationship between the introduction of GIS-based learning and student engagement in the learning process [3]. For example, several student research projects have used GIS to visualize data on noxious weeds, dead trees, security issues, and natural disasters [4]. The use of GIS in geological fieldwork allowed students to more easily understand the connections between geological structures compared to conventional teaching [5]. As a result of using GIS, students' spatial thinking and reasoning skills improved [6]. Geographic research skills are an important component of the Chinese secondary geography curriculum. However, in most secondary

schools in China, the learning process remains traditional and emphasizes rote learning rather than hands-on exploration, especially in the context of geospatial technologies, as it is also seen in South Africa [7].

Current research on this topic emphasizes the need to integrate digital technologies into the educational process, including geographic information systems (GIS), remote sensing, and automated data processing tools. This contributes not only to improving students' practical skills but also to adapting to the requirements of the modern labor market, where more and more attention is paid to working with large data sets and digital maps. For example, the development of geodata in curricula is aimed at developing skills in territory modeling and urban planning, which is important for increasing the competitiveness of graduates. In addition, the authors of this study emphasize the importance of involving students in geoinformatics at an early stage of education to increase their interest and prepare them for choosing geoinformation disciplines in the future. Successful STEM programs show that early exposure to the subject increases students' interest in continuing their education [8].

Another key challenge is to adapt the curricula to modern market conditions, which includes preparing specialists to work in the context of the digital transformation of the economy and changes in legislation. This includes the integration of new technologies, such as global navigation satellite systems (GNSS) and unmanned aerial vehicles (drones), which are critical for geodetic and land management work.

For example, in 2022, the Korean Ministry of Education revised the national curriculum with a focus on students' digital skills [9]. The secondary school subject of Korean Geography was transformed into a career-oriented discipline to help students with their professional development [10]. The study confirms that the use of geographic information technologies (GST) and new digital competencies meets the requirements of vocational education set by the Ministry [11].

The author of the article, Mary Farger, suggests using WebGIS not only as a tool for analyzing spatial data, but also for building powerful disciplinary knowledge (PDK). She considers the importance of an approach that emphasizes the use of geography knowledge and the development of spatial thinking skills. The study also presents a "curricular artifact" (learning object) built in ESRI ArcGIS Online based on the 2004 Indian Ocean tsunami, which demonstrates how integrating WebGIS into learning can develop an understanding of key geographic concepts such as space, place, and interconnection [12].

He, Wei & Chen, Mingze study the application of innovative technologies and artificial intelligence (AI) in urban design and planning. They analyze how AI, digital tools, and geodesy are transforming the modern urban environment, making it more sustainable, efficient, and convenient. The authors explore the functionality of technologies such as machine learning, neural networks, and other approaches used to predict trends, make decisions, and automate complex geospatial tasks [8].

The article by Wang, Chuanbing & Yang, Daihu & Xu, Huimin explores the implementation of a geographic information system (GIS) to explore the relationships between geographic elements and housing prices in school geography. Using data on apartment prices in Hefei, China, the authors developed GIS activities that help students better understand how different urban elements such as government offices, schools, hospitals, and roads affect price formation. The analysis showed that apartments closer to these facilities have a higher price. Students who had never worked with GIS before were given the opportunity to learn mapping and spatial data analysis skills. Students noted that the new approach made geography lessons more interesting, and GIS allowed them to move from passive learning to active research, providing more opportunities for self-study. It is noted that GIS tools have significantly increased students' interest, allowed them to visualize the relationships between elements of the urban structure and improved understanding of geographic factors impact on housing prices. However, the authors note the limited availability of equipment and short training time, which made it difficult to master GIS technologies more deeply [9].

However, the integration of digital technologies into education also has its challenges. One of the key barriers is the insufficient technical base in many educational institutions, which limits students' access to modern digital tools. Many universities and colleges face the problem of lacking the latest equipment,

licensed software, and high-speed internet, which is critical for the effective use of technologies such as GIS, remote sensing, UAVs, and BIM. This is especially true for educational institutions in remote regions, where limited funding is becoming a major obstacle to modernizing the educational process.

Moreover, preparing teachers for the introduction of digital technologies requires considerable effort. Not only do they have to learn new tools, but they also have to adapt their teaching methods to use digital solutions in teaching. For example, teachers need to learn how to work with software for creating spatial models, data analysis, and information visualization, which requires additional training, professional development, and advanced training that may not always be available due to limited resources. Many teachers face problems with mastering modern software solutions, which complicates the process of introducing digital technologies into teaching.

Another challenge is the adaptation of students to new teaching methods, as some of them may have difficulty mastering digital tools, which requires more intensive support from teachers. Students who have not previously worked with GIS or other digital technologies often need additional time to master the basics of working with such programs. In particular, beginners may have difficulty analyzing data, developing digital maps and spatial models, which can reduce their motivation to learn and slow down their progress in material learning. To ensure successful adaptation of students, it is important to introduce interactive learning approaches and use more intuitive software solutions.

There is also a cultural barrier associated with changing traditional teaching methods. The shift from passive information perception to active interaction with digital technologies requires a change not only in methods but also in the mindset of students. They have to learn to be more independent, proactive, and capable of independent research, which is not always common in the traditional educational system. This can create additional difficulties for students who are used to a more directive style of learning, where the main role belongs to the teacher.

In addition, there are ethical and legal aspects of integrating digital technologies. The use of digital data, especially when working with GIS and remote sensing, can lead to issues of data privacy and security. Students and teachers need to be aware of data usage legal norms and adhere to ethical standards when working with geospatial information, especially when dealing with personal or sensitive data.

During the COVID-19 period, which has become a real challenge for all areas of education, including the training of specialists in the field of geodesy, land management and urban planning, the integration of digital technologies into education has become crucial as quarantine restrictions and distance education have become a new reality. Digital technologies have made it possible to continue the educational process, supporting the continuity of learning and providing students with access to the necessary resources, tools and data even when their physical presence is limited.

The COVID-19 pandemic has forced higher education institutions to quickly switch to distance learning, and digital technologies have become the only way to continue training students in geodesy and urban planning. The use of geospatial data processing software (e.g., GIS, AutoCAD, Revit) has become not only an alternative to traditional methods, but also contributed to the expansion of students' competencies in remote analysis and modeling. Digital technologies have become indispensable in providing remote access to educational materials and practical tasks, which made it possible for students to continue their studies without significant losses in the quality of knowledge.

The transition to online platforms and digital tools has opened up new opportunities for more flexible learning. Students are able to access lectures, videos, interactive simulations, and online libraries at any time. Such changes made the learning process more adaptive, which is important for both students and teachers, especially during the pandemic, when the work schedule often changed due to unforeseen circumstances. The integration of digital platforms has allowed educational institutions to quickly adapt their curricula to new conditions, which has become a critical success factor in the pandemic.

The use of digital technologies in the training of specialists during the COVID-19 period has contributed to the development of new skills that have become necessary in the modern digital world.

Students not only mastered basic professional competencies but also developed skills in remote collaboration, communication, and problem solving in the online environment. This has become an important aspect of the training, as the labor market has also adapted to new realities, requiring specialists who can work effectively with digital tools in a remote format.

COVID-19 has stimulated the use of international online platforms that have allowed students and teachers to interact with colleagues from other countries, share data and research results. This was made possible by digital platforms that support geodata exchange, remote modeling, and project visualization. The integration of digital technologies into education during the pandemic has created a unique environment for international cooperation and access to the latest scientific developments in the field of geodesy and urban planning.

Moreover, modern digital solutions allow for a quick response to challenges related to infrastructure restoration and rebuilding, damage monitoring, efficient land management, and planning for future reconstruction and development of the affected areas, which is especially relevant for the Ukrainian context. Military operations in Ukraine have caused significant damage to infrastructure, including roads, buildings, bridges, and other facilities. The use of geographic information systems (GIS) and remote sensing (RS) in assessing the extent of damage and creating damage maps is an important tool for quick decision-making in the field of reconstruction. Digital technologies enable the creation of 3D models of damaged facilities, which helps in planning recovery efforts, prioritizing objects for reconstruction, and assessing the need for materials and resources. Digital tools increase the efficiency of geospatial data management, enabling specialists to respond to changes in the field more quickly and accurately.

Modern technologies play a crucial role in the training of specialists in the field of geodesy, land management and urban planning, contributing to more efficient management of territories and natural resources. Growing urbanization, climate change, and the need for sustainable development are driving the introduction of innovative solutions that improve the accuracy of data collection, optimize analysis processes, and allow for modeling future changes in the territory. Digital technologies such as geographic information systems (GIS), remote sensing (RS), LiDAR, blockchain, Internet of Things (IoT), artificial intelligence (AI) and machine learning, 3D modeling and virtual reality (VR), and BIM (building information modeling) are actively used in this area. GIS allows for the creation of interactive maps and terrain models for making land use decisions, while remote sensing technologies use satellite and drone imagery for accurate mapping. Lidar systems enable the creation of three-dimensional terrain models, and blockchain improves the transparency of land registries. IoT helps to monitor the condition of soil and natural resources, while AI automates data processing and predicts landscape changes. 3D modeling and VR technologies help visualize infrastructure projects, while BIM ensures efficient coordination of construction and facility management. All of these technologies not only increase the accuracy and efficiency of specialists' work, but also promote transparency and integration of spatial planning processes. The introduction of digital innovations in geodesy, land management and urban planning not only optimizes workflows, but also creates conditions for more efficient use of resources, sustainable development of territories and improvement of the quality of life in cities and rural areas (Fig. 1).

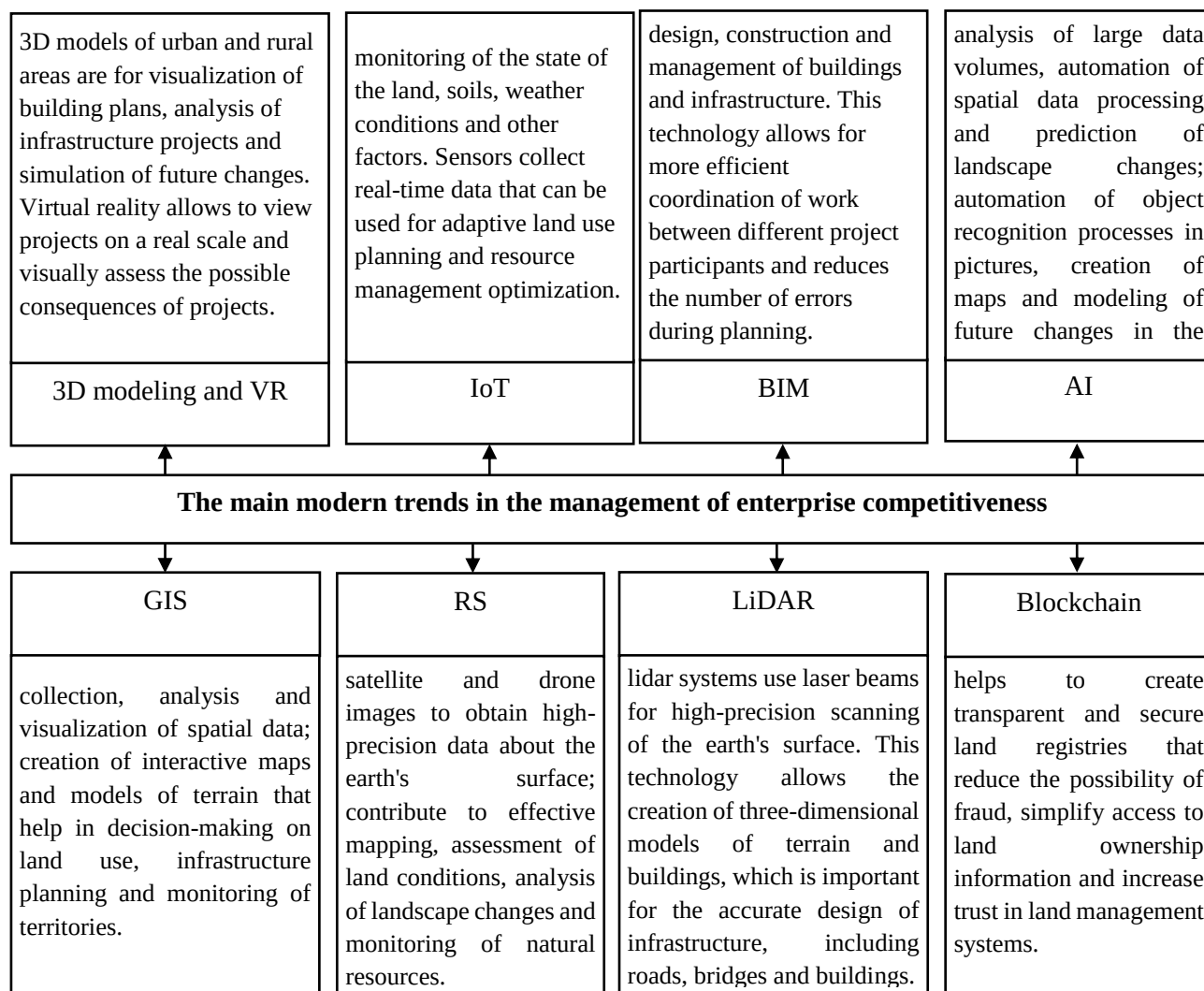


Fig. 1. Digital technologies used in the training of specialists in the field of geodesy, land management and urban planning

Modern information technologies are fundamentally changing approaches to education and work in the field of geodesy, land management and urban planning. They provide fast data processing, reduce resource consumption and increase the efficiency of planning and management of territories. Integration of digital tools into the educational process improves students' knowledge and skills, adapting them to the requirements of the labor market, which is increasingly focused on automation. The use of technologies such as GIS, BIM, drones, LiDAR, blockchain, IoT and artificial intelligence ensures transparency, process optimization and rational use of resources, contributing to sustainable development and improving the quality of life in urban and rural communities.

The strategic task of modern higher education in the context of innovative informatization of society is to accelerate positive changes, provide intellectual resources and scientific developments through the introduction of innovative technologies in all areas of activity, including the management of the educational process. The main goal is to create a comprehensive computer network to ensure quality education, which will allow for the implementation of individualized education in the long term through the use of modern technologies. The development of university infrastructure is a prerequisite for the effective use of innovations in education, including the introduction of computer equipment, network support, educational terminals and specialized software.

To achieve this goal, higher education institutions should develop a long-term informatization program that will include automation of the educational process management, improving its efficiency and

productivity. Such a program should facilitate the integration of feedback into the educational system, ensuring rapid changes in the content and methods of teaching, and optimizing the methodological and technical support for training specialists. The use of cloud technologies allows teachers to store teaching documents in the cloud storage and organize shared access to resources for students and teachers. This contributes to the operational control of student performance and expands the possibilities of managing the educational process, regardless of geographical distance.

Modern educational programs also emphasize the development of practical skills and project work, which allows students not only to master theoretical knowledge but also to participate in real projects, internships, and laboratory research. The development of distance learning provides access to interactive simulations, online courses, and training materials, making the educational process more flexible and adaptive. Given the complex problems of urbanization, climate change, and land management, training should also focus on an interdisciplinary approach that integrates knowledge of ecology, economics, sociology, and engineering. The use of artificial intelligence and big data analysis allows specialists to solve complex problems, such as predicting landscape changes, automating land management processes, and developing strategies for sustainable development of territories. Global cooperation also plays an important role, facilitating the implementation of international standards and the exchange of experience, in particular through participation in international projects and exchanges. In the context of post-conflict reconstruction, particularly in Ukraine, training should focus on methods of infrastructure reconstruction, damage monitoring, and designing resilient infrastructure.

The integration of digital technologies in the training of specialists in geodesy, land management and urban planning has significant prospects, but requires addressing a number of organizational, technical and social challenges. Despite these obstacles, the successful implementation of digital technologies in the educational process contributes to improving the quality of education and the competitiveness of graduates in the labor market.

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RESEARCH ON THEORETICAL ISSUES OF REAL ESTATE MANAGEMENT

A.Ya.Abdukhamidov

Samarkand State University of Architecture and Construction, Ph.D., Associate Professor

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Keywords: management, Housing Code, functional purpose, persons established, rule

Introduction

Real estate management is a management activity at the stages of the building's life cycle, covering such processes as design, construction, demolition and demolition of the facility, consulting on technical conditions for design, selection of engineering systems, development of a technical system, and operation of the facility.

Asset management is an activity to manage the financial flows of the owner of real estate.

Today, the field of professional real estate management, which in foreign practice is called surveying, is developing in the Republic of Uzbekistan. High efficiency of real estate management can only be achieved with a professional approach to it.

Surveying is an integrated approach aimed at managing facilities in a way that meets the interests of the owner of the property, its users and society as a whole.

Surveying implies consistency in the management and development of real estate. This approach places high demands on planning at all levels:

1. General; 2. Strategic; 3. Operational.

At the same time, a number of measures are being carried out (mainly feasibility studies), which allow for regular monitoring of the efficiency of the facility and, accordingly, finding ways to improve it. Management of real estate objects in accordance with an integrated approach is aimed at meeting the needs of the state to the maximum extent.

Currently, the most common types of surveying are facility management and property management, within which it is necessary to distinguish management by a specialized organization (management company), lease, trust management, concession management.

It should be noted that there are practically no legislative restrictions on the use of these management methods in any of the real estate sectors, however, over the past decade, there has been a gradual differentiation of individual approaches to managing real estate objects depending on the functional purpose of real estate (Table 1).

Let us consider the methods of managing real estate objects in more detail.

Management of a real estate object by a specialized company is often used in the residential real estate sector and is one of the three management methods provided for by the Housing Code of the Republic of Uzbekistan:

1. Management of the owners of buildings; 2. Management of a real estate object by a specialized organization (management company); 3. Management of a homeowners' association or consumer cooperative.

Table 1

Methods of real estate management from the point of view of functional purpose

Management method	Main functions according to the functional purpose of the property			
		Commercial buildings	Industrial buildings and structures	Socio-cultural facilities
Lease	Use of property in accordance with the contract; storage of property*; provision of services**; monitoring the safekeeping of entrusted property; timely payment of fees for the use of property, etc.			
Trust management		Generating income and increasing property value; maintaining property*; maintaining technical documentation; concluding contracts with repair and utility service providers; monitoring the quality of public services; searching for and attracting tenants; formulating rental policy; interacting with tenants; monitoring the safekeeping of entrusted property, etc.		
Privilege			Creation of an object; use of property in accordance with the contract; storage of property*; current and capital repairs**; modernization of the object**; control over the safety of the property of the entrusted person; timely payment of payment for the use of property, etc.	
Specialized organization management (management company)	Property management*; maintenance of technical documentation; conclusion of contracts with repair and utility service providers; quality control of public services.			
	registration, accounting and certification; information services to districts; payment management and debt collection, etc.	search and attraction of tenants; formation of rental policy; interaction-my tenant; optimization of economic data object; representing the interests of the property owner in inspections and state supervisory bodies, etc.	optimization of the economic data object; acting in the interests of the owner of the object, responding to state administration bodies; system protection of the territory, etc.	
Note: * technical inspections and bypasses; seasonal preparation for use; current and major repairs;				

A real estate management organization is a legal entity established to manage property, performs economic and operational functions related to sanitary engineering and provides maintenance services for apartment buildings.

The management organization provides intermediary services in the supply of resources such as electricity and water supply, carries out maintenance and repair work on the premises. It appears before resource supply organizations on behalf of all homeowners of the subordinate building as a single customer and as a single executor of communal services.

It should be noted that management organizations can have different specializations - only servicing or only management, as well as perform both of these functions.

The most common method of managing real estate of any functional purpose, both abroad and in Uzbekistan, is leasing. On the one hand, leasing real estate is one of the types of entrepreneurial activity, and on the other hand, it is an effective way to obtain the necessary buildings, structures, non-residential or residential premises for an appropriate fee. This indicates the dual economic nature of differential rent.

The right to lease real estate objects belongs to the owners and persons established by law. The landlord (the owner of the object or a person authorized by law or the owner) undertakes to lease the real estate object to the tenant for temporary possession and use or only for temporary use. Conclusions

According to the results of real estate management studies, when it comes to buildings that have been historically used by different organizations for different purposes, there is no alternative to trust management:

- First, only with trust management is the object principle of financing implemented: maintenance and operating costs are financed from its own income and subsidies from the budget are not required. The manager himself, as a rule, invests in the development of the object, but never requires a share of the property;
- secondly, the owner appears in the building;
- thirdly, only professional management allows the object to be developed; not randomly, but in accordance with a clear and long-term concept.

Nevertheless, trust management of real estate objects as a form of entrepreneurial activity in the Republic of Uzbekistan has not yet become widespread.

Concession is a form of public-private partnership involving the private sector on mutually beneficial terms in the effective management of state property or in the provision of services usually provided by the state.

Under a concession agreement, one party (concessionaire) undertakes to create and (or) reconstruct the property (real estate or real estate and movable property, technologically interconnected and intended for the implementation of the envisaged activities) specified in this agreement at its own expense.

The property rights that belong or belong to the other party (concession grantor) for the implementation of activities using (using) the object of the concession agreement, and the concessionaire is obliged to transfer the right of ownership and use of the object of the concession agreement to the concession holder for the period specified in this agreement.

Concession is often used in the management of state and municipal real estate, which requires large financial investments in measures to reconstruct and restore its technical and operational properties.

Thus, real estate as an economic category occupies an important place in economic theory. First of all, the land and real estate located on it are the wealth of the entire country and play a huge role in the life of society.

Secondly, real estate is an element of the process of reproduction that affects the stability of the economy and its growth.

Thirdly, real estate is one of the important indicators of the standard of living of the population, economic security and state stability. Consequently, real estate is an important element of economic relations.

Real estate objects need constant management to maintain them in a normal functional state. At the current stage of development of science in the field of real estate management, the main methods of real estate management are legal, economic, organizational and technical. The main methods of real estate management are lease, trust management, concession and management by a specialized organization. In this case, the choice of control method directly depends on the functional purpose of such an object.

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ANCIENT EGYPTIAN GARDEN-PARK ART

Shamsieva Rafoat Aslidinovna

teacher of "Interior Design", SamDAQU

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Abstract. *This article describes the history of the ancient Egyptian garden parks, their origin, and the composition of the Egyptian gardens. Such general concepts as the ancient Egyptian garden, an organic combination of religious, utilitarian and aesthetic functions are highlighted. The ancient Egyptians created an advanced irrigation system, built canals that supplied water to fields, and hydraulic structures that protected cities and settlements from floods during river floods. Magnificent palaces, majestic and durable temple complexes, pyramids, partially preserved to the present day, were built from valuable building materials - granite, limestone, sandstone, etc., which Ancient Egypt is rich in. Architectural monuments of the Ancient Kingdom (ABB.28-23 centuries BC) pyramids of the pharaohs and tombs of the nobility, temples with columns in the form of lotus, palm, papyrus (pyramids of Giza: Cheops, Chephren and Mycerinus. The Middle Kingdom (ABB.21-18vv) stone tombs decorated with paintings on traditional themes. The shapes of temples with two rows of columns and a high middle part. The New Kingdom (XVI-XI centuries BC) – temples consisted of a sanctuary, a closed hall with columns and a closed courtyard surrounded by a blank wall. Gardens were very common in Egypt. The houses of the nobility were always surrounded by gardens, which varied in location and wealth depending on the condition of the owner. Especially large and rich gardens belonged to the kings. Temples of the gods are usually surrounded by luxurious gardens, and funds for their creation often came from kings. For example, the Harris papyrus says that Ramses III ordered large and luxurious gardens to be laid out for the temples of the gods of Heliopolis. "For your temple, vol... I have built large gardens planted with grapes and cabbage soup... I have created alleys of olive trees in your city. I have provided these seedlings to gardeners and many technicians who will make the purest Egyptian oil needed to light the lamps in your luxurious temple. I gave you pistachio groves and ponds covered with lotuses, sitnikas, herbs and flowers from different countries — Tetmer, An-Haut — fragrant and pleasant to look at your magnificent face. Currently, interest in Garden and Park art is growing rapidly, which, along with new research in the field of Garden and park art, forces us to also assimilate the legacy of the past in order to be able to extract from this heritage everything that is valuable to us. The history of parks and parks of different eras and peoples is covered in the literature 1) not in the same detail. There is the most detailed information about parks and parks of the XVI—XVIII centuries. In this article, we will give a description of ancient Egyptian and Assyro-Babylonian gardens and parks based on the study of some literary and artistic monuments of Egypt and Assyria-Babylon.*

Keywords: Ancient Egypt, Gardens, Nile Valley, hanging garden, Assyria, temples, Pleasure Gardens, temples, compositional features, pyramids.

Introduction

Ancient Egypt was established as an independent state in the 4th millennium BC. The development of the state is closely linked to the Nile Valley, which carries its waters from south to north into the Mediterranean Sea. The valley was naturally dominated by tamarixes and palm trees, and the Nile by reeds, papyri, and lotuses. Hot and arid climate, hot winds, almost no precipitation, excluded the presence of tree-shrub plants in some important areas. The ancient

Egyptians created a developed irrigation system, canals supplying water to the fields, and hydrotechnical devices that protect cities and settlements from flooding during river floods. Of the valuable building materials rich in Ancient Egypt - granite, limestone, sandstone and others, magnificent palaces, durable temple complexes and pyramids were built, which partially survive to this day.

Ancient Egypt is characterized by huge monumental structures that glorify Pharaohs. The most ancient monuments are located in the delta of the Nile Valley, which is due to the fact that the beginning of Egyptian civilization was located in this very place. In Egypt, building materials used clay-covered reeds, Nile Valley clay bricks, and straw, while monumental structures used stone. Artistic forms and pictorial motifs were taken from nature by the Egyptians: the sun, Lotus, papyrus, Palm, Lions had a mainly religious purpose. Monuments of the architecture of the ancient kingdom (3rd century BC). 28th-23rd centuries) pharaonic pyramids and Noble tombs, Lotus-shaped, palm-shaped, papyrus-shaped pillared temples (Giza Pyramids: Cheops, Hephren and Mikerin. The middle kingdom (avv.21 18vv) rock tombs decorated with paintings on traditional themes. Forms of chapels with two rows of columns and a high middle part. New Kingdom (16th-11th centuries BC) – temples consisted of a sanctuary, a hall with an enclosed column, and an enclosed courtyard surrounded by an empty wall.

Analysis of thematic literature (Literature review).

On this topic, K.D Rahimov, A.S Uraiov, textbook “design of objects of Landscape Architect”. Tashkent; "Psalm-standard", 2015. - 224 b. Avtor: starshiy prepodavatel café " Das " Dobronravova E.A. Uchebnoe posobie. Chast 1. (STR. 114., TASI 2009). Rakhimov, K.D. Educational guide” modern directions of Garden-park design " / K.Rakhimov. Tashkent: "celebrity-press", 2017. Rakhimov K.D., Uralov A.S. Design of objects of Landscape Architecture. - T., 2015. S.S.Ojegov, A.S.Uralov, K.J.Rakhimov. Landscape architecture and design.. Samarkand, 2003. the study was carried out by Lar adaboyots and research.

Analysis and Results (Results and discussion).

At this time, the gardens are regular: it is usually a walled Square area, trees are planted along its perimeter, in the central part there is a vineyard; there are also rectangular pools for storing fish and waterfowl.

The ancient Egyptian Garden was characterized by an organic combination of religious, utilitarian and aesthetic functions.

In general, garden art was formed in Ancient Egypt with clear compositional and planning laws:

- regular plan, which involves the construction of the composition axial and the use of symmetry;
- formation of closed compositions;
- the presence of water bodies as an integral and often basic part of the garden;
- use of rhythm as a compositional tool;
- use of alleys and ordinary seedlings;
- the use of exotics in the assortment of woody plants.

Assyrian and Babylonian States (Mesopotamia) (8th-7th centuries BC). The main material for construction is fragile raw brick. The cities were created in a round, and later rectangular shape, surrounded by two-or three-ring fortress walls. Based on the location of Babylon, there are 8 wide streets leading to the city gates. From them, a wide street leads to a certain temple .

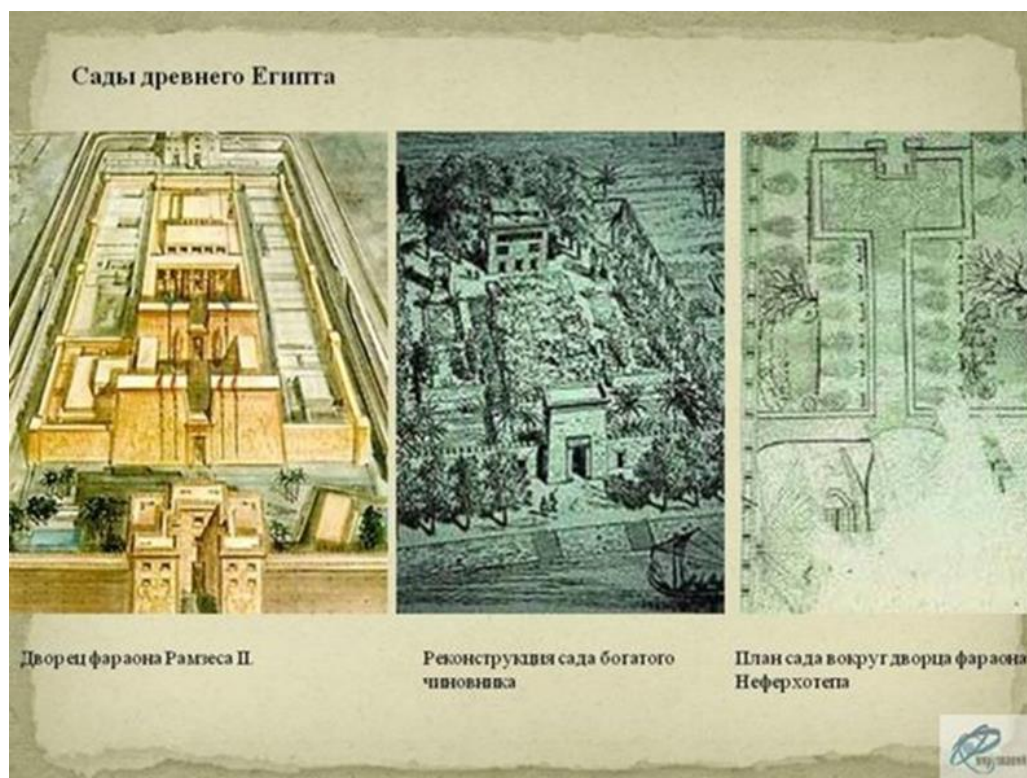


Figure 1. Ancient Egyptian Gardens

Temples are secondary elements of the composition. On the terraces, the construction was used for ziggurat temples in the form of towers with stairs. On their lower platforms, plants are planted in pits with plant soils along the perimeter.

In Ancient Egypt, the first "Pleasure Gardens" appeared in rich priests. Resting in the shade of trees and enjoying the taste of fruits, their images can be seen in ancient Egyptian frescoes.

A variety of fragrant flowers were grown in the priestly Gardens, the surface of the ponds covered with lotuses and nymphs. And in the era of Cleopatra, garden roses were often grown.

Everyone knows the Hanging Gardens of Babylon in Babylon. The structure consisted of 4 terraces raised one above the other. Trees were planted on the bottom two terraces, and shrubs and flowers at the top.

The "Hanging Gardens" of Semiramis were located in the courtyard of the palace of King Nebuchadnezzar and consisted of a series of high terraces: trees at the bottom, shrubs and flowers at the top. There were also extensive parks for hunting and horseback riding. (prototype of modern forest parks).

E. BC. Near the city of Borsippa, trees brought from other countries grew in it: cypress, cedar, maple, Willow, Poplar, Laurel, boxwood, fruit trees. There are also records of extensive parks or forests, mainly for hunting and horseback riding. These massifs are prototypes of modern forest parks.



Figure 3. A bouncy garden look that originated in Ancient Egypt.

The Hanging Gardens of Semiramis are one of the Seven Wonders of the world. A more correct name for this structure is hanging gardens (amitis Springs — Semiramis; according to the third sources — Amanis): this is how the wife of the Babylonian king Nebuchadnezzar p was named, for which Gardens were created. It is assumed that they were located in the ancient State of Babylon, near the modern city of Hill, Iraq. According to an alternative version based on the reopening of cuneiform inscriptions, they may have been located in Nineveh, the capital of the Assyrian Kingdom, and were built in the early 7th century BC. They are the wonder of the world that existed for the shortest time. The forces of King Nebuchadnezzar P of Babylon (605-562 BC) formed a military alliance with King Kiahra of the Medes to fight the main enemy of Assyria, who had twice destroyed the capital of the Babylonian state.

After their defeat, they divided Assyrian territory between themselves. Their military alliance was confirmed by the marriage of Nebuchadnezzar to the daughter of Amitis, King of the Medes. The dusty and noisy Babylon, located on a bare sandy plain, did not please the queen, who grew up in the mountainous and green mussels. To comfort him, Nebuchadnezzar P ordered the construction of Hanging Gardens. However, in historical science, the name of the gardens is incorrectly given to the Assyrian Queen Semiramis, who lived two centuries ago.



Figure 4. Semiramis Gardens

Gardens in ancient Egypt are common. Perhaps due to the vegetable gardens irrigated by the Nile River, as the prosperity of Egypt increased, they became luxurious complexes with ponds, flowers, statues, shady alleys of fruit trees. The gardens were built around the temples, palaces and castles of wealthy Egyptians. In addition, the tombs were known as Tomb gardens around them.

The history and development of ancient Egyptian gardens, like other aspects of ancient Egyptian life, was associated with the Nile River and the network of canals that fed it. The oldest Gardens in Egypt were flower beds and vegetable gardens, which belonged to temples that cultivated vegetables for certain rituals. During the Middle Kingdom, large garden complexes and pleasure gardens around the estates of wealthy Egyptians appear around the palace Tsartfarons.

The layout of the Egyptian Garden. Images of Egyptian gardens are found on tomb walls, some papyri, and sarcophagi. The composition of Egyptian Gardens was based mainly on the principle of geometric structures. Gardens were often laid out on level ground and rarely on terraces.

The gardens were rectangular and symmetrically planned. A distinctive feature of Egyptian Gardens is the pond, which is often located in the middle. The Pharaohs ' Palace Gardens may have many pools, including wide enough to row. The ponds were Artificial with banks covered with stone slabs. The pond was overgrown with aquatic plants, and fish and waterfowl were grown. Pavilions, gazebos and churches for domestic deities were built near the ponds.

Extensive gardens were built around the temples. There were twenty-six vegetable gardens, as well as an ancient Botanical Garden, which contained the inscriptions "beautiful flowers and strange plants of all kinds found in the blessed land conquered by His Majesty". Temple gardens often had alleys of fig palms and plantains (dedicated to the Temple Of Love Hathor). Sometimes rows of trees stretched for several kilometers, connecting the various temples. For trees planted

away from the river, it was necessary to dig wells up to ten meters deep. Flowers for religious ceremonies were grown in the gardens, as well as medicinal plants and spices such as cumin, licorice and coriander.

The palace gardens were very large, with many ponds and shady alleys for the rest of the Pharaohs. The rulers of Ancient Egypt were brought from other countries, planting unknown plants in Egypt in gardens. For example, Queen Hatshepsut sent an expedition to the Land of Punt, which, along with Ivory, myrtle and incense, brought myrtle trees dug with roots from the ground.

From the time of the New Kingdom, Pleasure Gardens were a common feature of the lavish estates of wealthy Egyptians. In the center was a rectangular pond, surrounded by a row of trees in succession. The edges of the ponds are inclined to stairs, allowing gardeners to collect water for irrigation. The pond is often surrounded by vineyards and is surrounded by walls or columns decorated with colorful paintings of people, animals and plants.



Figure 4. A bouncy garden look that originated in Ancient Egypt.

Conclusions and suggestions (Conclusion).

In conclusion, Ancient Egypt amazes, Awakens imagination, inspires. Gone are the days when knowledge of Ancient Egypt was preserved for a small group of insiders. Now every amateur enthusiast can get acquainted with the secrets of hieroglyphs, the history of a fascinating civilization, admire the magnificent buildings, mausoleums of the country on the Nile River.

In the Nile River valleys, spruce, fir, dates, figs, tamarix, Poplar, tar were grown. But this was not enough for complete comfort. It was necessary to create gardens that bring coolness and shade. "He is poor, he has no shadow" - this is what ancient texts say. The absence of shadow is one of the most severe plagues of the poor in this country. In hot climates, the organization of gardens required artificial irrigation. Figs, mulberries, almonds, dates, grapes were grown in a simple, rectangular network of canals that required the creation of rectangular, symmetrically laid Gardens.

A distinctive feature of Egyptian Gardens was the fact that vegetables and flowers occupy almost all the space between the trees. Egyptian gardeners were engaged not only in cultivation,

but also in forcing lilies, gillyflowers and roses of the Valley. Minyonet, poppy, cornflower, chamomile, lilies, chrysanthemums have grown in Egyptian Gardens. The lily flower was a symbol of hope and the brevity of life, from which fragrant oil was obtained. Flowers were widely used for artistic decoration of holidays, from which wreaths and garlands were woven. During the reign of Queen Cleopatra (69-30 BC), special attention was paid to the cultivation of various roses.

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**THE IMPORTANCE OF ENGLISH IN STUDYING LEADING COUNTRIES'
EXPERIENCES IN ARCHITECTURE AND URBAN PLANNING**

Davron Mamaraimov

SamSACU, Teacher of Foreign Languages Department

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Abstract. *The global nature of architecture and urban planning demands that professionals and students engage with diverse international perspectives to create sustainable, functional, and aesthetically appealing spaces. This article examines the critical role of English as a lingua franca in accessing, understanding, and integrating the architectural and urban planning innovations of leading countries. English enables students, researchers, and practitioners to engage with the vast majority of academic publications, technical standards, and case studies, which are often published in or translated into English. By analyzing current practices, design philosophies, and regulatory approaches from countries with advanced architectural and urban systems, this paper highlights how English proficiency facilitates interdisciplinary learning and collaboration. Ultimately, the paper argues that English fluency is essential for staying at the forefront of global architectural trends and implementing best practices that align with both local and international standards for urban development.*

Keywords: *English, learning and collaboration, translation, solutions, English facilitates, complex research, proficiency in English, research collaborations.*

Introduction

In today's globalized world, English has become the universal language of communication, enabling access to a vast wealth of knowledge and experiences across various fields. This is particularly true in architecture and urban planning, where leading countries share innovative designs, sustainable practices, and advanced technologies. By mastering English, professionals and students in these disciplines can explore pioneering ideas, collaborate with international experts, and apply global best practices to local contexts.

This essay discusses the significance of English as a tool for studying the experiences of leading countries in architecture and urban planning. It highlights how English facilitates access to academic resources, participation in global discussions, and the integration of cutting-edge solutions into local projects, ultimately contributing to the advancement of architectural and urban planning practices worldwide.

Access to Global Research and Innovation: English serves as a gateway to the vast pool of global research and innovation in architecture and urban planning. The majority of scholarly articles, research papers, and technical publications are written and published in English, making it the dominant language of academic and professional discourse. By understanding English, students and professionals can access cutting-edge studies, groundbreaking theories, and innovative solutions developed by leading countries in the field.

For instance, globally renowned architectural journals, such as *Architectural Record* and *The Journal of Urban Design*, primarily publish in English. Similarly, international conferences and forums like the World Urban Forum and the Venice Architecture Biennale use English as their main language of communication. Mastery of English not only allows individuals to learn from these resources but also enables them to participate actively, share insights, and contribute to global advancements in the discipline.

Moreover, many digital platforms, open-access repositories, and online courses provide valuable learning opportunities in English, equipping learners with the skills to tackle complex urban challenges. Access to such resources empowers architects and urban planners to adopt sustainable practices, innovative technologies, and culturally sensitive designs inspired by the experiences of leading countries.

English plays a pivotal role as a bridge to global knowledge, particularly in architecture and urban planning. It is the primary language used in most international academic publications, research papers, and case studies, enabling students and professionals to explore the best practices and experiences of leading countries.

By mastering English, individuals can access a wealth of resources, including books, journals, and online databases, that document innovative architectural designs, sustainable urban planning solutions, and advanced construction technologies. Prestigious institutions and organizations, such as the United Nations, the American Institute of Architects (AIA), and the Royal Institute of British Architects (RIBA), produce extensive materials in English, providing valuable insights into modern challenges and solutions in the field.

Additionally, English facilitates participation in global forums, workshops, and conferences, where experts from around the world share their expertise and discuss emerging trends. This access to international knowledge fosters cross-cultural collaboration and equips architects and urban planners with the tools to create impactful, context-sensitive solutions inspired by global experiences.

In essence, English serves as a key to unlocking the vast repository of global knowledge, empowering professionals to stay at the forefront of innovation and contribute meaningfully to the development of architecture and urban planning.

One of the most significant advantages of English in the field of architecture and urban planning is its role in providing access to diverse approaches and case studies from around the world. Leading countries often document their projects, innovations, and lessons learned in English, making these resources widely accessible to a global audience.

Through English, students and professionals can study how different nations address challenges like urbanization, climate change, and resource management. For instance, Scandinavian countries' emphasis on sustainable design, Japan's efficient use of limited urban space, and the United States' advancements in smart city technologies are well-documented in English, serving as valuable learning opportunities.

Moreover, English enables a comprehensive comparison of methodologies and practices. By analyzing case studies from diverse cultural and geographical contexts, architects and urban planners can identify adaptable strategies and avoid potential pitfalls in their projects. For example, studying Singapore's integrated urban planning model can inspire innovative approaches to land use and infrastructure in rapidly growing cities.

These insights not only enrich professional knowledge but also foster creativity, encouraging the development of unique, context-specific solutions that blend global expertise with local needs. Mastering English ensures access to this wealth of diverse experiences, driving innovation and excellence in architecture and urban planning.

Different countries contribute unique solutions shaped by their own cultures, resources, and challenges. For example, Japan's compact, space-efficient housing solutions or Scandinavia's sustainable urban strategies are regularly documented in English for a global audience. This

allows professionals worldwide to learn from these varied experiences and apply relevant aspects to their own regional or local contexts.

In the rapidly evolving fields of architecture and urban planning, technological advancements play a crucial role in shaping the future. English, as the global language of technology and innovation, serves as a vital tool for staying updated with these developments.

Many groundbreaking technologies, such as Building Information Modeling (BIM), parametric design tools, and smart city infrastructure systems, are developed and documented primarily in English. Software manuals, tutorials, and technical guides, including those for widely used tools like AutoCAD, Revit, and Rhino, are predominantly available in English, enabling professionals to master these tools effectively.

Moreover, leading technological research and discussions, such as those on sustainable construction materials, AI-driven urban planning, and digital twin technologies, are often published in English-language journals and presented at international conferences. Staying proficient in English allows architects and urban planners to keep pace with these innovations and integrate them into their work.

Additionally, English facilitates collaboration with global tech companies, research institutions, and interdisciplinary teams, making it easier to adopt and implement advanced solutions. For instance, learning from smart city initiatives in countries like the United States, Germany, or South Korea becomes possible through English, as detailed case studies and technical documents are widely shared in this language.

In conclusion, English empowers professionals to stay ahead in the technological race, ensuring their projects are efficient, innovative, and aligned with the latest advancements in the field.

In architecture and urban planning, international collaboration is essential for addressing complex global challenges such as rapid urbanization, climate change, and sustainable development. English serves as the universal medium for fostering such collaboration, enabling professionals, researchers, and institutions from diverse backgrounds to share knowledge and work together effectively.

English allows architects and urban planners to engage in global networks, participate in international projects, and exchange ideas with experts from leading countries. For instance, multinational initiatives like the United Nations' *New Urban Agenda* or the Smart Cities Mission often rely on English as the primary language of communication and documentation.

Furthermore, English facilitates the dissemination of research findings and case studies across borders. Professionals can contribute to and learn from international journals, conferences, and workshops, where groundbreaking studies and innovative solutions are shared. Platforms like the *International Journal of Urban and Regional Research* and global forums such as the *World Urban Forum* rely heavily on English for accessibility to a wider audience.

Collaboration also extends to academic partnerships and joint ventures. Many universities and institutions conduct exchange programs, dual-degree courses, and collaborative research projects in English, creating opportunities for knowledge sharing and mutual learning.

In summary, English plays a critical role in breaking down linguistic barriers, enabling seamless collaboration, and ensuring the global exchange of research and ideas. This enhances the collective effort to develop innovative and sustainable solutions in architecture and urban planning.

In the interconnected world of architecture and urban planning, English has established itself as a vital tool for accessing global knowledge, embracing technological advancements, and fostering international collaboration. By mastering English, students and professionals can explore the innovative experiences of leading countries, engage with diverse approaches, and contribute to global discussions on sustainable development and urbanization.

The ability to communicate in English not only unlocks vast academic and professional resources but also empowers individuals to participate in international projects and implement cutting-edge solutions in local contexts. It bridges cultural and geographical divides, ensuring a continuous exchange of ideas and best practices that drive progress in the field.

Ultimately, English serves as a cornerstone for growth and excellence in architecture and urban planning, enabling professionals to adapt, innovate, and shape a better-built environment for the future.

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DESIGN AND AESTHETICS IN CREATING HARMONIOUS URBAN SPACES (using street litter bins as an example)

Babakandov Otabek Nuritdinovich

Samarkand State Architecture-Construction University

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Abstract. *The article examines the state of the architectural environment today and what measures should be taken to avoid pollution, as well as the role of litter bins in this.*

Keywords: *landscape, outdoor environment, household waste, garbage, litter bins.*

Introduction

Street litter bins play an important role in maintaining cleanliness and order in urban spaces. However, in addition to their primary function, litter bins can also make an aesthetic contribution to the urban environment. In this article, we will look at how the design and aesthetics of street litter bins can contribute to the creation of harmonious urban spaces.

1. Integration with the environment: Waste bins can be designed to blend in with the architecture and surrounding landscape. For example, retro-style bins can be a great addition to historic areas, while modern glass bins can blend in beautifully with contemporary buildings.

2. Aesthetics and design: Street bins can be designed in a variety of styles and shapes to match the overall design of the urban environment. They can have interesting geometric shapes, original color schemes or decorative elements. This approach allows not only to provide a functional place for throwing out garbage, but also to create visual interest for residents and visitors of the city.

3. Using litter bins as art objects: Some cities and architects are treating street litter bins as art objects. They invite artists and designers to create litter bins with unique designs or sculptures. This practice helps transform ordinary litter bins into works of art and draws attention to the problem of trash in cities [1].

Street litter bins are an integral part of urban infrastructure. They not only perform an important function of waste collection, but can also be an element of design and aesthetics in the urban environment. The design of litter bins can be integrated with the environment, have original shapes and colors, and also serve as art objects [3]. This approach allows for the creation of harmonious and attractive urban spaces that will be pleasant for residents and visitors of the city.

In general, environmental design is the improvement of a place where a person lives. This can be an apartment, office and public spaces, such as an airport, train station, theater, etc., or a plot in front of a house, in the courtyard of offices or enterprises, the territory of public buildings [2]. The second option is a type of outdoor environmental design. It is usually designed by landscape designers and architects, who take into account many nuances when designing.

Recently, the appearance of cities has been changing very quickly, which is due to the following:

Rapid growth of cities.

A sharp increase in car traffic.

Pollution of the urban environment (figure-1).



Figure-1. Problems in the design of architectural environments

We are used to seeing only beautiful spaces around us, such as these clean beaches, shores, mountains... (figure-2).



Figure-2. beautiful city space

Many find it easier not to notice than to contribute to the action of collecting and eliminating garbage and, unfortunately, today we increasingly encounter natural - beautiful landscapes, disfigured by people. We see waste from human life, which accumulate in tons for half a century or more [5].

Household waste (consumer waste) can be in both solid and liquid, and less often in gaseous form. Solid household waste is the remains that we throw out of houses, institutions, offices and usually call garbage (food waste, plastic, paper, glass, leather, etc.).

Their number increases annually due to population growth and improvements in the quality of life of people (for example, packaging for goods, including those made of polymeric materials, aluminum cans, etc.). Liquid waste is mainly represented by household wastewater, gaseous waste is represented by emissions of various gases. "Plastic waste is splashed onto the beaches of Indonesia, settles on the bottom of the Arctic Ocean and gets through the food chain onto our plates," report experts from the United Nations Environment Program (UNEP) [4].

In 1950, the world's population of 2.5 billion people produced 1.5 million tons of plastic. In 2020, the planet's population exceeded 7 billion people, and the amount of plastic reached 300 million tons, which had devastating consequences for marine plants and animals [6].

If people don't stop using single-use plastics like plastic bags and plastic bottles, there will be more plastic in the world's oceans than fish by 2050 (figure-3).

Dutchman Tommy Klein, getting to his work by bike, rode past a terribly littered beach every day. At some point, he simply took the initiative into his own hands - and began collecting

a bag of garbage every day. No, it wasn't his garbage, and no, no one forced him - he just did it, thus setting an excellent example for others.



Figure-3. Ocean Pollution

From this it should be understood that before creating a landscape design of the external environment, it is necessary to consider the number of people and the amount of waste they throw out. When creating a beautiful landscape, it is necessary to put garbage bins in parallel and necessarily for different materials separately [7].

Also, you can use convenient, lightweight and multifunctional large-sized garbage containers in places where a large number of people accumulate, such as stadiums, circuses, amusement parks, concert halls or cinemas.

Such containers can be three compartments in one container or three different containers separately (figure-4).



Figure-4. Street litter bins

The Importance of Street litter bins in the Urban Environment. Street litter bins serve as an important element for collecting and disposing of waste, helping to maintain cleanliness and order on the streets. However, their importance is not limited to functionality alone. The design of bins can contribute to the formation of a characteristic style of the urban environment and create harmony with architectural features.

Innovative design for functionality and aesthetics. Modern street litter bins offer a wide range of design solutions, combining functionality with aesthetic aspects. The design of the bins can be adapted to the style and character of the city, emphasizing its uniqueness. From minimalist and modern forms to classic and artificially aged, the design of street bins can add a special accent to the urban landscape [8].

Innovative materials and technologies also play a role in creating bins with high functionality and durability. Engineers and designers are working on developing materials that are resistant to aggressive environmental influences and are also able to withstand different weather conditions. The result is bins that retain their appearance and effectiveness for a long time.

Functionality and ease of use. The main purpose of street bins is to provide an effective waste collection system in the urban environment. To do this, they must be easy to use for both citizens and cleaners. It is important to ensure that the bins are correctly located so that they are accessible and visible, eliminating the possibility of accidental dumping of waste near them.

Modern street litter bins can be equipped with separate compartments for different types of waste: plastic bottles, paper, food waste, etc (figure-5). This promotes more efficient recycling and disposal of waste, reduces the negative impact on the environment and promotes the development of a sustainable environmental culture.

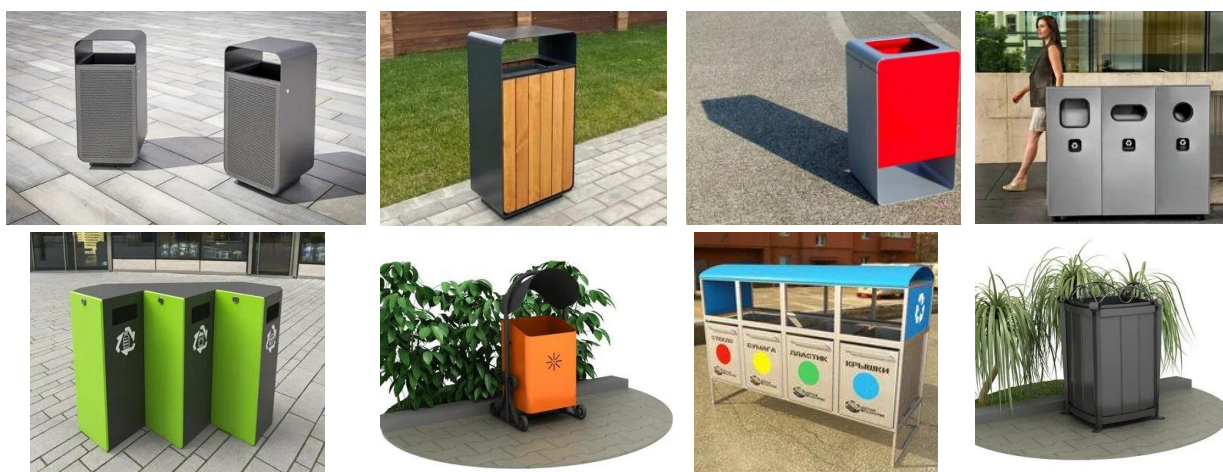


Figure-5. Modern street litter bins

Conclusion. The production of street litter bins is an important part of urban infrastructure that must be performed with high quality standards. The production of street litter bins is an important part of urban infrastructure. Street litter bins must be made of durable and long-lasting materials to ensure safety and durability. They must be designed in such a way that they are convenient to use and attractive to citizens.

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**TILES USED IN BUILDING ANCIENT MONUMENTS OF UZBEKISTAN, THEIR
TYPES AND COMPOSITION**

Djuraeva G.N.¹, Suyunova N.A.²

^{1,2}Samarkand University of Architecture and Civil Engineering

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Abstract. *In this article, the simple shapes of ceramic coverings are cut out of clay or molded, dried, dipped in a mixture of different colors, and then created by the method of cooking in humdons. Later, it became more complicated and from the second half of Проблемы архитектуры и строительства 2024, №2 11 the 14th century. became the main type.*

Keywords: memorial, tile, pattern, zoomorphic, raw brick, archaeological, majolica, mosaic, terracotta.

Introduction

In the architectural heritage of Uzbekistan, the direction of giving luxury to buildings was manifested from the first forms of urban planning. This process, which began in the ancient stone age in the wall paintings in caves and caves, was also perfected in the creation of the first settlements based on human labor. From the IV-II millennia BC, in the early stages of agriculture, the first emblematic patterns appeared in red and red colors, on ceramics. As a result of archeological excavations, the first simple patterns were found on earthenware, and then they were carried out in the interior and exterior styles of buildings. The archaic patterns on the original clay walls were different in different tribes. Broken curves, parallel lines, circles, snake traces, geometric shapes, later became more complex and became patterns in the form of zoomorphic (images of animals), plants, human (anthropomorphic) symbols. In the early Middle Ages (1st-6th centuries), economic, political, and cultural relations were also a place for masters who embodied the signs of culture formed in Central Asia, including the territory of Uzbekistan, Levant, Byzantium, Iran, India, and China. Cooperation with masters The rise of trade relations along the Great Silk Road, the strengthening of international cultural cooperation, also happened in the field of architecture and construction, there is no doubt. In the architecture of this period, the construction of castles was a common practice for every ruler and selfsufficient zamindars. Complexes of such constructions made of pakhsa, raw bricks, and castles, castles and palaces surrounded by thick protective walls were built in Khorezm, Kashkadarya, Zarafshan, Fergana valleys, Choch, and their interior appearance was enhanced in various ways. Along with wall paintings, the first ceramic decorative elements were also used. Ganch, stone, terracotta (baked clay), carved, cast decorative materials are also characteristic of early medieval architectural decorations, and in the 5th-8th centuries, their delicate and complex patterns were identified in Varakhsha, Panjikent. By the beginning of the 9th-13th centuries, the process of the formation of architectural decoration was influenced by the political and economic conditions that occurred in these periods. Arab armies gradually subjugated Central Asia to the caliphate over several centuries. At the end of the 9th century, the ruler of Bukhara, Ismail Somani, created a strong, independent state. From this period, there was development in art, science and culture. Architectural elements of different shapes made of baked brick and ceramic appeared in the buildings [1]. During the rule of the Somonites in the 10th century and the Karakhanids in the 10th-11th centuries, brickwork and carved ornaments reached a high level in architectural decoration. This trend, which started with the decorations of the palace of Termizshahs in

Afrosiyab, Surkhandarya, and the brick decorations of Ismail Somoni's mausoleum, reached a high level in Raboti Malik in Karmana, the minaret in Jarkurgan, Mirsaid Bahrom's mausoleum, Minorai Kalon, Vobkent minaret. The appearance of brick patterns, harmoniously combined with carvings and reliefs, was preserved in the Mag'oki Attori mosque, built in Bukhara in the 12th century. During this period, baked clay - terracotta carved decorative elements began to be widely used. During this period, it is observed that the process of creating terracotta, ceramic decorative elements with air color, turquoise and brown boards started. Such decorative ceramic tiles were found in the Kalon tower of the 12th century. Inscribed ceramic slabs were originally given one or two different colors in the embossed parts. Due to the invasion of Mongols in Central Asia, the use of glazed ceramic or glazed tile in architecture was somewhat delayed. The methods of production of glazed earthenware, formed in the pottery of the Karakhanid period, made it possible to create architectural glazed earthenware decorations during this period. Simple forms of ceramic coverings are cut out of clay or molded, after drying, dipping in a mixture of different colors, creating a colorful glazed ceramic coating by baking in steamers. Later, it became more complicated, and since the second half of the 14th century, glazed tile - architectural ceramic decoration has become the main type of architectural decoration in the territory of Uzbekistan. It was during this period that the tile-based techniques of enamel engraving and printed terracotta (ceramic) full-scale decorative majolica and mosaic appeared and developed rapidly. Masters identified several types of decorative material, which they called tiles. Tile is made of light, porous, flowable silicate clay, which is made by adding a small amount of clay and lime to fine white quartz sand. There is also an opinion that the name Koshin was given because it was invented by the craftsmen of the city of Koshan in present-day Iran. In Movarounnahr, there were tiles made in mortars using crushed glass quartz or quartzites (sand, sandstones). In Khorezm, a mixture of glue (shirach) and natural fine sand was used a lot [2]. Fragile, easy-to-work tiles are used for making Me'morchilik va qurilish muammolari 2024 йил, №2 сон 12 mosaics. The necessary flowers of the pattern are easily sawed in detail and cut with a knife. In the territory of Uzbekistan, in addition to dull colored bricks, tiles with a good glaze are also produced. The ceramic plate is coated with a substance that forms a vitreous layer of light-melting colored glaze. Clear coatings are made of glaze, dull, dull ones are made of colored enamel. Glaze includes materials and oxides that make up the vitreous coating. Lead glazes are made from the chemical addition of silica material to lead oxide obtained by burning metal in air (silica - white sand). But they will not be hard, they will break and cause small cracks (ceka) due to the different coefficient of expansion with the clay. They migrate quickly due to the acid in the air. Lead-free glazes are more durable. They can be used to create different colored tiles. For example, iron oxide gives yellow, red, brown, black and other colors when heated under different conditions and in additives. Copper oxide gives air color, blue and green colors. Sulfuric copper is used in the restoration of ceiling chandeliers as a creator of light effects. Tin oxide, when added to glaze, dulls the clarity of the color (lead-tin, dull glaze or enamel is used to cover the smooth surface of majolica boards). Chromium oxide gives a pure green color, and cobalt salts give a blue color. Instead of vinegar mixtures, natural clays, together with metal salts, were widely used. Additions of iron, calcium, magnesium, sodium, potassium give glaze dark colors. Types of local glazes based on alkali (alkaline plant ash) show a clear, bright, clean and shiny color. The color bouquet of the mysterious patterns looks different depending on the viewing angle and the intensity of each color depending on the change of light. These factors determine the individual skills of each

master tiler and tiler [3]. In Central Asia, especially Uzbekistan, the basis of architectural decorative coverings are square, rectangular, hexagonal ceramic tiles, and cut brick surfaces. They are covered with ganch mixtures on the surface of the structure. Muqarnas are baked in special forms and glazed to ceramics (pottery). There are also muqarnas elements cast from ganch, which are then covered with majolica boards. There are different types of majolica glazed tiles, such as ceramic tiles with different colored glazes, tiles with patterns on the surface, majolica tiles with black borders and enamel patterns or different colors with carved borders, and gilded tiles. Each glaze has its own firing point, at which point the color comes out clear and bright. Therefore, each color is cooked separately. Patterns printed on majolica boards of different colors are not baked at the same temperature. A board for each color is cooked separately in the oven. Therefore, it is convenient to type the coverings in a mosaic method. It is possible to make each pattern in different sizes, to achieve clean and bright colors. However, coating with majolica boards requires relatively little work, and the main thing is that it allows to increase the gloss of colors, free placement of patterns, and the artistry of decoration. An architectural decorative coating that does not blend into one another, it is the cutting of tiles to create a mosaic (koshinburush). A piece of tile cut into pattern lines on paper or a flat surface is sorted by color and ganch is poured over it. In the same way, pieces of tile are picked up and hardened with ganch on the surface of curved, arc-shaped surfaces drawn on the roof. A certain size piece of a tile mosaic hardened with a hammer is installed in a designated place on the wall. A welldefined general pattern will be imperceptible. Tilers used the khataba method when installing common tiles on the wall surface, that is, they leave an air gap between the wall surface and the tile - ganch coating. In this way, the grout that holds the tile hardens quickly, and the condensation in the air between the wall and the tile coating makes it easier to remove moisture, thus protecting the tile from spills. Unique schools of architectural decoration of Khorezm, Samarkand, Bukhara were formed in medieval tiling. Kashkadarya and Fergana schools also functioned independently. The representatives of these schools were not left behind by skilled foreign masters brought in during the Timurid period. Each of these schools has left its mark on the art of architectural decoration in Uzbekistan [4]. The first Khorezm pottery was discovered in the 14th century in the tombstone of Najmuddin Kubra in Old Urganch (20-40s of the 14th century), in the mausoleum of Khoja Allouddin (Khiva), and in the mausoleum of Mazlum Khan Suluv. The patterned majolica tiles of various shapes with delicate Islamic flowers are done with great skill. Dark blue, inky ground is coordinated with white, yellow, airy flowers, Arabic inscriptions, ornaments on four sides. Khoja Olovuddin's tombstone is an example of majolica decoration that has no equal of its time. During the period when Khorezm masters were perfecting the majolica decoration, Bukhara and Samarkand masters brought glaze to the traditional carved ceramic boards, bricks, elements (terracotta) and continued the tradition of porcelain tiles formed before the Mongol invasion and brought cast porcelain architectural decoration to a perfect level. This direction was preserved until the Timurid period. In the 30s-70s of the 14th century, the mausoleums of Qusam ibn Abbas, Khoja Ahmad, 1360, the mausoleum of Turkan Aqo and Tughluq Tekin, and the mausoleum of Buyonkulikhon in Bukhara, built in the Shahizinda complex of Samarkand, clearly showed the possibilities of this direction. In the tomb of Qusam ibn Abbas (XIX centuries), several types of covering of carved terracotta with cast glazes have been identified. The designs on the majolica tiles are embossed in a kundal style, and the geometric shapes are carved with glazed ceramic. The colors of the patterns in the mausoleum cave are different, and the flowers are similar to the patterns of the XI-XII

centuries. Glazed bricks were made along with cast glazed elements with relief patterns. Arabic inscriptions with carvings are very skillful, an example of which can be seen in the mausoleum of Khoja Ahmed (XIV century). By the time of the Timurids, high and diverse types, methods, and forms of architectural ceramics appeared. Patterned majolica is not only a set of boards that cover vertical surfaces: in addition to nimustuns, cornices, muqarnas, signs, the complex geometric shapes of arches, bagals, domes, domes under the dome are also covered with Проблемы архитектуры и строительства 2024, №2 13 boards of different shapes. In the period of Timurids, together with majolica, carved glazed tiles and cut tiles were used together. The practice of applying gold to the surface of printed, engraved, and embossed lacquered boards also appeared. The mausoleum of Tughlu Tekin (1375-1376) in Shahizinda, where the patterns on the top and bottom of the glaze are shown on the six-sided majolica board, was skillfully used. Koshinburush, which was widely used in the later periods, was also used in the architectural decoration of the Timurid period. More than 40 monuments of the historical architectural complex of Shahizin in Samarkand embody all types of the art of tiling in the heritage of Central Asian architecture and are considered the masterpieces of architectural decoration of the Timurid period. Various methods of ceramic coverings with an as-yet-unrevealed pattern of architectural ceramics have also been identified in archaeological excavations. Such ceramic coverings were found in the remains of the madrasa built in 1066 by ruler Tamgoch Bugrokhan from the Karakhanid dynasty. In the remains of the main entrance gate of the madrasa, there are also terracotta carvings in the same airy color. The types of carved ceramics with intricate designs and decorated with different glaze colors were used for the mausoleums of Khwaja Ahmad and 1361, which were built at the place where the first western road of the Shahizinda complex was completed. Glazed brick mosaics, carved tiles, glazed ceramic tiles, engraving fine lines of Arabic inscriptions on the carved clay - all these indicate the high level of artistic possibilities of the glazed ceramic material used in the architectural decoration of the Timurid period. Examples of tiling decorations from majolica to very fine cut tiles in the architectural decoration of the Timurid era are available in the mausoleums of the Shahizinda complex built during the Timurid era. These tile coverings determined the main directions of the further development of tiling in the architectural decoration of Central Asia [5]. Various covering decorations produced on the basis of architectural ceramics - ceramics are known in all regions of the world where pottery is spread, and are even called by the names of some countries. For example, porcelain made by Chin (Chinese) potters was brought to Western countries from ancient times through the Great Silk Road. Porcelain vases (porcelain vases) made in China were very delicate and sonorous. Even the box in one of the gardens built by Ulugbek was covered with porcelain. Porcelain shapes were made from white clay, with a certain amount of crushed stone powder (feldspar) added to the clay, which was called "kaolin" after the place where high-quality soil for the production of porcelain was found in the VI-VII centuries, and baked at a very high level of heat. Ceramics (porcelain) obtained in this way were leaky and brittle. In the 7th and 8th centuries, porcelain was painted with colored images and patterns (colored varnish) and then baked in a furnace. Of course, there was an attempt to produce such porcelain products in Central Asia. But the local materials here (clay, ganch) allowed to create monand ceramics of only high level porcelain. The use of angob instead of lacquer ensured that the glaze of the ceramic was clear and elegantly polished. That is, lead and tin glazes are processed by making the white color contained in the alkali liquid and applying it to ceramic, terracotta before glazing. Because the angob is white, when the main glaze (paints) are applied to

the surface of the tiles and they are baked again, they are resistant to any cold and moisture, they do not move, and the colors come out clean. Some experts say that the angob, angobling process, which originated in the ancient Greek pottery art of Western Europe, was supplanted by Middle Eastern potters in the 9th century and replaced by angob, enamel, and vitreous glaze. There is also an opinion that the rough and uneven places are covered with white or colored liquid clay before firing, that is, the liquid and thin layer of white clay forms a smooth layer during the firing of the clay, which is called angob in French. The development of this direction has given rise to lacquer coatings, glaze, enamel. The word glaze (glaze) comes from the German word "glas" - "glass", and it refers to the vitreous layer on top of the ceramic. Glazed ceramic gives a decorative look while maintaining an atmospheric effect. Glaze (glaze) is a vitreous layer formed by adding oxide (salt) of sodium, potassium, magnesium, silicon, aluminum and other metals. Stiff glazes are glass (quartz - white sand) glazes, which are called enamel in Europe, that is, heavy metal oxides are mixed. About the preparation of such secrets, Abu Rayhan Beruni also writes about one of the ancient methods in his book "Mineralogy": "Enamel baked by taking feldspar and rock crystal in a certain proportion and adding tin salt and soda (soda) comes out colorful like precious stones. It depends on the added mineral dyes. For example: lead oxide (suric) emits a yellow color, copper oxide - green, burnt gold, dark red, and cerdalic (precious stone) powder emits a purple color. Eastern masters knew well the secrets of producing blue colors. Among the monuments of Shahizin, there is a type of tile made by adding white sand, feldspar, and partially ganch to faience - clay. Produced in abundance by the potters of Faenza, Italy, this type of glaze was widely used as household and building materials. The ceramic tiles produced in Metlach, Germany are typed on the floor and called metlach tiles. Faiencebased ceramics were produced as early as Ancient Egypt. Beruni testifies to the fact that Chinese porcelain has been known for a very long time. Ceramics (from the Greek word keramos - clay), porcelain - porcelain (China) made on the basis of a mixture mined in special white clay deposits. Kaolin (the name of one of the mines in China) clay (which is also available near Angren in Uzbekistan) has the ability to bake clear, thin, water-resistant porcelain with images of various colors. Along with the development of pottery and porcelain production, architectural tiles also improved. By the 15th century, the main direction of coating various tile elements in majolica and mosaic methods appeared in architecture. The decoration of wall surfaces and architectural forms (domes, column heads, surfaces, arches, bagals, muqarnas, sharafa, etc.) reached a high level in the architecture of the Timurid era [6]: Me'morchilik va qurilish muammolari 2024 йил, №2 сон 14 1. Cast glazes on carved terracotta - a ceramic plate with a carved pattern, a variety of ceramic architectural forms (concave, relief, circular, arcuate surfaces, muqarnas pieces, among them). This method, where the colors are the same or several colors are given separately, has been known since the 12th century. It was highly developed in the 14th century; 2. The width of Islamic, girih, epigraphic flowers is 1.5-2 mm. thin elements are also created up to the thickness. Carved terracotta (ceramic) architectural elements given a completely different color glaze; 3. Majolica tiles, a method of covering the ceramic surface with tiles in different colors. These boards are baked on a ceramic or faience base and then patterned. Since the glaze is given in different compositions, the colors are baked separately. Although such majolica was prepared by Eastern and Western masters in different ways, it was only during the Timurid period that it was widely used to cover large surfaces on the exterior of buildings; 4. The production of majolica tiles with relief, carving or embossed color separation (that is, they do not flow into each other) is observed in the constructions of the Timurid period;

5. During this period, gold plating on majolica patterns was also practiced; 6. Brick glaze mosaics - the art of making mosaics from glaze bricks given one type of color. The further development of this method led to koshinburush; 7. Although the art of mosaic covering from cut tile began in the Middle East, Iran, it was not as prominently displayed in architectural decoration as in the period of Timurid architecture in terms of covering large surfaces, fineness, clarity of colors, and skill of dialing. Some experts associate the method of glazing thin tiles and then cutting them to make a mosaic with the practice of craftsmen from the city of Koshan, Iran. However, during this period, the art of tiling was at a high level in Movarounnahr. Koshinburush - the method of cutting, cutting, and sawing flower and pattern lines from tile boards to cover the Shahizinda complex in Samarkand, Tuman Aqo's mausoleum (1405), was the peak of the development of this method. It is surprising that even very delicate flower paths are cut. Although the cut tile mosaic method requires a lot of work, the colors are clear, clear, and have a high artistic effect when they are selected with jeweler's precision; 8. The skill of using various techniques, such as brick glaze mosaic, glaze majolica, carved brick, ganch, wood carvings, and the harmony of the tiling method, skillfully used in the external decoration of buildings, reached a high level in the architecture of this period, and determined the stages of the development of architectural decoration in the following centuries. During the 16th-18th centuries, in the architecture of Khorezm, Bukhara, Shahrisabz, Kokand, Samarkand, various methods of architectural porcelain continued to be used. In Kok, brick-glazed mosaic, in Samarkand, majolica, koshinbrush methods, and in Khorezm, the use of majolica continued successfully. The famous madrasas of Samarkand in Registan Square, Sherdor, Tillakori madrasas, many buildings in Khiva, Kokan Horde, many buildings in Bukhara are evidence of this process; 9. Many years of research are being carried out in order to restore and repair architectural ceramic tiles. Especially with ceramics, although there are tests and experiments to find the ancient quality of paints and glazes, to restore the original colors, and to determine the methods of replacing the coatings, the originality of the repaired architectural ceramic coatings remains a problem. Violation of technologies is observed during tile baking, color preparation, and installation; 10. The main reasons: it is necessary to prepare this raw material in the same quality as the old potters and tilers. The choice and method of fuel for cooking tiles in Khumdons can also be appropriate if it is done in ancient technology. Each type of tile requires its own preparation. Installation in place, i.e. coverings in the ancient way, khataba - cavity, using compounds for gluing to the surface in accordance with the experience of folk masters, makes the practice of modern repair science effective.

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THE ROLE AND IMPORTANCE OF ENGINEERING FIELDS IN MODERN CONSTRUCTION AND ARCHITECTURE

Parmanov.J.T.

Samarkand State University of Architecture and Construction named after Mirzo Ulugbek

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Abstract. *A physicist uses his knowledge and experience to study and apply the interaction of physical processes in the fields of engineering, construction, architecture, assembly work, mechanics.*

Keywords: *engineer; civil engineer, design engineer; mechanical engineer; development engineer; computer engineering specialist; applied mechanics specialist.*

Introduction

Physics teacher Provides training and education of students, taking into account the specifics of teaching the subject "Physics". Conducts lessons, additional optional classes, leads subject circles. Develops a thematic work plan on the subject, ensures the implementation of the curriculum. Participates in methodological work, uses the most effective forms, methods and teaching aids. Analyzes student activity, ensures compliance with educational discipline. This forms the skills of independent work of schoolchildren, stimulates their cognitive activity and motivation for learning. Achieves a strong and deep assimilation of knowledge on the subject, the ability to apply knowledge in practice. Equips and prepares the classroom. He studies and takes into account the individual characteristics of students in his work, participates in working with parents. Physics is one of the most complex subjects in the curriculum in higher education institutions, as it is a constantly changing scientific field. Therefore, a physics teacher must keep up with all the innovations in the world of science, familiarize himself with new discoveries, technical achievements and inventions.

The main task of a physics teacher is to teach children to understand the world around them, the processes taking place in their daily lives.

Physics is one of the most important and ancient sciences. With its help, a huge number of different processes are studied. Therefore, specialties related to physics will remain relevant for a long time. Physics is a fundamental science and is used in many areas of activity.

Physicist-engineer-A profession associated with knowledge of physical phenomena and constant practice. In this profession, it is necessary to know all mechanical processes, since this work is associated with the introduction of equipment and new technologies in various enterprises. In the event of the invention of a new technology in any research, you will have an incredible career and success. There are many directions in this field.

Physicist- A scientist whose research is mainly devoted to physics. Physicists work on a wide range of problems, from subatomic particles to the behavior of the universe.

The subject of the professional activity of a physicist is a field of science and technology that includes a combination of tools, techniques, methods and techniques for obtaining complete and reliable information about the nature and quantitative patterns of physical processes in the world, existing and new technical systems for various industries.

A physicist is engaged in the study of objects of the world and the laws of their interaction. He considers objects as physical bodies, and their interaction as physical phenomena. Conducts

physical research through experiments, builds mathematical models of physical phenomena, describes the main properties of the world. The study of physical phenomena allows physicists to discover general laws and use them for progress. For a physicist, observation and curiosity, perseverance and the desire to learn new things, patience and critical thinking, a penchant for experimentation, interest in nature and the ability to be creative in science are important. The profession requires a specialist to make mainly intellectual expenditures. The activity is associated with the analysis, comparison and interpretation of data, the development of new solutions.

The engineering profession has always been the basis of world development. The level of technical equipment determined the superiority of one civilization over another until the beginning of our era. And today it is technical innovations that ensure the development of civilization.

Today, engineering professions are the most numerous professions of highly qualified workers. More than a third of specialists with higher education in our country are engineers. An engineer participates in the production of all material goods of society - from food and consumer goods to complex computers and space rockets. A modern engineer is a highly cultured specialist who is well versed in modern engineering and technology, economics and production organization, who can use engineering methods to solve engineering problems, and at the same time has the ability to invent. The work of an engineer is a link between scientific discoveries, developments and their practical application. Engineers manage production sites at industrial enterprises, construction, agriculture and other industries, work in design bureaus, laboratories and research institutions, are engaged in the organization, planning and economics of production. They participate in the design and development of design technologies, industrial equipment, machines, production control systems, production automation, business, management processes. They study the causes of breakdowns and production failures, test products, determine their quality, etc. For full and high-quality work, an engineer needs mathematical and technical skills; analytical thinking; concentration; abstract thinking; a penchant for research; the ability to draw.

Design engineer- Engineering specialty, the activity of which is necessary for the creation and creation of a final (target) product from existing material production products and resources. Creates new objects of material culture from existing resources, organizes and technically equips the work of other people.

Design engineers create, check and edit drawings, calculate the design of structures, participate in the coordination and protection of the project, and carry out technical and architectural supervision over its implementation. The designer's duties include testing and commissioning experimental products and parts planned for future use. Develops preliminary designs, technical and working structures and products of varying complexity, organizes production processes for the production of parts and assembly machines, conducts research in the field of design, determines the technical level of manufactured products, calculates the economic efficiency of implemented projects, and draws up technical documentation for the developed design.

Mechanical engineer- a specialist with higher technical education in the field of design, construction and operation of technological equipment.

A mechanical engineer designs, builds and operates mechanical equipment, machines, devices and apparatus, automatic lines, means and systems of complex mechanization and automation of production, organizes and conducts their installation, commissioning, testing. He develops, plans and organizes technological processes, selects optimal conditions for their

implementation. His duties include planning and repairing machines, reconstructing existing ones and preparing technical specifications for the construction of new plants. In the field of agricultural production, a mechanical engineer supervises machinists, manages all mechanization of agriculture.

The main purpose of a mechanical engineer's activity is to design mechanical equipment and technological processes and organize technical maintenance of equipment.

An engineer of this specialty is a highly qualified specialist with in-depth knowledge of the theoretical foundations of electrical engineering, the theory of automatic regulation, industrial electronics and computer engineering.

Civil Engineer Works in general construction and specialized construction, construction and installation, commissioning, service, design, engineering and scientific organizations.

Carries out production and technological, organizational and management, design and research work in the field of construction. These specialists solve problems related to the design and construction of buildings and structures, water supply and sewage systems and devices, roads and pipelines, power lines and communications and other objects.

During their professional activities, a civil engineer calculates, designs and develops building structures, foundations and foundations, underground parts in various soil conditions. Develops and implements projects for the organization of construction and construction work using technologies for the production and installation of building structures, established mechanization and modern methods of labor.

A civil engineer supervises construction, installation and commissioning work, monitors their quality, exercises technical supervision over design decisions and the implementation of construction and installation works. In construction, he is engaged in the regulation of labor and budget, providing engineering support for the accounting of the self-financing brigade, draws up work orders and estimates of wages and salaries.

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USING INTERACTIVE AND NON-STANDARD TEACHING METHODS TO INCREASE THE EFFECTIVENESS OF LANGUAGE LEARNING

Kultayeva F. E.

Senior teacher of the Department of Foreign Languages,
Samarkand State University of Architecture and Construction

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Abstract. *Some pedagogical methods of interactive teaching models and methods of teaching languages which are used at the English lessons are considered in the article. The article is suggested some exercises which help to get to know members of the group closely and to establish inherent group communications for the effective learning and vocabulary activities. For the purpose of developing creative specialists' ability are provided using communication possibilities in the teaching process on the professional basis.*

Keywords: *Interactive methods, presentation, computer training, group discussion, business and role play, quiz, different classroom activities, try to find answer, choose the correct answer.*

Introduction

The 21st century is an age of rapid information and communication development, technological progress and market relations. To feel "comfortable" in this "space," future specialists need not only to acquire a specific volume of professional knowledge and skills but also to develop and enhance their speech and thinking skills for communication, particularly in a professional context, with a team. Alongside political issues, language problems are no less relevant today. Recently, there has been significant interest in educational models, pedagogical innovations, technologies, and methods that prove to be the most effective for this purpose. Questions about the organization and intensification of the educational process are constantly in the spotlight. How can we provide the necessary amount of information in the least amount of time? How can we ensure deep understanding? Since foreign language teaching is a specific instance of education as a whole, answers to these and many other questions can be found in the underlying methodology.

In the current conditions of educational service market development and the demands of the information technology era, teaching in English should combine established directive practices with a modern, innovative, and interactive learning model. It is essential to remember that learners must be ready to consciously use the language in real-life communication situations. By "consciousness," we refer to a principle of communicative linguistics through which learners acquire knowledge of facts, definitions, and laws, comprehend conclusions and generalizations, express their thoughts correctly in speech, and independently apply their knowledge in practice.

There is no doubt that organizing such education for non-philology students, who have a modest number of hours allocated for foreign language study at university, can be quite challenging. Therefore, the communicative goal of education must be clearly defined: why do we speak or write? The process of information transfer is based on the principle of interaction between the teacher and the student. It requires greater activity from the learner and their creative rethinking of the information received. It is advisable to use some non-traditional methods of conducting lessons, incorporating poems and riddles, songs and charades, rebuses and crosswords,

as well as tests. All the aforementioned points, as practice shows, are highly effective in enriching students' vocabulary.

In classes focused on developing oral speech, the following problematic situations are interesting: relationships between children and parents, friendships, money issues, norms of behavior in society, the role of technical progress, youth problems, politics, economics, etc. Interactive game lessons offered for communicative language study involve exercises categorized into seven sections based on their objectives: exercises to create an atmosphere for group work; exercises to identify common interests; exercises that support group coherence; linking exercises; stimulating exercises; exercises that encourage creativity; and concluding exercises.

Here are some exercises on the topic: "**Environmental Pollution.**"

Task I. Vocabulary Match. Objective: Enhance vocabulary related to environmental pollution. Instructions: Match the terms with their correct definitions.

Pollutant

Ecosystem

Biodegradable

Greenhouse gases

Acid rain

Sustainability

Definitions:

- a. A gas that contributes to the greenhouse effect by absorbing infrared radiation.
- b. A substance that contaminates the environment.
- c. Rain that has been acidified by pollutants, leading to environmental harm.
- d. Capable of being decomposed by biological processes.
- e. A community of living organisms and their physical environment interacting as a system.
- f. The ability to be maintained at a certain rate or level without depleting resources.

Answers: 1-b, 2-e, 3-d, 4-a, 5-c, 6-f

Task II. Discussion Questions. Objective: Encourage critical thinking and discussion about environmental pollution. Instructions: Discuss the following questions in pairs or small groups.

1. What are the main sources of environmental pollution in your community?
2. How does pollution affect human health and wildlife?
3. What measures can individuals take to reduce their carbon footprint?
4. Discuss the role of governments and organizations in combating environmental pollution.
5. Can you think of any successful initiatives that have reduced pollution in your area or globally?

Task III. Choose the correct answer. Learner A asks a question and Learner B answers it.

A: 1. Why is pollution very important problem now?

B: a) Because, air, water and soil are necessary for existence of all living things, but polluted air can cause illness, and even death.

b) People lived in uncrowded rural areas and did not have pollution — causing machines.

A: 2. Can food on polluted soil, be grown?

B: a) Yes, it can.

b) No, it cannot.

A: 3. What are the problems with automobiles and industrial enterprises connected with environmental pollution?

B: a) *Automobiles are polluting the air but they provide transportation for the people. Factories pollute the air and the water but they provide jobs for people and produce goods.*

b) *We need automobiles and factories polluting environment*

Task IV. Add the missing parts of the sentences from the text and complete the sentences below with the words

1. Air, ... and ... are necessary for existence of all living things

2. Environmental ... spoils the natural beauty of our planet.

3. Fertilizers and pesticides are important for growing crops but they can ruin...

4. Scientists and engineers can find the ways ... pollution from automobiles and factories.

a) *soil*

b) *to reduce pollution from automobiles and factories*

c) *water, soil*

d) *pollution*

Check your answers below (see example)

1	2	3	4
<i>c</i>	<i>d</i>	<i>a</i>	<i>b</i>

Home assignment: Research and Presentation

Objective: *Develop research and presentation skills.*

Instructions: *Choose one type of environmental pollution (air, water, soil, noise, etc.) and prepare a short presentation (5-10 minutes) covering the following points:*

Definition and causes of the chosen type of pollution.

Effects on the environment and human health.

Case studies or examples of areas affected by this type of pollution.

Solutions or strategies to mitigate this type of pollution.

Your personal opinion on what actions should be taken.

These exercises help group members get to know each other better, set the tone for group work, quickly establish intra-group relationships and norms of behavior, stimulate and encourage collaboration, and provide new perspectives on the world by applying new, unconventional methods. (Games and Warm-ups: A Guide for Group Work. Based on the Warmups guide prepared by Nell Warren Associates Inc.) Educational games make the process of learning a foreign language more interesting and engaging. A sense of equality and an atmosphere of enthusiasm allow students to overcome shyness, stiffness, language barriers, and fatigue. Through games, for example, pronunciation can be practiced effectively, vocabulary and grammar can be activated, and not only listening skills but also the art of speaking can be developed.

However, when using games in lessons, educators need to constantly keep in mind several important points:

1. The choice of game format should be pedagogically and didactically thought out and justified; it should involve as many learners as possible.

2. The game should match the language abilities of the learners.

3. Language games should be conducted exclusively in the foreign language.

4. The teacher should remember that games are not a substitute for systematic study and intensive training.

5. Games should be used in a measured, appropriate and planned way, bearing in mind that they build on material already learnt.

Performing in the classroom is the strongest motivator for learning a language, it helps to create a natural language environment. For example, the use of theatrical games in lessons has proven effective, primarily for developing skills and abilities in preparatory oral speech. Students in colleges or universities, especially in the early stages of learning, are also keen on the opportunity to try their hand at acting. In modern practice, active learning methods are becoming increasingly common: training sessions, programmed and computer-based learning, group discussions, analysis of specific practical situations, and business and role-playing games.

There are also so-called non-standard lessons, such as press conference lessons, conference lessons, birthday lessons, spinning lessons, and matchmaking lessons. It is advisable to conduct these in advanced classes or student groups, as they require significant preparatory work from both the teacher and the students themselves. In this context, the strategy for verbal interaction is defined by the communicative goals of the participants, such as: requesting, obtaining, or conveying significant information; discussing an issue in order to make a decision; convincing a partner of something; proving the correctness of one's position; evaluating a fact, situation, or object; expressing or defending one's opinion; supporting or refuting a partner's opinion; proposing and justifying a hypothesis; indicating a way to solve a problem; stating a fact; summarizing; making a conclusion; supporting an idea; expressing doubts or criticism regarding the discussed issue; telling about an event, person, object, etc.

Teaching a foreign language based on interactive methods fosters team cohesion, promotes sociability, creates a favorable psychological climate in the learning group, and ensures a solid grasp of knowledge. The process and results of learning gain personal significance for all participants and allow learners to develop the ability to solve problems independently. Time moves forward, new technologies enter our lives, and therefore, resting on one's laurels means losing much. Utilizing the resources of a computer lab or university classroom for foreign language lessons is a powerful motivator that must be leveraged in teaching practice.

In conclusion, it is important to note that foreign language lessons provide educators with significant opportunities to spark interest and create motivation for further study. The importance of learning a foreign language today cannot be overstated. For practicing teachers, there are limitless opportunities for creative exploration of new forms and methods to activate the learning process.

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EFFECTIVE METHODS OF TEACHING ENGLISH FOR CONSTRUCTION STUDENTS

Muslim Khayitboev

SamSACU, Teacher of Foreign Languages Department

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Abstract. *This article discusses the importance of teaching English to construction students and examines various effective methods for enhancing their language proficiency. The focus is on addressing the specific needs of construction students who require functional English skills for their future careers in the construction industry. Different pedagogical approaches, including task-based learning, communicative language teaching, and industry-specific vocabulary acquisition, are explored. The paper also highlights the role of technology in facilitating language learning and the importance of integrating real-world construction contexts into the curriculum.*

Keywords: *English for Specific Purposes, construction students, task-based learning, communicative language teaching, industry-specific vocabulary, language acquisition*

Introduction

The construction industry is a dynamic and technical field that demands effective communication and collaboration among professionals from diverse backgrounds. In this context, English serves as the universal language for technical documentation, cross-border collaboration, and access to global resources. For construction students, mastering English is not only a tool for academic success but also a key to thriving in their future careers.

However, teaching English to construction students requires a specialized approach. Unlike general English learners, these students need to focus on industry-specific terminology, technical communication, and real-world applications of the language. The challenge for educators is to design a curriculum that is engaging, practical, and tailored to the unique needs of construction students.

This article explores effective methods for teaching English to construction students, emphasizing techniques that combine technical knowledge with language skills. By integrating contextual learning, interactive activities, and modern teaching tools, educators can create an environment where students gain both language proficiency and confidence for their professional journeys.

1. The Importance of English for Construction Students. Construction students must acquire a level of English proficiency that enables them to navigate the technical and interpersonal aspects of their future roles. English is used extensively in technical documentation, blueprints, safety guidelines, contracts, and negotiations. Moreover, construction workers often collaborate with international teams, making communication in English indispensable. For students to be competitive in the job market, it is necessary for them to develop both technical and communicative English skills.

2. Task-Based Learning (TBL) is an effective approach for teaching English to construction students. This method focuses on using tasks that reflect real-life activities related to the construction field. Tasks such as interpreting blueprints, writing construction reports, and participating in meetings allow students to practice English in authentic contexts. Through TBL, students not only improve their language skills but also develop a deeper understanding of the specific vocabulary and structures used in construction.

3. Communicative Language Teaching (CLT) emphasizes interaction as the key to language learning. In the context of construction, CLT encourages students to engage in realistic conversations that they may encounter in the workplace, such as discussions about project timelines, safety protocols, or site management. Role-plays, simulations, and group activities can be used to provide students with opportunities to practice communication skills in a low-pressure environment. This method helps students gain confidence in using English to express themselves in professional settings.

4. Focus on Industry-Specific Vocabulary. A key challenge for construction students learning English is mastering the industry-specific terminology. Construction language is filled with specialized terms that may not be found in general English courses. Therefore, a focus on vocabulary acquisition is essential. Teachers should introduce technical terms and phrases related to construction materials, machinery, tools, project management, and safety regulations. Flashcards, word lists, and interactive exercises can help reinforce the vocabulary, ensuring that students can confidently use these terms in their professional roles.

5. Integrating Technology in Language Learning. Incorporating technology into language learning can significantly enhance the effectiveness of English instruction for construction students. Online platforms, language learning apps, and virtual simulations can provide students with interactive exercises, quizzes, and real-world scenarios to practice their language skills. Moreover, platforms like YouTube and podcasts can expose students to authentic English used in construction contexts. By using multimedia and digital tools, students can practice English at their own pace, reinforcing what they learn in the classroom.

6. Incorporating Real-World Construction Contexts. Teaching English in the context of real-world construction situations is one of the most effective methods. Teachers can incorporate case studies, project scenarios, and construction site visits to expose students to the practical applications of English in construction. This not only reinforces their language skills but also helps them understand the daily demands of their future professions. For example, students can be tasked with analyzing a construction project's documentation, interpreting building codes, or drafting safety protocols—all while using their English language skills.

7. Effective assessment and feedback are crucial for ensuring students' language development. Regular formative assessments such as quizzes, oral presentations, and project reports allow teachers to monitor students' progress and identify areas for improvement. Constructive feedback should focus not only on grammatical accuracy but also on the appropriate use of industry-specific vocabulary and communication strategies. Peer feedback can also be an effective tool, encouraging collaboration and improving learning outcomes.

In conclusion, teaching English to construction students requires a tailored approach that incorporates both general language skills and industry-specific knowledge. Methods such as Task-Based Learning, Communicative Language Teaching, and a focus on vocabulary acquisition, combined with the use of technology and real-world contexts, provide an effective framework for developing students' English proficiency. As the construction industry continues to grow and globalize, equipping students with the language skills necessary to navigate this field will play a crucial role in their professional success. Effective English language teaching can bridge communication gaps, enhance safety, and improve the overall quality of construction projects.

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STUDY OF SOLUTION OF MAGNESIUM CHLORATE-GUANIDINE SULPHATE-WATER SYSTEM

Shakarov Norboy Jumayevich¹, Nomirov Maxsidin Nosirovich², Ergashev Isomiddin
Shodiyarovich³

^{1,2,3}Samarkand State University of Architecture and Construction

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Abstract. *The solubility of magnesium chlorate, guanidine sulfate in aqueous solutions was studied visually-polythermally in the temperature range of 52.4 to 40 °C. X-ray phase, infrared spectrum, derivatographic analysis methods were used to determine the composition and structure of the substances formed in the system. A structural formula corresponding to the new substance guanidine chlorate was proposed.*

Keywords: *Magnesium chlorate, guanidine sulfate, guanidine chlorate, visual-polythermal method, solubility diagram, X-ray phase, infrared spectrum, derivatographic analysis methods.*

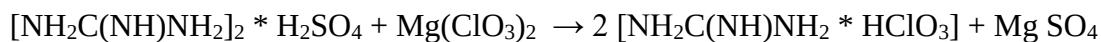
Introduction

The main direction of the development of the economy of our republic is to achieve a sharp increase in the national economy and, on this basis, to further improve the well-being of the population through the rapid development of all sectors and the acceleration of scientific and technological progress.

Cotton growing in our country is one of the important branches of agriculture. To date, the primary goal of technical progress in cotton growing is mechanized harvesting, and in its implementation, chemical cleaning of cotton from leaves using chemical means before harvesting plays an extremely important role. Without this important agrotechnical measure, it is impossible to achieve high productivity of harvesters and success in cotton growing at the current stage of cotton cultivation.

Harvesting the cotton crop without leaving it for the rainy days of autumn is an important agrotechnical measure, and high-quality chemicals that ensure the artificial shedding of cotton leaves are widely used. Although they are highly effective in defoliation, they pollute the cotton crop due to dried leaves, reduce the germination rate of seeds, and spray large amounts of chemicals due to the small percentage of the active substance in the working solution, which has a negative impact on the ecological situation. Guanidine salts have an antiauxin effect, under their influence the “aging” process in plants accelerates, cotton leaves turn yellow and fall off. The creation of defoliant from chlorate chemicals and guanidine salts, which are used in low doses, have a high efficiency, and have a gentle effect on cotton leaves, is one of the urgent problems that our growers are waiting for a solution. To this end, the subject of our scientific research was the study of magnesium chlorate and guanidine sulfate salts in aqueous solutions by the visual-polythermal method.

$\text{Mg}(\text{ClO}_3)_2 - [\text{NH}_2\text{C}(\text{NH})\text{NH}_2] \cdot \text{H}_2\text{SO}_4 - \text{H}_2\text{O}$ The system was studied by conducting nine internal sections in the temperature range from -52.4 to 40 °C (Table 1). The polymer diagram of the system shows ice, guanidine sulfate, $\text{Mg}(\text{ClO}_3)_2 \cdot 6\text{H}_2\text{O}$, $\text{Mg}(\text{ClO}_3)_2 \cdot 4\text{H}_2\text{O}$, $\text{Mg}(\text{ClO}_3)_2 \cdot 2\text{H}_2\text{O}$ and $\text{MgSO}_4 \cdot 2\text{H}_2\text{O}$ the concentrations of the initial substances and the system temperature values that characterize the formation are determined. Due to the low solubility of $\text{MgSO}_4 \cdot 2\text{H}_2\text{O}$, the precipitation limit occupies the main part of the diagram, that is, the following exchange reaction occurs.



$\text{MgSO}_4 \cdot 2\text{H}_2\text{O}$ the guanidine chlorate compound was isolated in the solid state by isothermal evaporation of the precipitated solution and analyzed for its identification by chemical elements and X-ray phase analysis. Thermal properties were studied using thermogravimetric methods, and the structure was studied using infrared spectral methods. $\text{NH}_2\text{C}(\text{NH})\text{NH}_2 \cdot \text{HClO}_3$ the results of chemical analysis for the substance represented by the formula %.

	ClO_3^-	C	H	N
Calculated	58,31	8,29	4,16	29,24
Experimentally determined	58,19	8,36	4,39	29,27

According to X-ray diffraction analysis, guanidine chlorate has a crystalline structure, and in the X-ray diffraction pattern, diffraction reflexes at 3.74; 2.94; 2.42; 2.16; 1.64; 1.50; 1.44; 1.39; 1.34 Å are observed, which are not characteristic of the original substances.

The thermal stability of the guanidine chlorate salt was studied. During heating, three thermoeffects occur, which correspond to the processes of liquefaction and decomposition of the substance. The decomposition occurs in two stages: the first is exothermic, the second is endothermic. Thermogravimetric analysis of the dori and togram shows that the mass change during the decomposition of the substance by exothermic and endothermic effects is 100%, that is, the substance decomposes without residue.

Secondary and tertiary points of the magnesium chlorate-guanidine sulfate-water system

Table 1.

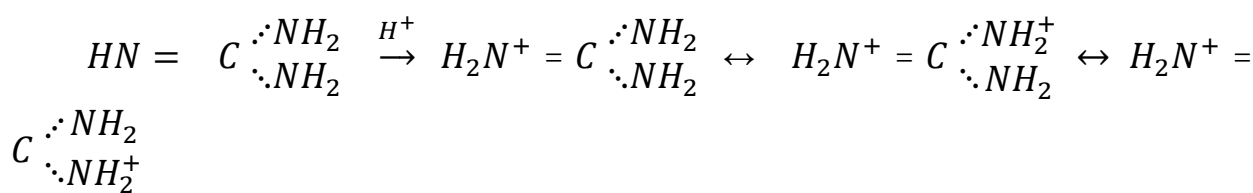
Mg (ClO_3) ₂	2 $\text{NH}_2\text{CNHNH}_2 \cdot$ H_2SO_4	H_2O	Crystallization, t°C	Solid phase
-	57	43,0	-12,0	Ice + $\text{Mg}(\text{ClO}_3)_2$
36,9	-	63,1	-52,0	Ice + $\text{Mg}(\text{ClO}_3)_2 \cdot 16\text{H}_2\text{O}$
42,0	-	58,0	-21,7	Ice + $\text{Mg}(\text{ClO}_3)_2 \cdot 12\text{H}_2\text{O}$
45,4	-	54,6	-7,5	Ice + $\text{Mg}(\text{ClO}_3)_2 \cdot 6\text{H}_2\text{O}$
2,6	56,0	41,4	-13,6	Ice + $\text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
2,3	60,6	37,1	-3,0	$\text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4 +$ $[\text{NH}_2\text{CNHNH}_2]_2 \cdot \text{H}_2\text{SO}_4$
1,8	69,0	29,2	23,0	-//-
5,6	52,0	42,4	-15,8	Ice + $\text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
11,4	35,6	53,0	-13,6	-//-
19,0	19,0	64,0	-18,0	-//-
28,2	5,0	66,8	-31,6	-//-
34,8	2,6	62,6	-48,7	-//-
7,3	46,7	46,0	-14,3	Ice + $\text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
48,0	10,0	42,0	8,0	$\text{Mg}(\text{ClO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{NH}_2\text{CNHNH}_2 \cdot$ H_2SO_4
46,8	6,8	47,4	1,2	-//-

43,2	3,0	53,8	-16,4	$\text{Mg}(\text{ClO}_3)_2 \cdot 12\text{H}_2\text{O} + \text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
45,0	4,2	50,8	-8,0	$\text{Mg}(\text{ClO}_3)_2 \cdot 12\text{H}_2\text{O} + \text{Mg}(\text{ClO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
41,6	2,5	55,9	-22,1	$\text{Mg}(\text{ClO}_3)_2 \cdot 16\text{H}_2\text{O} + \text{Mg}(\text{ClO}_3)_2 \cdot 12\text{H}_2\text{O} + \text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
36,6	2,0	61,4	-52,4	$\text{Ice} + \text{Mg}(\text{ClO}_3)_2 \cdot 16\text{H}_2\text{O} + \text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4$
4,4	55,0	50,6	-16,8	$\text{Ice} + \text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4 + [\text{NH}_2\text{CNHNH}_2]_2 \cdot \text{H}_2\text{SO}_4$
5,4	69,2	25,4	35,8	$\text{NH}_2\text{CNHNH}_2 \cdot \text{H}_2\text{SO}_4 + [\text{NH}_2\text{CNHNH}_2]_2 \cdot \text{H}_2\text{SO}_4$
7,4	68,0	24,6	40,0	-//-
3,4	69,6	27,0	28,0	-//-

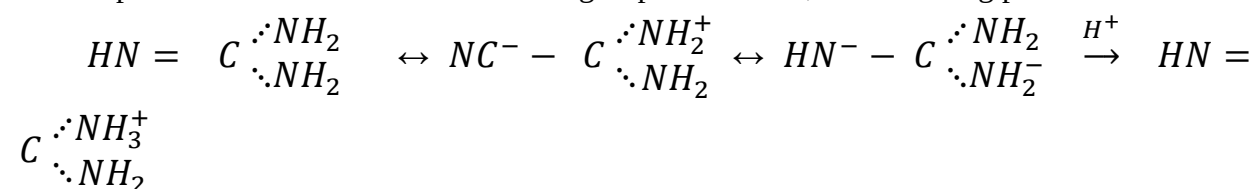
Guanidine exhibits strong basic properties. It contains amino and imino groups. In order to determine to which group the chloric acid proton is attached, the infrared spectra of the guanidine chlorate compound were studied.

The intense absorption lines belonging to the γ ($\text{C}=\text{N}$ -) group around 1510 cm^{-1} disappear from the infrared spectrum of guanidine chlorate. Based on this information, the imino group in the guanidine molecule is protonated (i.e., it binds an H^+ ion), and as a result, the electrons of the $\text{C}=\text{N}$ bond are delocalized.

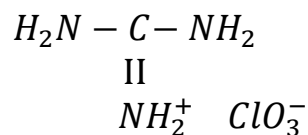
In fact, the protonation of the imino functional group practically forms a stable cation. Since it stabilizes the positive charge in the hybrid ion, all nitrogen atoms participate in it in an equal contribution.



If protonation of the amino functional group is observed, the following process occurs:



A small number of nitrogen atoms participate in stabilizing the charge of the resulting hybrid ion, that is, the degree of stability of the hybrid ion is high. In the infrared spectrum of guanidine chlorate salt, there are absorption lines around 3400 and 3270 cm^{-1} , which correspond to the frequencies of the asymmetric $\gamma_{\text{As}}(\text{NH}_2)$ and symmetric $\gamma_{\text{s}}(\text{NH}_2)$ vibrations of the amino group. This means that the amino groups in the guanidine chlorate molecule are preserved unchanged. Based on the data obtained, the following structural formula can be proposed for guanidine chlorate



Conclusion. The results of the study show that when using 70% magnesium chlorate and 80% guanidine sulfate solutions, the yield of guanidine chlorate formation is high, and its share in the product composition is 58.27%. It can be used in the preparation of a working solution of cotton defoliant.

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ADVANCED METHODS OF TEACHING ENGLISH TO ECONOMISTS

S.O.Boboyorov

Teacher, Department of Foreign languages, SamSACU

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Abstract. *Teaching English to economics students requires a tailored approach that integrates language learning with subject-specific knowledge. This article explores innovative methods to enhance the learning process, including content-based instruction, project-based learning, and case studies. Examples and practical applications are provided to demonstrate how these techniques foster both linguistic proficiency and economic expertise.*

Keywords: *English for Specific Purposes (ESP), Economics education, Content-based instruction, Project-based learning, Case studies, Simulation-based learning, Language acquisition, Economic terminology, integrated teaching methods, advanced language teaching techniques.*

Introduction

The role of English in the modern economic landscape cannot be overstated. As the global economy becomes increasingly interconnected, proficiency in English has become a critical skill for economists, whether they are engaging in academic research, participating in international conferences, or negotiating trade agreements. However, teaching English to economics students requires a specialized approach that goes beyond traditional language instruction.

Unlike general English learners, economics students need to master a lexicon filled with technical terms such as "inflation," "monetary policy," and "elasticity of demand." They must also develop the ability to analyze economic data, articulate complex ideas, and navigate sector-specific literature—all in English. Consequently, conventional teaching methods, which often focus on basic communication skills, fail to address the unique challenges faced by this group.

This paper explores advanced methods of teaching English to economists, emphasizing the integration of language learning with economic content. Techniques such as content-based instruction, project-based learning, case studies, and simulations will be examined in detail. These methods not only enhance linguistic competence but also prepare students to excel in their professional and academic pursuits.

The following sections will delve into these innovative approaches, providing practical examples and highlighting their effectiveness in bridging the gap between language proficiency and economic expertise.

Content-Based Instruction (CBI) is an approach that integrates language learning with the teaching of subject-specific content. In the context of teaching English to economists, CBI focuses on using economic topics and materials as the medium for language instruction. This method enables students to acquire both linguistic skills and a deeper understanding of economic concepts simultaneously.

Key features of Content-Based Instruction.

1. **Integration of Language and Content:** Students learn English by engaging with real-world economic topics, such as trade policies, market structures, or fiscal policies.
2. **Contextual Vocabulary Development:** CBI provides students with the opportunity to learn economic terminology in context, making it easier to understand and use accurately.

3. **Critical Thinking and Analysis:** Students develop analytical skills by discussing, debating, and solving problems related to economics while practicing their language skills.

Example: A Lesson on "Supply and Demand"

Objective: Enhance students' ability to discuss supply and demand dynamics in English while building relevant vocabulary.

Pre-lesson vocabulary building:

- Students are introduced to terms such as *equilibrium price*, *demand elasticity*, and *market shortage*.
- Activities include matching definitions, completing sentence gaps, and discussing examples from their local economy.

Reading activity:

- Students read an article from *The Economist* or another source discussing a real-world market scenario, such as rising fuel prices.
- They underline key terms and phrases while noting how they are used in context.

2. **Discussion task:**

- Students form small groups to debate the factors influencing supply and demand in the given example. For instance: *Why are fuel prices rising? What could governments or producers do to stabilize the market?*
- The discussion is guided by prompts and supported with sentence structures like "One reason for the increase in demand is..." or "A possible solution could be..."

3. **Writing task:**

- Students write a short essay summarizing the key points of the article and their discussion. This helps reinforce both language and content learning.

Benefits of CBI in Teaching English to Economists

• **Relevance:** Students find the lessons engaging as they align with their academic and career goals.

• **Practicality:** Learning economic terms and concepts in English prepares students for real-world applications, such as reading international journals, writing research papers, and presenting at conferences.

• **Efficiency:** CBI allows simultaneous development of English proficiency and subject knowledge, saving time and effort.

By focusing on authentic materials and meaningful tasks, CBI equips economics students with the tools they need to excel both linguistically and professionally.

Case studies are a highly effective method for teaching English to economists because they combine practical, real-world economic scenarios with language learning. This approach encourages students to engage in critical thinking, problem solving, and discussion, all while practicing specialized vocabulary and communication skills.

Key Features of Case Studies in Language Learning

1. **Real-World Relevance:** Students analyze actual economic events or issues, making the learning experience more meaningful and applicable.

2. **Integration of Skills:** Case studies require reading, writing, speaking, and listening, fostering comprehensive language development.

3. **Collaboration and Discussion:** Group work promotes peer learning and helps students articulate their ideas in English.

Example: The 2008 Financial Crisis

Objective: Teach students how to explain economic crises, identify causes and effects, and discuss policy responses in English.

Other Simulation Ideas for Teaching English to Economists

1. **Stock Market Simulation:** Students act as traders, analyzing market trends and making investment decisions.
2. **Economic Policy Debate:** Groups represent different political parties, proposing and defending fiscal policies.
3. **Corporate Meeting Simulation:** Teams present and evaluate business strategies for market expansion.
4. **World Bank Loan Application:** Students role-play as developing nations seeking loans and as bank officials reviewing proposals.

By immersing students in realistic scenarios, simulation-based learning develops not only their English proficiency but also their ability to think critically and act decisively in economic contexts. This dual focus prepares them for success in global academic and professional environments.

Teaching English to economists requires a specialized approach that combines language acquisition with the mastery of economic concepts. Traditional teaching methods are often insufficient to address the unique challenges faced by economics students, who must navigate technical terminology, analyze data, and communicate professionally in English.

Advanced methods, such as content-based instruction, project-based learning, case studies, simulation-based learning, and technology integration, provide effective frameworks for meeting these challenges. By immersing students in real-world scenarios, fostering critical thinking, and encouraging active participation, these methods not only enhance linguistic proficiency but also equip students with the skills needed to excel in their academic and professional endeavors.

The adoption of these innovative strategies creates a dynamic and engaging learning environment. Students gain confidence in using English in diverse contexts, from academic research to policy discussions and business negotiations. This dual focus on language and content prepares them for success in a globalized economy where English is the key to unlocking opportunities.

By bridging the gap between language and economics, educators can empower students to thrive in their future careers and contribute meaningfully to the international economic landscape.

Conclusion

In conclusion, teaching English to economists involves a multifaceted approach that integrates both language learning and the development of subject-specific expertise. As the global economy continues to rely on English as a universal language, it is essential that economics students not only master technical vocabulary but also become adept at using English in real-world economic contexts.

The advanced methods discussed with content-based instruction, project-based learning, case studies, simulation-based learning, and the integration of technology—offer effective strategies for achieving this goal. These methods provide students with the opportunity to engage with authentic economic materials, participate in meaningful discussions, and apply their language skills to solve practical problems. Through these approaches, students can bridge the gap between language proficiency and economic expertise.

Ultimately, by combining linguistic development with practical economic applications, educators can prepare students for the demands of an interconnected global economy, where effective communication and deep economic understanding are essential for success.

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BILINGUALISM IN BOBURNAMA

Jurayev Shahriyor Ibodulloyevich

Head of the Department of Uzbek Language and Literature,
Samarkand State University of Architecture and Construction (SamDAQU),
Doctor of Philosophy (PhD) in Philological Sciences

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Annotation. *The article highlights how the Turkic-Uzbek and Persian-Tajik peoples have lived together in harmony and friendship since ancient times, particularly from the era of the Baburids to the present day. It examines the skillful use of Persian words in Baburnama and analyzes them by grouping them into categories. The work also presents examples of Persian-Tajik sentences and poems.*

Keywords: *comparative analysis, lexeme, word formation, prose and poetry, borrowings, lexical layer, artistry, bilingualism, and zullisonayn (mastery of two languages).*

Introduction

Historically, our homeland, located at the bustling crossroads of the Great Silk Road, has been in continuous contact with numerous cultures and languages. These social connections between the two peoples have evolved and strengthened in our era. The history of the peaceful coexistence and friendly relations between the Persian-Tajik and Turkic-Uzbek peoples is deeply rooted and continues to this day. The ties between the two peoples are vividly reflected in artistic creativity, as seen in both Turkic and Persian-Tajik literature.

According to B.G. Gafurov, such closeness between the Tajik and Uzbek peoples began in the 6th century CE [Gafurov, 1952:110]. Even a form of bilingualism developed in these two languages, with the earliest forms of this bilingualism tracing back to the Achaemenid and Kangju states (7th-6th centuries BCE and 3rd-2nd centuries BCE, respectively) [Askarov, 2007: 34, 64, 164-183, 228, 262].

In our era, these social connections have been further perfected and solidified. Thus, the history of the Tajik and Uzbek peoples living together in harmony and their longstanding friendly relations, dating back to ancient times, continues to this day. These connections are also reflected in artistic creativity in both Turkic and Persian-Tajik literature. Through mutual understanding, multifaceted spiritual and educational cooperation has developed, leading to respect and attention between the peoples. Tolerance has always been a key virtue of our national character. During his reign, Zahiriddin Muhammad Babur treated all ethnic groups equally. In one of his decrees, he wrote:

"Let all nations, whether Turk, Tajik, Arab, Ajam, Hindu, or Persian, citizens and soldiers alike, who rely on the kindness of the sovereign, unite in gratitude for our enduring reign, observe the decrees without deviation, and act accordingly. This decree, bearing the great sovereign's signature, was issued in the year 932 AH (1525 CE) on the 24th of Jumada al-Awwal" [Z.M. Babur, 1990:360].

Therefore, there has always been a need for learning multiple languages in our land. It is known from history that our ancestors were multilingual. Great scholars such as Abu Rayhan al-Biruni, Abu Ali ibn Sina (Avicenna), Abu Nasr al-Farabi, Mahmud Kashgari, Alisher Navoi, and Zahiriddin Muhammad Babur mastered and created in multiple languages. For instance, Abu Nasr al-Farabi is reported to have known over 70 languages. In addition, bilingualism and the tradition

of creating poetry "like sugar and milk" have been an ancient custom in our land. The tradition of bilingualism has existed in all regions of the country, including Tashkent, Jizzakh, Kashkadarya, Surkhandarya, Navoi, Fergana, and Andijan. In Baburnama, Babur describes his native Andijan in this way:

"Andijan... its people are Turkic. In the city and market, there is no one who does not speak Turkic. Their speech is as proper as written language" [Z.M. Babur, 145].

The issue of bilingualism is also clearly evident in Baburnama. In this article, we propose studying the Persian-Tajik words in Baburnama by categorizing them as follows:

1. Numerical terms:

"There was a great quantity of gold and silver tools and equipment. The cloth bearers and cannon bearers were countless" [Z.M. Babur, 1990:41].

2. Characteristics and qualities:

"Boysungur Mirza's affection for Tahir Muhammad increased. Boysungur Mirza also liked Khisravshah. The grapes, melons, apples, and pomegranates, in fact, all the fruits, were excellent" [Z.M. Babur, 1990:43].

3. Descriptive circumstances:

"A Friday mosque was built inside the Iron Gate fortress. Stones were mostly brought from Hindustan, and stonemasons worked there... After raiding the Zardak desert, they brought back over one hundred thousand sheep and about three thousand camels... They were in a state of turmoil and confusion about what to do, whether to stay or leave, for several days" [Z.M. Babur, 1990:54].

4. Place names:

"To the south of Samarkand lies the Garden of Plane Trees, near the fortress. To the lower side of Samarkand are the Garden of Winds and the Garden of Paradise" [Z.M. Babur, 1990:44].

In Baburnama, several Persian-Tajik sentences and poems are also included:

*"And with the intention of battle and confrontation, they turned toward the victorious army camp. The warriors of Islam, who are like the trees of the garden of bravery, formed ranks as straight as cypresses. The tips of their helmets, resembling cypress tops, reached the heights of the sun. Their ranks were like the iron-clad 'Alexander's Wall,' their foundations as strong as the prophetic shariah. The steadfastness of their strength was 'as if they were a well-compacted structure,' and their success was in line with 'they are on guidance from their Lord, and they are the successful'" [Z.M. Babur, 1990:203].

Or: From that sentence, Mawlana Shihab derived the chronogram for this verse:

"Humayun was the heir to his kingdom..."

"Separation from you, I knew, would destroy me,

Otherwise, I could have left this city." [Z.M. Babur, 1990:259].

Or: "Kill today, since it is possible to kill,

When fire rises high, the world burns.

Do not let him pull the bowstring,

When an enemy can be pierced by an arrow." [5, 105].

In Baburnama, Persian-Tajik words, phrases, sentences, poems, ghazals, as well as rubaiyat, are used throughout:

"In this breach, there is no anxiety or fear,

As long as it aligns with the will of the king and firm religion.

His banners spread like the wide heavens,

Each letter like the 'Indeed, We have granted victory' verse." [Z.M. Babur, 1990:293].

In conclusion, it is a historical fact that, since ancient times, Uzbek writers have created works in Persian-Tajik, and Tajik writers in Turkic-Uzbek. They have maintained mutual connections and were never indifferent to each other's works. On the contrary, the tradition of writing in two languages has developed as a shared literary heritage. Interestingly, the customs, religion, living conditions, behavior, clothing, appearance, and culture of the two nations are remarkably similar. The comparative interpretation of the materials of these two languages from a linguistic perspective is also an ancient tradition.

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WASTEWATER TREATMENT BY CHEMICAL METHOD

Sh.J.Shakarov.¹, M.N.Nomirov.², I.Sh.Ergashev.³, J.T.Parmanov⁴

^{1,2,3,4}Samarkand State University of Architecture and Construction

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Abstract: During the oxidation process, toxic impurities in water are converted into non-toxic substances as a result of a chemical reaction, and they can be separated from the water. When wastewater is treated with oxidizing agents, a large amount of reagent is consumed. Therefore, this method is used when it is not possible to treat wastewater pollutants by other methods. Oxidation with ozone allows you to simultaneously decolorize water, eliminate various tastes and odors, and neutralize water. Wastewater can be cleaned of phenol, petroleum products, hydrogen sulfide, arsenic compounds, surfactants, cyanides, dyes, carcinogenic aromatic hydrocarbons, and pesticides by ozonation.

Keywords: Wastewater, waste, ozonation, cyanides, phenol, petroleum products, hydrogen sulfide, arsenic compounds, surfactants, dyes, carcinogenic aromatic hydrocarbons, pesticides, iron sulfate, sodium bisulfate, hydrogen, sulfur dioxide, pyrite ash.

Introduction

Nowadays, as the population of our planet increases, the need for clean water is increasing day by day. Due to the increase in waste and wastewater, the level of pollution of water bodies is increasing. As a result, water bodies are becoming unusable. Solving this issue positively is one of the most important problems facing countries around the world. If this problem is not solved in a timely and sensible manner, the consequences will worsen day by day.

Therefore, if scientific and technical achievements are not used to protect water bodies, if the relevant regulatory bodies do not strengthen their work in this regard, and if industrial and urban wastewater treatment facilities do not work, the health of the population and the need for clean water will always be at risk.

Water plays an important role in the main processes occurring in nature, as well as in human life. In industry, water is used as a source of raw materials and energy, a coolant or heater, a solvent, an extractant, a means of transporting raw materials and materials, and for a number of other needs.

In this regard, paying great attention to the protection of the nature and water bodies of the Republic of Uzbekistan, the Supreme Council of the Republic approved the Law on State Sanitary Supervision on July 3, 1992. This law attaches great importance to the protection of the environment, water bodies, and providing the population with clean water.

Therefore, filters, coagulants, various reagents, disinfectants (chlorine, calcium chloride, ozone), and physical methods are used to improve water quality. Maintaining the taste of water depends on the choice of water sources. The smell and unpleasant taste of water are eliminated by its treatment.

The degree of harmfulness of wastewater depends on the nature and composition of pollutants (toxicity) in it. Salts of heavy metals, cyanides, phenols, hydrogen sulfide, carcinogens and other similar substances lead to high levels of toxicity and odor change of wastewater.

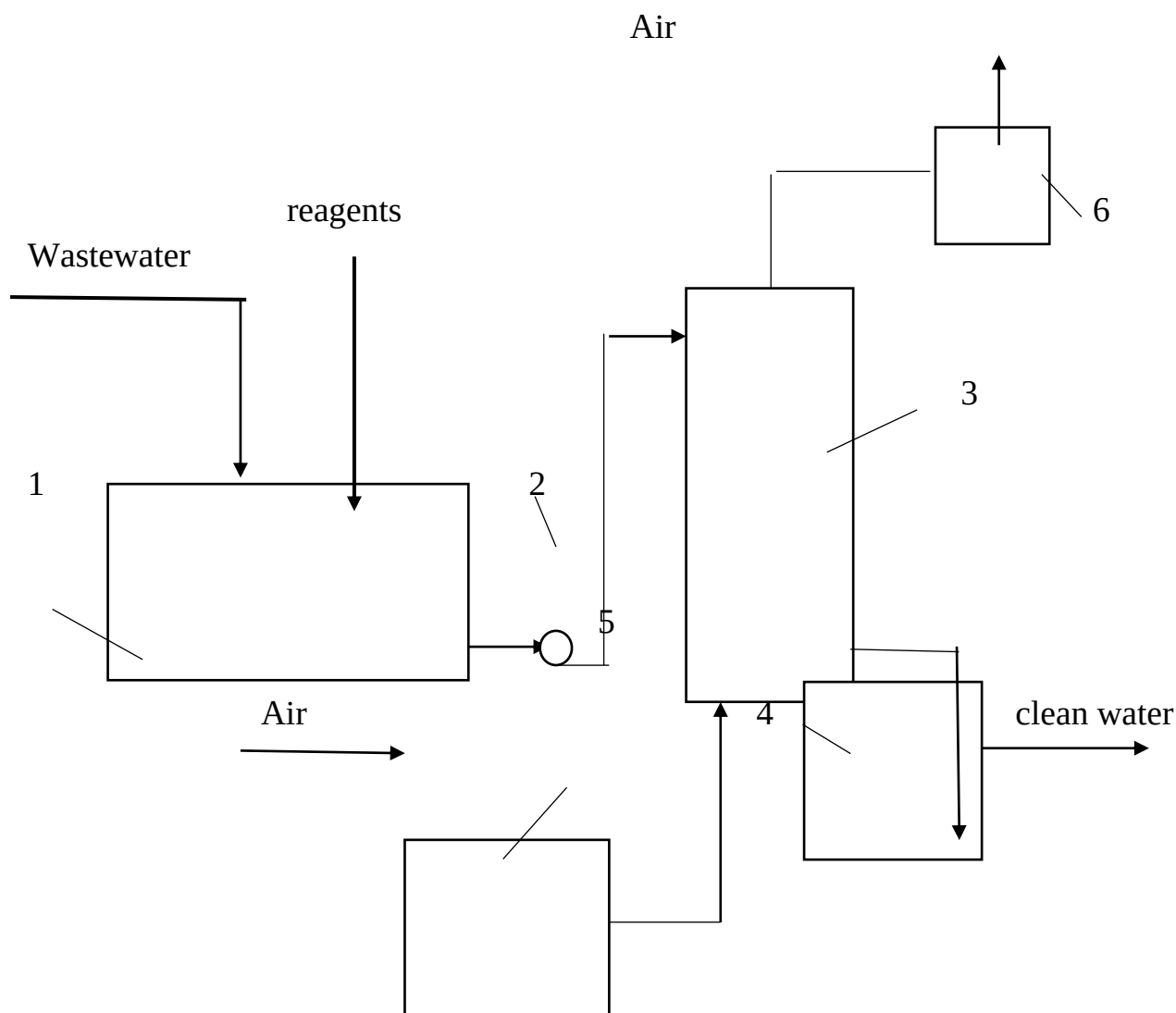
Therefore, ozonation of wastewater (passing O₃ gas) is of great importance. In this case, by passing ozone gas through water, wastewater can be clarified, its taste improved, odor removed and neutralized. During the ozonation process, wastewater can be cleaned of phenols, petroleum

products, H₂S, arsenic compounds, PAV (surfactants), cyanides, dyes, carcinogenic aromatic hydrocarbons, pesticides.

Ozone is an allotropic modification of oxygen, a light purple, highly toxic gas. Its specific odor has a concentration of $1 \cdot 10^6$. It is located in the upper stratosphere of the atmosphere, molecular weight — 48, density — 2.154 g/l. Pure ozone is explosive (in industry it is obtained from oxygen - 0.2), a highly toxic compound. The maximum permissible concentration in the air of the working zone is 0.0001 mg/m³. The neutralizing effect of ozone gas is due to its high oxidizing properties, that is, it has the ability to easily give atomic active oxygen ($O_3 = O_2 + O$). Ozone oxidizes all other metals except gold, converting them to oxide form. It decomposes very slowly in dry air. When wastewater is treated with ozone gas, organic compounds in it decompose, and bacteria die (1000 times faster than chlorine).

The solubility of ozone in water depends on the pH of the solution and the substances dissolved in it. If there are acids and neutral salts dissolved in water, ozone dissolves well. If there are alkalis in water, ozone dissolves more difficultly.

The effect of ozone can proceed in three different directions: through the action of atomic oxygen in ozone, through the complete binding of ozone to the oxidizable substance, and through the catalytic action of atomic oxygen in ozone. When ozone is passed through wastewater (dispersion), 2 main processes occur: oxidation and disinfection. At the same time, the water itself is saturated with oxygen. Ozone is supplied to wastewater in the form of an ozone-air or ozone-oxygen mixture. The ozone content in the mixture is about 3%. Now let's look at the technological scheme (Figure 1).



1 - Fig. Wastewater ozonation disinfection device:

1-mixer; 2-pump; 3-reactor; 4-collector; 5-ozonator device; 6-gas purification unit.

In this scheme, after mixing the wastewater with reagents in the 1-mixer, it is fed to the 3-reactor using the 2-pump.

Here, under the influence of ozone gas prepared in the 5-ozonator, the substances contained in the wastewater are oxidized and become harmless. The purified water is sent to the next stage through the 4-collector. The gases formed during the purification stage are neutralized in the 6-block and released into the atmosphere. In this case, the residual amount of ozone used in the purification process must also be retained.

Because ozone gas is actually a strong poisonous gas. Therefore, this gas leaving the reactor is passed through activated carbon (Fig. -2). 450 g of activated carbon is used to produce 1 kg of ozone.

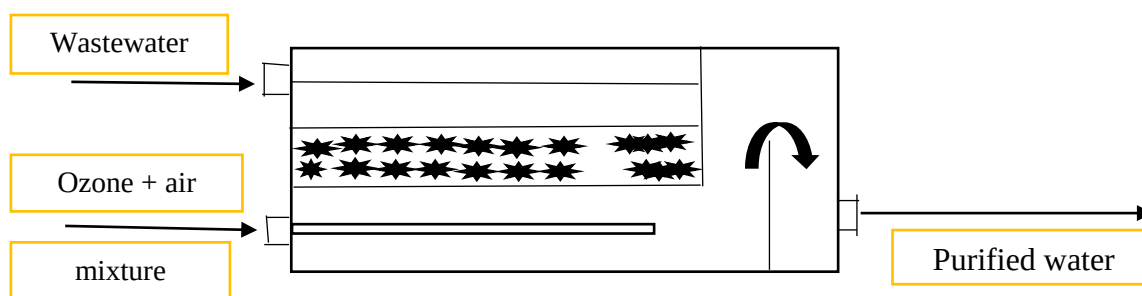
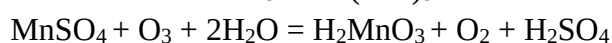
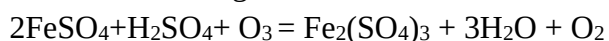


Figure 2. Nozzle ozonation device.

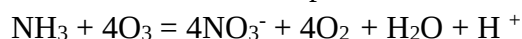
Ozone oxidizes both inorganic and organic compounds in wastewater. For example, with iron and manganese compounds, the following reactions occur:



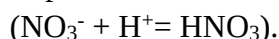
The oxidation of hydrogen sulfide proceeds as follows:



The oxidation of ammonia proceeds as follows:

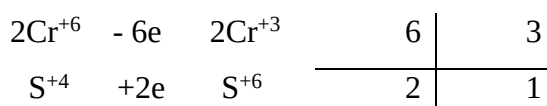
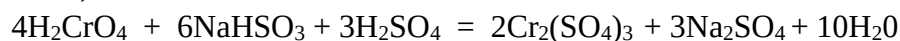


This produces nitric acid:

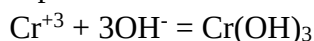


Reduction method: Wastewater treatment by the reduction method is used when there are rapidly reducible substances in the water. This method is widely used to separate mercury, chromium, arsenic from wastewater, and they are separated from water using settling, filtration or flotation methods.

If organic compounds of mercury are found in water, the organic substance is first decomposed, then mercury cations are reduced to the metallic state. To purify wastewater from hexavalent chromium, it is first converted to trivalent based on the reduction process, and then chromium hydroxide is precipitated in an alkaline medium. Activated carbon, iron sulfate, sodium bisulfate, hydrogen, sulfur dioxide, pyrite ash, etc. are used as reducing agents. In practice, sodium hydrosulfate (bisulfite) is most often used.



This reaction occurs rapidly, and the environment is around pH=3-4. Alkaline reagents are used to precipitate trivalent chromium.



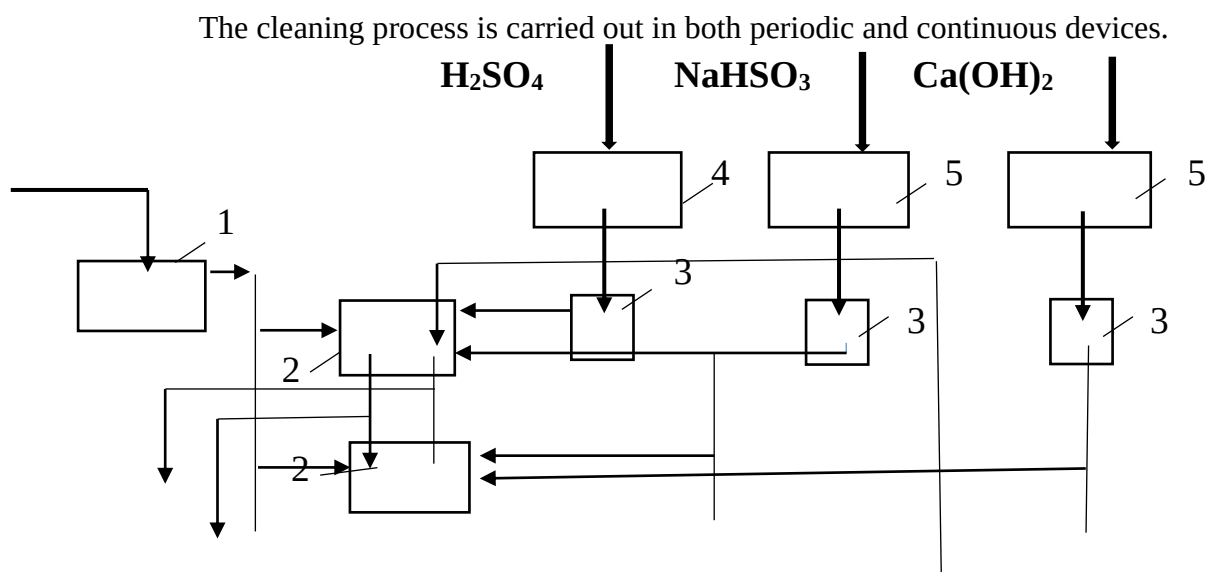
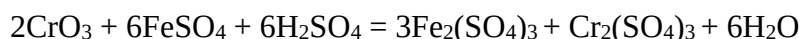


Figure 3. Scheme of the chromium recovery device:

- a) 1-tank; 2-reactors; 3-meter; 4-6-tanks; b) 1-concentrator; 2-mixer; 3-tanks for neutralization and sedimentation.

In a batch plant (a), wastewater is pumped from tank 1 to reactor 2 using a pump. If the wastewater pH is greater than 3, sulfuric acid is added from tank 4 until the medium pH is 2.5-3. After that, sodium bisulfite is added and stirred vigorously for 30 minutes. Then alkali and polyacrylamide are added and precipitation is carried out. In continuous plants, wastewater is first fed to tank 1, then to the mixer and neutralizer. The bisulfite solution is fed to the mixer after the medium pH decreases to 2.5-3. At the end of the mixer, calcium hydroxide (lime water) is added to increase the pH of the medium. The cleaning process lasts 30 minutes. A precipitate forms in the neutralizer. Polyacrylamide is added to the solution to accelerate precipitation. Good results can be achieved when using iron sulfate as a reducing agent. The process can be carried out in both alkaline and acidic environments.



The consumption of FeSO_4 depends on the pH of the medium and the concentration of chromium. The favorable conditions of the process are as follows: $T=20^\circ\text{C}$, $\text{pH} = 7$, the consumption of FeSO_4 is 1.3 times higher than the target.

Conclusion. 1. Oxidation with the participation of one oxygen atom.

2. The combination of an ozone molecule with the substance being oxidized to form ozonides.

3. Catalytic acceleration of the oxidizing effect of oxygen in ozonized air.

4. The mechanism of the ozone decomposition reaction is very complex, since the rate of destruction is influenced by many factors. These factors include the conditions of the transition of ozone from the gas phase to the liquid phase, the ratio between the partial pressure of the gas and its solubility in an aqueous solution, and the kinetics of ozone oxidation of impurities in water.

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THE IMPORTANCE OF STUDYING PHYSICS AND THE IMPORTANCE OF PHYSICS IN THE MODERN WORLD

J.T.Parmanov¹, I.SH Ergashev²

^{1,2}Samarkand State University of Architecture and Construction named after Mirzo Ulugbek

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Abstract .We offer physics courses that meet a variety of goals students may have in studying physics, take elective courses to improve scientific literacy, meet requirements for a science or engineering major, or pursue a degree in physics or engineering physics. get a degree. Physics courses reveal the mathematical beauty of the universe on a subatomic to cosmological scale. physics usually focuses on the general nature of the natural world through mathematical analysis. Society's dependence on technology represents the importance of physics in everyday life.

Keywords: MRI, radiotherapy, medicine, Magnetism, electricity, conductors, television, computers, telephones, home technology, aircraft and telecommunications, matter, energy, Electromagnetic radiation, magnetic resonance imaging (MRI) scanners, X-rays, cancer treatments for radiation therapy, ventilators for patients whose lungs cannot breathe.

Introduction

Electricity- One of the greatest discoveries that mankind has made is electricity, by understanding the physics properly we have been able to turn it into something useful for electricity, which is just a giant pile of electronics. By creating a voltage difference in something as simple as a battery, we can move the electrons that are the basis of electricity, the moving electrons that make the circuits that allow the operation of radios, televisions, lights and electronics, any other electronic device. Travel and transport - Due to the main role of transport in the modern era, one of the most successful fields of application of the statistical physics of complex networks is the study of traffic dynamics, but most of the works consider transport networks as isolated systems, which is contradictory. many complex networks with nontrivial interdependence. To simulate a real scenario, we use a multilayer network system to create a two-layer traffic model, where the upper layer provides a higher traffic rate than the lower layer. Medicine-Physics has made very important contributions to health since the advent of medicine 5,000 years ago, and these contributions are many and varied, including diagnostic X-rays, nuclear medicine, clinical PET scanning, magnetic resonance imaging, magnetoencephalography has been and remains so. , MRI, radiotherapy treatment, minimally invasive surgery, interventional magnetic resonance imaging, photonics, scaling theory, and high-intensity focused ultrasound with a combination of diagnosis and treatment. These contributions will increase as the molecular mechanisms of disease are better understood and new technologies allow the in vivo investigation of these molecular processes, with medical physicists, for example, playing an important role in ensuring patient safety, safe and effective implementation. new health technologies based on physics. Technology-Physics has helped us expand medical technology, for example we have examples from physics: magnetic resonance imaging (MRI) scanners, X-rays, radiation therapy for cancer treatments, and ventilators for patients with lung failure. For example, GPS satellites. To account for the relativistic effects of the motion of their companions, they agree on their internal time, Einstein's special theory of relativity describes the period dilation experienced by rapidly vibrating objects. The importance of physics in everyday life. Physics is heavily involved in the application of physics in everyday life and in a number of activities we do

and things we use in our daily lives. Physics is a natural science because it deals with things like matter, force, energy and motion. Since all of these are related to tasks related to everyday life, we can say that physics explains how the universe works, how the Earth moves around the sun, how lightning ignites, how our refrigerator works, and many other things. will try Physics - When we study and discuss physics, we focus a lot on energy, which is a fundamental element of science, to better understand this connection, it is useful to refer to a strong working definition of physics. Physics is an important mechanism for educating chemists, engineers and computer scientists, and for professionals in various physical and biomedical sciences, in the same way that it enhances and refines our knowledge of other methods, such as earth, agricultural sciences. , chemical, biological, and ecological, as well as astrophysics and cosmology, important categories of evidence for all parts of the world. Physics helps to improve the quality of life by providing the basic concepts needed to develop new methods and techniques for use in medicine, such as: CT scan, Magnetic resonance imaging, Positron emission tomography, Ultrasound imaging, Laser surgery.Importance of Studying Physics-Physics is second only to mathematics in the purity of its principles, physics describes how the natural world works through practical mathematical formulas, it deals with the fundamental forces of the universe and they range from galaxies and planets to atoms and depending on how it interacts with matter. quarks and everything in between. All other natural sciences derive from physics, chemistry is mainly applied physics and biology is applied chemistry, the theory of physics is responsible for advances in electronics that lead to advances in modern computers and electronic media. Everything around us uses energy in one way or another, our cars, planes and trains burn fuel, our computers require power supplies and Wi-Fi connections, and our telecommunications devices are powered by cellular signals. more needs have arisen and this may be alternative energy sources such as nuclear energy or fossil fuels or hybrid energy.Why Studying Physics is Important-The purpose of physics is to understand how things work from first principles, we offer physics courses that match the different goals students may have in studying physics, we take elective courses to improve scientific literacy, meet requirements for a science or engineering major or have a degree in physics or engineering physics. Physics courses reveal the mathematical beauty of the universe from the subatomic to the cosmological scale, and the study of physics reinforces quantitative thinking and problem-solving skills that are valuable in fields beyond physics. Students studying the fields of Physics or Engineering Physics are able to consider developments in science and technology, academia, government or the private sector. Careers can focus on basic research in astrophysics, cosmology, particle physics, atomic physics, photonics or charged matter physics, or more rigorous research in areas such as renewable energy, quantum information science, materials engineering, biophysics or medical physics. Majors may also include management in teaching, medicine, law (especially intellectual property or patent law), scientific writing, history of science, philosophy of science, science policy, energy policy, government, or technical fields. Physics and engineering majors are excellent preparation for almost any career because they teach students to analyze complex problems and give them a strong quantitative foundation that can be applied in any technical field.Importance of Physics in Modern World-Physics covers all aspects of our life, it includes the study of matter, energy and their interactions, as it is a branch of science that includes all other sciences and other sciences depend on concepts and methods developed through physics. Other disciplines such as chemistry, agriculture, environmental and biological sciences use the laws of physics to better understand the nature of their research, physics usually focuses on the

general nature of the natural world through mathematical analysis. Society's dependence on technology represents the importance of physics in everyday life. . Many aspects of modern society would not be possible without the important scientific discoveries made in the past, and these discoveries were the foundation upon which today's technologies were developed. Discoveries such as magnetism, electricity, conductors, and more have created modern conveniences such as television, computers, telephones, and other business and home technologies, as well as modern vehicles such as airplanes and telecommunications. all based on physics concepts.

Electricity- One of the greatest discoveries that mankind has made is electricity, by understanding the physics properly we have been able to turn it into something useful for electricity, which is just a giant pile of electronics. By creating a voltage difference in something as simple as a battery, we can move the electrons that are the basis of electricity, the moving electrons that make the circuits that allow the operation of radios, televisions, lights, electronics, and any other electronic device.

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Technology-Physics has helped us expand medical technology, for example we have examples from physics: magnetic resonance imaging (MRI) scanners, X-rays, radiation therapy for cancer treatments, and ventilators for patients with lung failure. For example, GPS satellites. To account for the relativistic effects of the motion of their companions, they agree on their internal time, Einstein's special theory of relativity describes the period dilation experienced by rapidly vibrating objects. The importance of physics in everyday life. Physics is heavily involved in the application of physics in everyday life and in a number of activities we do and things we use in our daily lives. Physics is a natural science because it deals with things like matter, force, energy and motion. Since all of these are related to tasks related to everyday life, we can say that physics explains how the universe works, how the Earth moves around the sun, how lightning ignites, how our refrigerator works, and many other things. rganadi. When we walk, we actually move like an inverted pendulum, when we put our foot, it becomes our axis, and our mass is located in the abdominal cavity, describing the shape of an arc.

Fresh Egg-One way to know if an egg is rotten or good to eat is using the float test, to test whether the egg is good or not, the egg is placed in a glass of water, the fresh egg remains at the bottom. .glass, an egg that sinks but stays in the middle of the glass may be a little bigger but still cooked and eaten, because if the egg floats it's bad, all this is achieved thanks to the importance

of Physics. Climate-Climate is related to the absorption of solar radiation and the subsequent redistribution of this energy through radiative, advective, and hydrological processes. Soap is lighter than water, so when you make a bubble, the water is attracted to the bottom of the bubble and pushes the soap up, due to gravity, the surface tension of the water is greater than that of the soap. the top of the bubble shrinks as more soap accumulates at the top, and soon the internal pressure is sufficient to break the bubble and cause it to burst.

Used as Wifi-Radio waves Electromagnetic radiation, these waves transmit data by vibrating at different frequencies, these waves are very important for Wi-Fi to work well, electromagnetic waves are the same as mechanical waves have characteristic components, such as wavelength, amplitude, frequency, but do they behave very differently.

How physicists think about physics-Philosophy of physics begins with the findings of particular theories in physics and then deals with the consequences and puzzles that arise, such as the nature of determinism or the relative or fixed nature of space or time. people are better suited as philosophers. Einstein was first and foremost a philosopher, and his colleagues deserve some credit for that, and relativity was a metaphysician until he provided ways to incorporate it into physics.

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LINGUISTIC STATUS OF PAREMIOLOGICAL UNITS

Ibragimova Lola Asatullo kizi

Samarkand State Institute of Foreign Languages

Department of Second Foreign Languages

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Abstract: *This article explores the linguistic status of paremiological units, particularly proverbs, within various linguistic and cultural contexts. It examines the role of proverbs as a universal phenomenon in folklore across different languages, emphasizing their cultural and communicative significance. The study addresses the semantic and structural characteristics of proverbs, their stability in syntactic form, and their complex interplay with related linguistic phenomena such as sayings, aphorisms, and proverbial phraseologisms. The article highlights various theoretical perspectives, including A. Nosirov's categorization of proverbs within the phraseological inventory and the influence of communication contexts on their pragmatic meanings. Additionally, the article explores the broader implications of proverbs in reflecting national worldviews and their linguistic functions, such as their ability to convey both literal and figurative meanings. The article further discusses the complexities of translating proverbs across languages and their retention of meaning and structure in different cultural settings. Finally, the work draws on the ideas of notable linguists like A.I. Kunin, Georges Milner, and Sh. Balli to present proverbs as fixed expressions with a symmetrical relationship between form and meaning, underscoring their importance in the structure of language.*

Keywords: *Paremiological units, proverbs, phraseologisms, cultural significance, structural characteristics, pragmatic meaning, communication context, interlingual translation.*

Introduction

The fact that proverbs, as examples of folk literature, are a common phenomenon across all linguistic cultures can be demonstrated by their widespread study in various countries. Proverbs in different languages are collected and analyzed systematically in terms of their genesis, semantic and structural characteristics, and their role in speech activities. However, the linguistic status of proverbs and their relationship with related phenomena are viewed differently. The primary cultural source of language units is considered to be the realities that encompass the surrounding world. These include myths, legends, comparative imagery, folklore, customs, religion, literature, and historical and political events [Ashurova D.U., "Vneshinvestprom," Tashkent – 2021.-38, 48].

Most researchers place proverbs within the folklore system, acknowledging their distinct status, while some interpret proverbs and sayings as units of the linguistic phraseological inventory. For example, in A. Nosirov's doctoral dissertation, proverbs, along with sayings, wise words, aphorisms, and popular expressions, are all categorized as proverbial phraseologisms (Nosirov 2016). Indeed, proverbs, like other fixed phraseological units, exhibit a characteristic of stability in their structure, meaning that altering their syntactic pattern presents challenges.

It is well-known that the famous phraseologist A.I. Kunin defined fixed forms as expressions that have specific characteristics and are used in speech in a ready-made form, making it difficult to imagine them fulfilling any communicative-pragmatic function outside of the discourse context. The meaning of a proverb, however, is shaped by the speaker's communicative goals and the specific context, thus giving rise to pragmatic content and allowing for the expression of ideas in an indirect, expressive manner.

Proverbs, sayings, and aphorisms, as syntactic structures, are recognized by many scholars as having textual characteristics. Most of these researchers propose calling such units "communicative phrasemes" (Granger, Paquot 2008:42). In these structures, the speaker conveys broader ideas through a specific phraseme. By using such phrasemes, the speaker aims to persuade the listener of the validity of the message.

Moreover, the use of the term "multiword expressions" for these structures also indicates their status as complex speech units (Milder 2010). Proverbs are linguistic constructions that reflect the linguistic worldview and carry national cultural significance, with their worldview presented in an image-based form.

These structures are syntactically complete and semantically meaningful, reflecting the lifestyle of the community. According to A. Nosirov, the Western term "proverb" is derived from the Latin word "proverbium," which expresses concise ideas based on the life experience and surprising conclusions of the people. The scholar also notes that the Uzbek term "maqol" (proverb) originates from the Arabic word "qavlun" (to speak), referring to expressions of folk wisdom, and is used as a term for one of the aphoristic genres, meaning "a well-said word, phrase, or expression" (Nosirov 2016:15).

It is known that in phraseological units, meaning shifts occur, and the individual words may lose their original literal meanings or become entirely figurative. In contrast, words in proverbs often retain their core meaning, or both the literal and figurative meanings can coexist.

The figurative meaning in a proverb is related to its full context and defines its value, allowing communication participants to understand its meaning. Proverbs are often used to explain current situations by referencing past events. At the same time, the original meaning of a proverb is so clear that communication participants can easily understand and translate it. This can be observed in the interlingual equivalents and adoption of proverbs across languages.

Foreign linguists emphasize that proverbs are fixed combinations that carry meaning and retain their structure and form. They stress that the meaning and form of proverbs should not be overlooked. This is evident in the definition provided by French scholar Georges Milner in his article "The Structure of Proverbial Fixed Expressions," which states: "The distinguishing characteristic of proverbial expressions is that they have a symmetric link between meaning and form. That is, the meaning of a proverbial expression performs a specific function, and the symmetry of the form reflects the symmetry of the meaning" (Milne 1964:54, cited in Nasriddinova 2019:16).

The phraseologist Sh. Balli, who described the structural and semantic peculiarities of phraseological units, noted the equivalence of such units to words. However, the syntactic structure of a proverb does not fully align with the criteria established by French linguists. The structural-semantic peculiarity of a proverb lies in its ability to express a complete thought, which complicates its inclusion in the phraseological inventory of language.

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AGAINST RELIGIOUS BELIEFS IN STUDENTS STRENGTHENING OF IDEICAL IMMUNITY

Rahimova I.R.

Teacher of the Department of Foreign Languages,
Samarkand State University of Architecture and Construction

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Abstract. *Ideological thinking of students in socially useful labor activities can develop. One of the effective ways to increase students' ideological immunity is "Young ideologues" club. A critical attitude towards foreign ideologies various pedagogy by experienced professors and teachers for development methods are used.*

Keywords: *young ideologues, ideology, immunity, criticism, work.*

Introduction

Strengthening ideological immunity in young students, making them global, cultural and plays an important role in making it stable in relation to ideological features. Students ideological thinking can be developed in socially useful labor activities. For this reason, socially useful work, regardless of intelligent, organizational or physical work, of course, to society must be necessary.

One of the effective ways to increase students' ideological immunity is "Young ideologues" club. In this club, objective assessment of political events, lack of ideas and It aims to teach people how to deal with aloofness from politics. To foreign ideologies experienced professors and teachers to develop a critical attitude different pedagogical methods are used by For example, in the "skip forward" method The leaders of the club discuss current issues of the country and abroad discuss their events. This method allows students to freely express their thoughts and their own helps to form views, especially immunity against religious beliefs is of great importance in strengthening.

The activities of the "Yosh Mafkurachilar" club must be free, as otherwise, students may superficially understand ideological concepts. When setting the goals of the club, it is important to choose socially significant and relevant tasks that should align with the knowledge and experiences of the students. Ideological education for students should be directed towards developing a sense of responsibility towards the people, the homeland, the nation, their family, and their higher educational institution. For example, meetings on the topic "The deeds of our ancestors will always remain in our hearts" can be organized with the participation of students and their parents. During the meeting, students learn about national history and family values by creating their family tree. Connecting family history to national history helps students better understand the history of both the family and the country, and assists them in realizing the importance of national freedom.

Repetition plays a significant role in enhancing the clarity, emphasis and flow of conjunctive structures in English. Below are the key types of repetition employed with conjunctions to improve the expression of ideas:

Students develop their sense of filial duty and positive work assessment through family historical monuments, speeches at parent meetings, and excursions to enterprises. This fosters a sense of national duty and personal responsibility. Improving socially useful labor and ideological-educational processes prepares students for active social engagement in the interests of society. The harmony between ideological and labor education helps students develop

constructive traits. In the modern era, combating religious-extremist movements, such as organizations like "Hizb ut-Tahrir," is of significant importance for students. In this process, it is necessary to rapidly and deeply explain political situations relevant to the time in order to strengthen students' immunity against foreign ideologies. This helps them respond effectively to the threats of various ideologies and enhances their political awareness.

It is important to carry out active propaganda and ideological defense work at every higher educational institution. Necessary materials for both propaganda and counter-propaganda should be prepared and continuously updated. Experienced teachers and leading course students should actively participate in this process as part of propaganda teams. Through the curricula of humanitarian and natural sciences, students must gain a broad and reliable understanding of Uzbekistan's socio-economic situation, national policies, and the advantages of democracy.

It is important to teach students to have a negative attitude towards certain ideas and actions in the process of developing their ideological immunity. In this process, organizations of the "Youth Affairs Agency" should take the lead. Prevention should be the main focus, and measures should be taken by the leadership to prevent the influence of foreign ideologies.

Teachers face a number of problems, including:

- ☐ Explain the essence of ideological immunity to students;
- ☐ Encourage the use of various scientific manuals to gather information on this topic;
- ☐ Study the issues of counter-propaganda and develop skills for clear speech;
- ☐ Teach methods of ideological debate and struggle.

The development of historical and scientific concepts plays a crucial role in pedagogical research, particularly in the methodology of teaching social sciences. According to experts, international historical concepts are considered among the most difficult terms to master. If these concepts are not introduced to students in Uzbek in a timely and accurate manner, they may fail to fully understand the true meaning of the historical terms, leading to superficial learning. This can result in memory fatigue and these concepts may be quickly forgotten. Therefore, presenting historical and scientific concepts in a clear and understandable way is essential to enhance the effectiveness of students' learning.

Studying works of art undoubtedly plays a crucial role in shaping ideological thoughts. Art, especially theater, supports the purification of the spirit and the fostering of spiritual growth. In the late 19th and early 20th centuries, intellectuals chose to express their ideas through education and theater. According to art scholar U. Saidov, theater has a unique significance in this process, and works of art play an important role in purifying the heart. Therefore, it is essential to deeply study works of art, promote ideological thoughts through them, and achieve educational goals.

Theses play a significant role in making presentations more reliable and visual. They should primarily be provided with short and clear questions to attract the audience's attention. This approach helps increase the effectiveness of the presentation.

In today's world, strengthening students' ideological immunity plays a crucial role in stabilizing society and preparing future leaders, specialists, and socially active citizens. It is essential for students to gain not only academic knowledge but also a solid ideological foundation to understand their duties to society and the state. Therefore, various pedagogical approaches and methods should be used to foster this ideological immunity. **1. The Importance of Education in Shaping Ideological Immunity**

The foundation of ideological immunity lies in education. Students need to acquire not only academic knowledge but also the ability to analyze social processes, political situations, and economic circumstances. This broadens their worldview and equips them to withstand various ideological influences.

In the educational process, it is crucial to introduce students to national values, historical heritage, and the cultural and moral foundations of their people. This helps students understand and value concepts such as patriotism, devotion, and the defense of the national interest, which are vital for their ideological immunity.

2. The Role of the "Young Ideologists" Club and Expanding its Influence

The "Young Ideologists" club is an effective platform for strengthening students' ideological immunity. In this club, students can express their opinions freely, analyze current political events, and discuss critical issues. Teaching students to critically assess political events and engage in ideological debates strengthens their immunity against foreign ideologies and fosters independent thinking.

The club's activities must allow students to critically analyze social and political issues, offer constructive criticism, and engage in ideological struggle. This environment encourages students to explore different viewpoints and form their own opinions, which is essential for developing their ability to withstand foreign ideological influences.

3. Understanding Historical and Cultural Heritage

Fostering ideological immunity in students also requires deepening their understanding of their nation's historical and cultural heritage. Students who have a strong understanding of their national history are better equipped to withstand foreign ideologies and understand their role in the world. Engaging students in activities like exploring their family history or creating family trees helps them connect personal identity with national heritage and understand the importance of national freedom.

Teaching students the significance of independence and their nation's historical journey enables them to appreciate the value of freedom and take responsibility for preserving their nation's sovereignty.

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STRUCTURAL AND SEMANTIC ANALYSIS OF PAREMIOLOGICAL UNITS IN ENGLISH AND UZBEK LANGUAGES

Ruzikulov Elyor Shukurovich

Teacher of Samarkand State Architectural and Civil Engineering University

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Abstract. *This study focuses on the structural and semantic analysis of paremiological units (PUs) in the English and Uzbek languages. Paremiological units, which include proverbs and sayings, serve as linguistic artifacts that encapsulate the cultural, moral, and social values of a community. The research examines the syntactic structure, stylistic devices, and grammatical patterns of PUs in both languages, highlighting similarities and differences. It also delves into their semantic components, exploring literal and figurative meanings, cultural specificity, and universality. This comparative approach aims to identify how PUs reflect the worldview and cultural identity of English and Uzbek-speaking communities, contributing to a deeper understanding of cross-linguistic and cross-cultural dynamics.*

Keywords: *Paremiological units, Proverbs, Structural analysis, Semantic analysis, English language, Uzbek language, Cross-cultural linguistics, Figurative meaning, Cultural connotations, Syntactic patterns, Stylistic devices, Comparative linguistics, Universal themes, Culture-specific proverbs, Idiomatic expressions.*

Introduction

Paremiology, a specialized branch of linguistics, explores proverbs and sayings as a reflection of a nation's cultural and spiritual heritage. Every language possesses a unique set of paremiological units (PUs), representing its speakers' worldview, traditions, and values. English and Uzbek, as two linguistically and culturally distinct languages, provide a fascinating domain for studying the structural and semantic peculiarities of PUs. This paper aims to analyze the structural and semantic features of PUs in both languages, shedding light on their similarities and differences.

Paremiology, a distinct subfield of linguistics, examines proverbs and sayings as expressions of collective wisdom, cultural identity, and societal values. Proverbs, being short, metaphorical statements, encapsulate life experiences and moral principles of different communities. English and Uzbek, two languages from entirely different linguistic families, offer a rich field of study due to their unique cultural, historical, and social contexts. This paper explores the structural and semantic characteristics of paremiological units (PUs) in English and Uzbek, emphasizing their similarities and differences while highlighting the cultural underpinnings embedded within them.

In English, Structural Features of Paremiological Units are often simple, concise, and follow fixed syntactic formulas. Their brevity makes them easily memorable and widely applicable. For instance:

Practice makes perfect. Time is money. These proverbs typically adhere to straightforward grammatical constructions, often consisting of a subject-verb-object pattern or parallel clauses.

In Uzbek, PUs tend to exhibit greater structural complexity. They often consist of multiple clauses, joined by conjunctions, and include repetitions or rhymes for rhythmic and mnemonic purposes.

For example: *Chiroyni so‘z emas, ish ko‘rsatadi.* (Beauty is shown by deeds, not words.)
Yaxshi ot yomon aravani tortar, yaxshi odam yomon elni boqar. (A good horse pulls a bad cart, a good person leads a flawed community.)

The complexity reflects the oral traditions of Uzbek culture, where proverbs were designed to be recited and passed down through generations.

Both languages utilize stylistic devices to enhance the aesthetic appeal of PUs.

In English, stylistic devices such as parallelism (*Easy come, easy go*), alliteration (*Haste makes waste*), and metaphor (*Don't cry over spilled milk*) are prevalent.

In Uzbek, rhythmic patterns, rhyme, and alliteration are more prominent.

For instance: *Eshakning o‘limi – yo‘rg‘ada, odamning o‘limi – urishda.* (The donkey's death is in trotting; a man's death is in battle.)

These devices make Uzbek PUs more musical and reflective of oral poetic traditions.

Semantic analysis involves studying the meanings, cultural connotations, and underlying values of PUs.

English PUs often encapsulate universal truths, making them applicable across cultures.

For example: *A friend in need is a friend indeed.* – Highlights the value of true friendship.
Where there's a will, there's a way. – Stresses determination and persistence.

These proverbs focus on general life principles, transcending cultural and geographical boundaries.

Uzbek PUs, **while** containing universal themes, are deeply rooted in specific cultural and historical contexts.

For instance: *Dehqon yerda – el yerda.* (If the farmer prospers, the nation prospers.)
Qo‘shning tinch – sen tinch. (If your neighbor is peaceful, you are peaceful.)

These proverbs reflect the agricultural and communal lifestyle of Uzbek society, where harmony and cooperation are highly valued.

English PUs tend to maintain a fixed structure and are less prone to modification.

Uzbek PUs, being part of a rich oral tradition, are often flexible and adaptable in speech.

While English proverbs are widely recognized due to their universal appeal and global reach, Uzbek proverbs offer insights into Central Asian culture, highlighting its distinct moral and philosophical outlook.

The structural analysis of PUs involves examining their grammatical forms, syntactic patterns, and compositional elements.

The structural aspect of PUs was analyzed by:

Syntactic Analysis: Identifying common sentence structures (e.g., simple, compound, or complex sentences) in PUs from both languages.

Grammatical Features: Examining tense, mood, and word order patterns specific to English and Uzbek PUs.

Stylistic Devices: Highlighting the use of parallelism, rhyme, rhythm, alliteration, and other stylistic elements that enhance memorability and aesthetic appeal.

The semantic component of PUs was investigated through:

Literal vs. Figurative Meaning: Comparing the literal and implied meanings of PUs in both languages.
Cultural Connotations: Exploring culture-specific themes and values embedded in the PUs, such as family, agriculture, and communal harmony in Uzbek PUs, versus individualism and pragmatism in English PUs.

Universal Themes: Identifying proverbs that transcend cultural boundaries and reflect shared human experiences.

A comparative framework was employed to:

Highlight structural similarities and differences between English and Uzbek PUs.

Explore the cultural and historical factors influencing the thematic focus of PUs in each language.

Assess the role of PUs in fostering cross-cultural understanding.

The findings were validated by consulting linguistic and cultural experts in English and Uzbek, ensuring accuracy in interpretation and contextual understanding.

The study analyzed how PUs in English and Uzbek serve as tools for cultural expression and linguistic creativity, emphasizing their importance in language learning and intercultural communication.

By integrating structural and semantic analyses, this study offers a comprehensive perspective on the linguistic and cultural significance of paremiological units in English and Uzbek languages.

The paremiological units in English and Uzbek serve as linguistic windows into their respective cultures, illustrating distinct worldviews and societal values. While English proverbs tend to focus on universality and individualism, Uzbek proverbs emphasize collectivism, tradition, and morality. The structural conciseness of English PUs contrasts with the rhythmical richness of Uzbek PUs, reflecting differences in literary and oral traditions. A comparative study of these proverbs enriches cross-cultural understanding and highlights the intricate interplay between language, culture, and thought. The paremiological units in English and Uzbek languages serve as cultural and linguistic mirrors, reflecting the ethos and worldview of their respective societies. While English proverbs prioritize brevity and universality, Uzbek proverbs are imbued with cultural specificity and poetic richness. Understanding the structural and semantic features of these PUs not only enriches linguistic knowledge but also fosters cross-cultural appreciation. A deeper exploration of these proverbs reveals the intricate interplay between language, culture, and thought, underscoring the universality of human wisdom and the diversity of its expression.

Paremiological units, as concise and metaphorical expressions of human wisdom, play a vital role in language and culture. Their structural simplicity contrasts with their semantic depth, offering insights into both universal human experiences and culture-specific values. By analyzing PUs, linguists and cultural scholars gain a better understanding of how language shapes thought and reflects societal norms.

Understanding PUs not only enriches linguistic knowledge but also promotes cultural appreciation, underscoring the shared wisdom of humanity and the diversity of its expression.

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DESIGN SOLUTION FOR THE ORGANIZATION OF INTELLECTUAL CENTERS FOR PEOPLE WITH DISABILITIES

Abbosova M.S

Teacher of the Department of Interior design,
Samarkand State University of Architecture and Construction

<https://doi.org/10.5281/zenodo.14535277>

Abstract. *This article explores the significance of establishing Development Centers for people with disabilities within neighborhoods. It delves into existing literature to analyze the benefits of such centers, discusses methods for their establishment, presents potential outcomes, and offers conclusions and suggestions for future initiatives.*

Keywords: *Development Centers, Disabilities, Community Empowerment, Inclusion, Accessibility, Support Services.*

Introduction

Inclusion and accessibility are fundamental principles of a thriving society. However, people with disabilities often face barriers that hinder their full participation in community life. Development Centers tailored to their needs offer a promising solution. These centers serve as hubs for empowerment, providing support services and fostering a sense of belonging within neighborhoods.

Existing literature highlights the transformative impact of Development Centers for people with disabilities. Studies indicate that such centers promote social integration, enhance skill development, and improve overall well-being. Moreover, they facilitate access to specialized resources and promote a sense of community cohesion. However, there remains a gap in understanding the specific methods for establishing these centers within neighborhoods.

Establishing Development Centers requires a multi-faceted approach. Collaboration between local government, community organizations, and disability advocacy groups is crucial. Initial steps involve conducting needs assessments to identify specific requirements and preferences of the target population. Securing funding, designing accessible infrastructure, and recruiting qualified staff are essential components. Additionally, partnerships with educational institutions and healthcare providers can enrich the range of services offered.

Results:

Establishing Development Centers for people with disabilities in neighborhoods is a fantastic idea! These centers can serve as inclusive spaces where individuals with disabilities can access a variety of services and support tailored to their needs. Here's how such centers could benefit the community:

- **Accessibility:** By having centers located within neighborhoods, individuals with disabilities can access services more conveniently, reducing transportation barriers.
- **Community Integration:** These centers can serve as hubs for community engagement, fostering interactions between people with disabilities and their neighbors. This can help break down social stigmas and promote inclusivity.

Community integration centers play a vital role in promoting inclusivity and breaking down social stigmas associated with disabilities. By serving as hubs for community engagement, these centers provide opportunities for people with disabilities to interact with their neighbors in meaningful ways.

Furthermore, community integration centers can offer resources and support services tailored to the specific needs of people with disabilities, empowering them to participate fully in community life. By promoting collaboration and cooperation between individuals of all abilities, these centers contribute to the creation of a more cohesive and supportive community where everyone feels valued and included.

- **Comprehensive Support:** Development Centers can offer a range of services including education, vocational training, therapy, and recreational activities, providing holistic support for individuals with disabilities to thrive.

Comprehensive support services offered by development centers can significantly enhance the quality of life for individuals with disabilities. Let's break down some of these services:

Education: Development centers often provide tailored educational programs designed to meet the specific needs of individuals with disabilities. This may include special education classes, individualized learning plans, and adaptive teaching methods to ensure that each person can learn and grow to their fullest potential.

Vocational Training: Equipping individuals with disabilities with vocational skills is crucial for their independence and integration into the workforce. Vocational training programs offered by development centers can include job readiness training, skill-building workshops, and hands-on experience in various industries.

Therapy: Therapy services, such as physical therapy, occupational therapy, and speech therapy, play a vital role in helping individuals with disabilities improve their motor skills, communication abilities, and overall well-being. Development centers often have qualified therapists on staff who can provide these essential services in a supportive and nurturing environment.

Recreational Activities: Engaging in recreational activities not only promotes physical health but also fosters social connections and enhances overall happiness and quality of life. Development centers may offer a variety of recreational programs and activities, such as sports, arts and crafts, music therapy, and community outings, tailored to the interests and abilities of individuals with disabilities.

By offering a comprehensive range of services encompassing education, vocational training, therapy, and recreational activities, development centers can create a supportive and inclusive environment where individuals with disabilities can thrive and lead fulfilling lives.

- **Empowerment:** By providing resources and support tailored to their needs, Development Centers can empower individuals with disabilities to maximize their potential and independence.

Empowerment through tailored resources and support is crucial for individuals with disabilities to achieve their fullest potential and independence. Development Centers play a vital role in this process by offering specialized assistance, training, and guidance to address the unique needs and challenges faced by people with disabilities.

Furthermore, empowerment goes beyond just providing resources; it also involves fostering a supportive and inclusive environment where individuals feel valued, respected, and empowered to advocate for their own needs and rights. By promoting self-advocacy and self-determination, Development Centers can help individuals with disabilities become active participants in their own lives and communities.

Ultimately, the goal of Development Centers is to empower individuals with disabilities to become independent, productive members of society, capable of realizing their dreams and contributing to the world around them.

- **Family Support:** Families of individuals with disabilities can also benefit from these centers by accessing support groups, counseling services, and educational resources to better understand and support their loved ones.

- **Advocacy and Awareness:** These centers can also serve as advocacy hubs, raising awareness about disability rights and promoting policies that support inclusion and accessibility.

- **Employment Opportunities:** Development Centers can collaborate with local businesses to create job placement programs and vocational training opportunities for individuals with disabilities, promoting economic independence.

Conclusions and Suggestions:

In conclusion, Development Centers play a vital role in fostering empowerment and inclusion for people with disabilities within neighborhoods. To maximize their impact, ongoing support from stakeholders and policymakers is imperative. Future efforts should focus on replicating successful models, enhancing accessibility, and promoting public awareness. By investing in these initiatives, societies can strive towards a more equitable and compassionate future.

In summary, the establishment of Development Centers represents a tangible expression of community solidarity and commitment to diversity. Through concerted efforts, we can build a world where everyone, regardless of ability, has the opportunity to thrive and contribute meaningfully to society.

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NEGATIVE EFFECTS ON THE ENVIRONMENT WHEN BURNING FUEL IN HEAT SOURCES

Mahmudov R.¹, Abdurashidov R.R.²

^{1,2}magistr

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Abstract: *The proposed article is devoted to the issues of the release of harmful emissions into the atmosphere during the combustion of various types of fuels in heat sources and their harmful effects of the Earth's ozone layer, flora and fauna.*

Keywords: *fuels, gas, oil, coal, harmful emissions, thermal power plant, thermal power centers, soot, oxides, thermal energy consumption, ecology.*

Introduction

There is an inextricable relationship and interdependence of the conditions for ensuring heat and energy consumption and environmental pollution. The interaction of these two factors of human activity and the development of production forces attracts gradual attention to the problem of the interaction of thermal energy and the environment.

At an early stage of the development of thermal power engineering, the main manifestation of this attention was the search in the environment for resources necessary to ensure heat and energy consumption and stable heat supply to enterprises and residential buildings. In the future, the boundaries of the problem covered the possibilities of more complete use of natural resources through the exploration and rationalization of processes and technology, extraction and enrichment, processing and combustion of fuel, as well as the improvement of thermal power plants.

With the growth of unit capacities of blocks, thermal power plants and thermal power systems, specific and total levels of heat and energy consumption, the task arose of limiting polluting emissions into air and water basins, as well as more complete use of their natural dissipation capacity.

At the present stage, the problem of interaction between thermal energy and the environment has acquired new features, spreading its influence over vast territories, most rivers and lakes, huge volumes of the Earth's atmosphere and hydrosphere.

The intensive development of energy, industry and transport inevitably causes an increase in the consumption of hydrocarbon fuels, which, in turn, increases the amount of its combustion products released into the atmosphere.

According to long-term monitoring, the amount of environmentally harmful chemical compounds, substances and elements of fuel combustion products released into the atmosphere doubles every 12-14 years, and therefore the problem of atmospheric pollution by fuel combustion products belongs to one of the global problems of our time.

Atmospheric pollution occurs due to imperfections in the design of equipment, engine systems and installations, violations of operating rules, low technical culture and environmental illiteracy of maintenance personnel, and sometimes as a result of accidents or emergencies.

The toxicity of flue exhaust, exhaust and exhaust gases released into the surrounding air depends mainly on the quality, grade and type of hydrocarbon fuel burned, the conditions of the organization of its combustion process, the technical condition of heat engines and fuel combustion plants.

For example, the use of low-grade fuels, on the one hand, helps to reduce the current operating costs of purchasing fuel, and on the other hand, increases the amount of environmentally hazardous pollutants released into the atmosphere.

The intensity of atmospheric pollution with harmful gases is determined by the concentration and number of pollution sources per unit area; the type, capacity and mode of use of fuel combustion plants and heat engines; the design and condition of their fuel systems, the condition and level of operation of technical means; the availability of systems and devices that qualitatively and quantitatively reduce harmful emissions into the atmosphere, and other factors.

Studies of flue exhaust gases of fuel combustion plants show that in their composition the main pollutants of atmospheric air are carbon oxides (up to 50%), sulfur oxides (up to 20 percent), nitrogen oxides (up to 6-8%), hydrocarbons (up to 5-20%), soot, oxides and derivatives of mineral inclusions and impurities of hydrocarbon fuels.

The greatest air pollution comes from installations running on hydrocarbon fuels (gasoline, kerosene, diesel fuel, fuel oil, coal, natural gas, etc.). The amount of pollution is determined by the composition, the object of the fuel burned and the organization of the combustion process. In recent years, the amount of emissions from transport accounts for about 70% of all emissions into the atmosphere.

The combustible substance of fuels consists mainly of 3 elements: C, H, S. Gorenje - rapid connection of oxygen with these combustible elements, accompanied by the release of heat.

There are three types of fuel: solid, liquid, and gaseous.

The main solid fuel is coal. The characteristics of coal vary from deposit to deposit. Coal is divided into classes and groups according to the established carbon content and the heat of combustion attributed to the mass without mineral impurities. Solid fuels include oil shale, anthracite shale, wood, products of its processing, solid waste.

Liquid fuels include refined petroleum products, coal tar, shale oils and industrial waste of various compositions. Liquid fuels are sprayed into small droplets to increase the surface available for interaction with oxygen. These droplets easily evaporate at the temperature of the furnace and are in a gaseous state during gorenje.

Gaseous fuels are easily mixed with air and do not require preparation. Natural gas, mainly methane, is the predominant gaseous fuel currently in use.

Other gaseous fuels are petroleum gases, blast furnace gas, acetylene, hydrogen and satellite gases, the composition of which depends on the type of production.

During the operation of boiler equipment, various harmful compounds enter the environment, which have a negative impact. These include:

- Gases are the main source of emissions of harmful substances into the environment. During the operation of boilers, compounds of carbon dioxide and carbon monoxide, sulfur oxide, and nitrogen enter the air.
- At large thermal power plants, liquid compounds are released, which enter the sewer system and local reservoirs.
- Emissions include soot, ash and unburned coal particles remaining after the use of solid fuels.

The most popular fuels are coal, gas and fuel oil. Each of them has its advantages and disadvantages.

- Fuel oil allows you to get the maximum amount of energy, so the amount of fuel used is significantly less. In addition, it is not explosive, and is relatively safe to use. However, fuel oil cannot be considered as an environmentally friendly fuel. When it is burned, a huge amount of harmful substances is released into the air, so today most cities purposefully refuse to use this type of fuel in favor of other energy resources.

- Natural gas attracts with its cheapness, ease of transportation and great energy potential. However, this fuel is extremely explosive, and any malfunction of the equipment can lead to a serious accident.

- Coal is the most economical type of fuel in most regions of Russia. Its environmental friendliness is also attractive – compared to other energy resources, it turns out to be the cleanest and most efficient source of heat and energy.

The largest amount of gross emissions among the city's energy sources is emissions from thermal power plants. This fact is explained by the fact that a larger number of consumers in the city are supplied with heat from the CHP (heat and electric central). But for a number of some pollutants (soot and CO), boiler houses emit more than CHP plants. This is due, among other things, to different coal burning technologies.

"Flue gas purification (for solids – soot, benzene (a)pyrene, inorganic dust) is carried out at the CHP plant more effectively. This is explained by the use of more advanced exhaust gas purification systems (a significant part of CHP emissions is cleaned in electric filters, which are much more efficient than cyclones and battery cyclones). Also, lower emission values for solids are associated with more efficient and complete combustion of fuel at a thermal power plant," the city's Heat Supply Scheme says. – Despite the fact that the load of consumers connected to boiler houses is less than 30% of the total heat load of consumers in the city, gross emissions of inorganic dust (a substance for which the maximum permissible concentration is observed in the atmosphere of the city) account for about 40% of the gross emissions of energy sources in the city."

In conclusion, it can be noted that the operation of thermal power plants (thermal power plants) also causes irreparable damage to public health, since when burning coal they emit especially dangerous volatile compounds of mercury, arsenic, selenium, lead, cadmium, zinc and others. Once in the human body, they accumulate in organs, leading to the development of malignant tumors and mutagenic effects, reducing resistance to infections. The smallest particles with a diameter of up to 0.1 microns can penetrate into the capillaries even through the skin. The distance to which ash particles are carried and their deposition to the ground together with atmospheric precipitation depends on the physical properties of the ash and weather conditions. People living at a distance of 1-2 km from the thermal power plant are most affected.

The steady increase in the intake of toxic substances into the environment primarily affects the health of the Earth's population, worsens the quality of agricultural products, reduces yields, affects the climatic conditions of certain regions of the world, the state of the Earth's ozone layer, leads to the death of flora and fauna.

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VIEWS OF MODERN ARCHITECTS ON THE CONCEPT OF "STYLE"

Kushmanov Tulkin Fayzullayevich

Samarkand State University of Architecture and Construction

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Abstract. *This article presents various ideas related to the concept of style and explains its meaning.*

Keywords: *Concept of style, history of style concept, content of style concept, criterion of style concept, category*

Introduction

When it comes to the social sphere, or in the literature on art history, and in particular on the history and theory of architecture, the word "style" is often used. The concept of the word "style" is widely used to distinguish the elements that make up a system of a particular creative field. We try to analyze and define the views surrounding this concept, which has penetrated so widely into our social life. It seems expedient to begin to shed light on the concept of style from its beginning. The word "style" came from ancient Greek philology, and metal or bone sticks presumed to write on a special writing boards are called by the same name ("stylos"). From Greek to Latin it became "stylus" and began to express a way of expressing an idea [1, p.82]. The founder of the science of art, I.I. Winkelman was the first who introduced the word into art history. In his "History of Ancient Art", published in 1763, he tried to create a model for the study of art history using the category of "style." Since then, 260 years later, knowledge about style has increased, and as a result, style has begun to be viewed as an important artistic and aesthetic category. Since then, architectural forms of a particular period have been identified using stylistic criteria.

Different views on the concept of style

However, we encounter many ambiguities when it comes to give a clear definition of the concept of style. It is also emphasized by the famous art historian, academician V.V. Vinogradov's words: "In the fields of art, literature and linguistics, it is difficult to find an unstable and subjectively vague term with many meaningful and contradictory ideas, such as the term style and the concept of style" [2, p.7]. B.R. Vipper's words in this regard can be quoted: "Style is an objective view of art" - "it is a complex, diversified and wide concept" [3, p.5]. Indeed, views on the concept of style are very dissimilar, and there are many contradicting opinions about its existence, content and essence. In the late nineteenth and early twentieth centuries, the concept of style became one of the main categories of architecture. The new movement, called "Jugendstil" in Germany and "Ar Nuvo" in other European countries, was one of the most advanced in the theory and practice of architecture, and even promoted contemporary art in the Netherlands at the time called the movement "De Stijl". However, from the 1920s onwards, modern European architects began to deny the notion of style. Walter Gropius, with his words, "We tried to find a new way, not style," led to a change in the positive attitudes of many young architects at the time. [4, p.143]; Moreover, today one of the founders of functionalism Le Corbusier's says these words "Style is a lie" [5, p.10], "Architecture has nothing in common with different styles" [6, p.237], caused the style as a symbol of eclecticism. These words were a judgment on the long-standing dominance of the concept of style in architectural theory and history. Our research has shown that some modern architects today have a negative view of the concept of style: "I can say to those

who deny me that I am - without style - an eclectic architect, because the buildings which I build are diverse. In fact, I always start from scratch, I always look for a clear answer to every given task, and for me, each building must be unique ... "Jean Nouvel expressed his views on style [7, p. 254].

"The problem of modern architecture is not a question of the style of curved or simple surfaces, but of their synthesis," said Hans Hollein, who also expressed his Furthermore, today the architects who make up the architectural elite have expressed denials about style, which We list below:

Dominique Perrault "I am not interested in style, I am interested in the experience of creating environments" [9, p.34];

Antonio Citterio, I do not look at my work in terms of "style". Each project must be unique - the building can not "work" in the same way in Milan, as well as Moscow " [9, p. 170];

David Chipperfield, "I don't think I have my own style, I'm not interested in complicated things, more important to me is the typology of objects and how they "work" [9, p.188];

Rafael Vinoly "The architect does not have to serve the "style", it does not matter whether you build in Africa or in Sweden, a good project is expected everywhere. If it is approached to our profession better, if they think deeply about each project, they will not be able to duplicate each other" [9, p.157];

Willian Alsop, "Style is an illogical category in modern architecture where anything can happen in general" [9, p.202];

Peter Zumthor, "I'm not copying anything, I'm not repeating anything - I'm working on it now. It's not a style, without any hesitation it's not style. It's about exploring the universe and creating your own things by yourself. Creating things for today " [10, p.138];

N.V. Lyzlov said, "Style is either for critics or for epigons. It is a method of classification, not creation " [11];

Peter Eisenman, "Like other architects, I don't have a single basic idea. For instance, Richard Meyer builds his buildings the same method no matter where they are located. My works are logical and they are based on a certain logic. So I can't build the same building in Santiago, Berlin and Phoenix, Arizona. That's why, I don't have style. The buildings of Frank Gary and Michael Graves look similar. And mine is not " [12];

Rem Koolhaas, I feel both inside and outside the architectural "dictionary" at the same time. Our company OMA also participated in the creation of this dictionary. That's why, we always try to get out of it. Sometimes, in order to get out of the "dictionary" or style, it is necessary to take it to the point of irrationality " [13, p. 116-117];

Odile Decq, "I always look at my projects as a new adventure, a new history, and therefore I don't think they fit any style" [14];

Bjarke Ingels said: "Rem Koolhaas was an important discovery for me. He taught me not only to design beautiful buildings, but also to use architecture in political, social processes, to express different forms independently. For him, the main thing in architecture is not aesthetics. It is judged by the idea, not the architectural style " [15].

After these reflections, following questions will arise: is style losing its importance as an important category of art and architecture ?, Is the concept of style necessary or not? It can be answered to this question like this: the concept of style remains an important criterion in the analysis of architectural objects. It is good that some architects strive for originality, but the idea

that the creature “I” is outside the concept of style leads to confusion in defining the principles of architectural development, so it is very important to use the concept of style in the study of modern architecture. "There might not be no stylistics in art," said Hause. Brexit interprets it differently. According to his opinion, art cannot exist without style, because art creates a “lifestyle” rather than a “protocol of life”. He notes, “If we study the work of Dutch artists who call themselves realists, it becomes clear that their style is the style of their contemporaries. They took it all from life ”[16, p.238].

On the importance of the concept of style in art, journalist E. Barabash also said: "The presence of style in works of art today is very relevant, without it there can be chaos in art" [17].

It can also be given A.G. Rappaport's views on the role of style in modern architecture. “Traditions without style are meaningless. Over time, in the endless world of art, images lose their original appearance and this leads to chaos. The elimination of such disorder depends on the existence of common directions. Style once served as such a direction. The experience of modern architecture has shown that without style, not only the category of “architectural form” but also the concept of beauty loses its significance. There is no form or beauty without style. ”[18]. S.P. Zavarixin also agreed with these points. “Unlike the hero of the fairy tale, today's authors, more clearly, creators, have not three but thirty-three paths, from which one can either walk forward to the radiant peak of technological progress or crawl backwards and, in general, go to any creative destination. Architecture, devoid of a single stylistic direction, begins to grow in various forms on the basis of the principle of "writing as it is heard."

It is clear that the analysis of modern architecture, the determination of its development paths, cannot be done without style. Style is the core that connects all the elements that make up an architecture. According to T.Yu. Bystrova, “Style embodies the “heart of culture” which is impossible and understandable in artificial creation” [20, pages 246-247].

The content of the concept of style

To understand the essence of style, we pay attention to the definition in the encyclopedic dictionary of the former Soviet Union: "Style is a system of images, a means of artistic expression, a unity of creative methods aimed at the realization of a single idea" [21, p. 1269]. Well-known scientist R.S. Mukimov described style as follows: “Architectural style is an artistic category that accumulates in the course of a certain historical development and distinguishes the culture of one historical period from another, expressing the integrity of compositional methods, architectural forms” [22, p. 6].

Gathering all the ideas about style, scientists have given the following complex and weighty definition: Style is the artistic structure of a work felt by the senses, its original model ("heart", content, source of style - the artist's impressions of life and incomprehensible directions) is the principle of revealing all aspects of the work of art (internal structure, external expression of the content and historical originality) on the basis of "[23, p.226].

conclusion

No matter how contradictory the ideas about the concept of style may be, it remains a complex category that helps to analyze the socio-economic life of a certain period, the art of reflecting people's thinking, and architecture in particular. It is possible to analyze the extent to which the content of art forms is expressed through stylistic criteria. So, as art develops and evolves, the definition of the concept of style also changes and improves accordingly.

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ORNAMENTAL PLANTS USED FOR FLOWER ARRANGEMENTS

Khakberdiyev Alikhuja Muratovich

*Teacher, Samarkand State Architectural and Civil-Engineering Institute, Samarkand,
Uzbekistan*

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Abstract. *Through this article, future landscape designers will get acquainted with one, two and perennial flowers for flowerbeds from small floral arrangements. Information is given on what area and soil to plant flowers for flower beds, the correct choice of flowers depending on their flowering sequence, parameters (low, medium and high).*

Keywords: *Decorative plants, flowers, annuals, biennials, perennials, low-growing plants, medium-sized plants, tall plants*

Introduction

Each landscape designer chooses a set of flowers individually when designing small floral arrangements for any area. The decision-making process is primarily influenced by the taste of the beauty lover.

For every florist, experience plays an important role in growing this or that flower. In the process of planting flowers, due to factors such as improper location of the area, improperly selected soil, making mistakes in their cultivation can not achieve the desired result.

In small flower gardens, annual, biennial and perennial plants are selected for the selection and planting of flowers. Some people use annual flowers as a floral composition because they get tired of the monotony quickly. However, there are varieties of biennials and perennials that people never get bored of enjoying their beauty.

The purpose of this article is to study the morphological and biological characteristics of annual, biennial and perennial flowers and to introduce the types of flowers that can be used in the composition of a small flower bed.

The task is:

- ☐ Collection of material on the composition of the flower bed and the annual, biennial and perennial plants used in it;
- ☐ Targeted use of special literature and magazines, normative documents, analog drawings and projects, the Internet on the topic of the assignment;
- ☐ Systematic approach to the task, the tasks to be performed independently;
- ☐ to complete the work on the basis of compositional tools, geometric and harmonic laws;
- ☐ to develop skills and apply professional skills to apply the acquired knowledge and skills in life.

Annual flowering plants for flowerbed composition. It is hard to imagine most of the landscape architecture objects without flower beds. Flower beds are divided into several types, including flowerbeds, mixborders, rabatkas, modular flower beds, and more. The plants used for such flower beds are divided into annual, biennial and perennial.

It is convenient to plant in flower gardens from simple and bright annual plants, because with such plants can decorate flower arrangements every year at will. Annuals can be planted from seed or as seedlings.

By planting flowers that bloom at different heights and at different times in the flowerbed, it is possible to create a beautiful flower garden that blooms all summer long.

Low-growing annuals: These plants can be planted at the edges of flower beds, around trees, shrubs and perennials, to create curbs in sidewalks. Spaces in the flower bed based on perennial flowers can be filled with low annuals.

Among landscape experts, the most popular and popular flowers from low-growing plants are:

Velvet is distinguished by its orange-brown flowers throughout the summer. The height of this flower reaches a height of 20 cm. Velvet blooms not only in the sun but also in the semi-shade for a long time.

Lobelia is a low, spreading shrub, a small flowering plant in dark red (white), white, blue or purple. This flower loves moist soil and sunny places. The plant looks great when used in flower beds, curbs and hanging vases.

Iberis blooms mainly in white. It grows up to 30 cm in height, prefers soft local soils and can grow in partial shade. This plant can be planted in rabatka, flower beds and aisles.

Petunia is a favorite plant and comes in a variety of colors, such as the color of the rainbow. Petunias bloom from the first months of summer until the cold weather. She loves the heat and the sun. It is best to buy this plant as a seedling or as a flowering plant.

Medium-sized annual plants for flowerbed composition: Two-horned *Mattiola* grows to a height of 50 cm and blooms all summer long. The plant is famous for its beautiful flowers, which begin to emit a divine scent after sunset.

Cosmos is one of the most difficult-to-find plants. This flower grows up to 80 cm in height and the flowers are dark red, pink or white. It is a fragile plant with delicate leaves and grows well in partial shade.

Dimorphoteka flowers are well suited for compositions of annuals as well as perennials. This flower should be planted in sunny, open ground. Any floral composition made of *dimorphoteka* gives the atmosphere a festive mood.

Zinnia blooms in white, pink, purple, yellow or red throughout the summer until frost. This flower loves light and loves fertile soil. In most cases, the plant is planted in groups on flower beds and lawns.

Salvia is a beor plant that grows up to 30 cm in height and is compatible with many plants in the flower garden.

Annual tall plants for flowerbed composition: Tall plants in groups give good results in the flower bed. These little evergreens are super-easy to grow and will look good with almost no care.

Among the tall flowering annuals are:

Amaranth is a plant that has 30 different appearances. In late summer, it blooms in red, in the form of candles, grows up to 150 cm in height, and the leaves of some species are also reddish.

Malva flowers grow up to 130 cm in height. This beor plant blooms all summer long with beautiful flowers. Its large flowers are reminiscent of a bell. *Malva* is planted in the form of seedlings in May.

Kleshivina (castor oil) or the tree of paradise is an exotic plant that is absolutely barren. It is famous for its large leaves that resemble the leaves of a palm tree. In a short time, the cliché grows to a height of 150 cm.

Datura or durman- the plant blooms with white, blue, red and yellow flowers resembling bells. Grows up to 1 meter tall in fertile soil. *Datura* blooms all summer long until frost. Annual beor (non-selective) flowers that turn from a seed into a beautiful flower in one season tend to be

noticeable. With proper care of such flowers, it is possible to prolong the flowering period and spread the sea of people with its bright colors, giving a beautiful view of the environment.

Two year old flowers for flowerbed composition. This type of flowering plant is propagated from seed and transplanted after it has grown. An important aspect of biennial plants is that they do not bloom from the first year.

Bell - The plant can grow up to 80 cm in height. The flowers can be fluffy or simple. Available in Pink, White, Blue, or Purple.

The plant likes sunny places, but does not like windy places. They are planted at a depth of 0.5 cm in May-June. By the end of summer, they are transplanted to a permanent place according to the scheme 30x40 cm. Flowers appear only in 2 years.

Three different colored purple. Another name is *Viola Vitrokka*. Plant height - no more than 30 cm. The color of the flowers can be single or multi-colored (up to 2-3 colors). Seedlings are planted in March (probably February). The plant is dormant and blooms throughout the summer when grown in a sunny area.

Turkish clove is a beautiful plant that grows up to 60 cm in height. Looks great in the flowerbed. On the dark green stems are formed dense caps of inflorescences of different colors. The plant is grown from seed and transplanted in the second year. It grows well in both sun and partial shade.

Perennial flowers for flowerbed composition. Although many landscape experts do not bloom all summer long, they are resistant to the cold weather of winter, easy to care for and maintain, can be propagated in parts, and through them give a modern look to small flower beds prefer perennials that give.

Small flower beds made of perennials should be chosen not only on the color of flowers, but also on the parameters of the plants. For example, planting tall, bright, and large-flowered perennials in the center of the flowerbed; against the background of tall plants should be planted perennials of medium height; it is advisable to fill the edges and cavities of the flower bed with low-growing plants.

Low-growing perennials. The main difference of low-growing flowers is that they can be axed not only to the edges of small flower beds, but also to create patterns or drawings on small ponds, rocky cliffs or lawns.

Dwarf aster is one of the most popular plants in gardens. Its flowers come in many colors and bloom until July, when the weather is cold. Astra prefers regular watering and partial shade.

Balsam - with its bright flowers decorates flower beds and curbs until the cold weather. This plant also grows well in the shade and in any soil.

Alissumis a yellow and blue flowering plant with a collection of flowers that blooms until late autumn. This flower can be planted in flower beds and on rocky cliffs. The plant prefers dry, fertilized soil and requires regular pruning of dried branches.

Aquilegia loves moist soil, so it is recommended to plant it near water bodies. Does not require large amounts of fertilizer.

Tall perennials. *Peony* is a thick shrub that grows quickly and blooms in different colors. Requires regular watering, timely feeding, pruning of dried branches. Peonies amaze people not only with their beauty but also with their wonderful aroma.

Astilbais a fluffy flower of different colors that grows even in the shade. It is one of the longest flowering perennials (up to ten weeks). Astilbes grow in different sizes depending on the navigation. The plant loves regular watering and feeding.

Gladiolus is an onion-like perennial plant that blooms from July to September. Some varieties of plants bloom within 70 days after transplanting. Because gladiolus comes in a variety of colors, it can be used to create unusual compositions for flower beds. It is recommended to transfer this flower to partial shade.

Chamomile is a plant that will not leave indifferent people who love flowers. With proper care, they bloom for a long time and grow fast. It is recommended to plant chamomile in sunny areas, as it does not grow well in the shade.

Delphinium is the king of blue flowering plants. Today there are varieties of this flower that bloom in pink, yellow and orange. The plant blooms in June and July, requires no care and tolerates cold winter weather well. Properly selected flowers for the flowerbed will make it beautiful and bright, and will bloom from spring to the cold days of winter. A flower lover who knows the types and varieties of flowers well can create an unusual and beautiful flower garden.

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METHODOLOGICAL STUDY OF LEXICAL DERIVATIONS IN GERMAN LANGUAGES

Abdurakhmanov Sirojiddin Muminkulovich

PhD, Samarkand State Institute of Foreign Languages

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Abstract. *This article is about the methodological study of lexical derivations in German languages. In German languages word formation, morphemic structure and their teaching methods are discussed. The grammar, lexicology, and word formation of German languages have been studied from a historical perspective. Examples of word formation through lexical derivations are given.*

Keywords: *derivation, affix, inflectional language, analytic language, amorphous language, ablaut, lexical derivation, word formation, inflection, morphemics, affixation.*

Introduction

The question of the use of language in speech is of great importance at the present stage of development of linguistics, because this issue has long been on the agenda of our linguistic research. The main reason for this was that linguists have focused on the study of the complex system of language and its specific problems, and therefore the issues of interpretation of speech linguistics were of secondary importance. However, in the current era of linguistics, there is a need to study the use of language in speech, the role of the human factor and the cognitive-pragmatic problems associated with it. This, in turn, increases the prestige of the study of derivation patterns of language development. No serious monographic research has been conducted in Uzbek linguistics on the study of lexical derivation phenomena in German languages. In linguistic interpretations of the phenomenon of word formation, another phenomenon is distinguished by its study under the term derivation. The term was introduced to linguistics in the 60s of the last century by the Czech linguist E. Kurilovich. It should also be noted that most linguists interpret lexical derivation as synonymous with word formation. Lexical derivation requires word formation. Their basic principles are common.

At present, the issue of word formation is considered as an object of study in a number of branches of linguistics, including morphology, lexicology, dialectology, grammar. If we look at the science of linguistics, as well as the history of the development of German languages, the phenomena of word formation and those directly related to this process generally express a certain similarity in each language. But the cognitive meanings in word formation, the lexical, grammatical, and semantic features of artificial words, are radically different in each language. Therefore, there is still controversy about the phenomenon of word formation. While in most scientific works the phenomenon of word formation in languages is considered as an integral part of grammar, some scholars interpret word formation as a purely lexical phenomenon and consider it as an object of study of lexicology.

It should also be noted that a good knowledge of the internal laws of the methods and means of word formation also plays a worthy role in the study of a foreign language.

Of course, each branch of linguistics is inextricably linked. Also, most branches of linguistics are directly concerned with the study of word and word-formation phenomena.

According to Wikipedia, ancient German languages appeared about two thousand years and formed over the next millennium as the language of the people living in the area from the western

part of the Baltic Sea to the southern sides of Mount Harz. 750, German languages spread around the coasts of the North Sea and the Baltic Sea, as well as on the island of Jutland and the southern parts of the Scandinavian Peninsula. The people who spoke this language occupied the right and left banks of the rivers Vistula, Elbe, Oder, Rhein. In the first and third centuries AD, the Germanic peoples spread to the east and south, approaching the Danube territories of the Roman Empire. [de.m.wikipedia.org/wiki/] The work of Cornele Tacitus, the mayor of Rome, entitled "Germany" contains information about the dialectal differences between the ancient German languages under the influence of internal phonological changes. [Cap. 43,1] The period that lasted from 3 to 7 centuries and was known in history as the "Great Migration of Peoples" and the socio-economic relations that emerged between people during this period, especially property relations. The change paved the way for the emergence of a number of local languages belonging to the German languages. Thus, German languages began to differ from other Indo-European languages both phonetically and morphologically.

In the morphology of German languages, for example, there has been a shortening of consonants in the variation of noun cases. In particular, in the first millennium BC, the existing eight consonant forms in Indo-European languages were reduced to six in German languages. There was also a singular, binary, and plural form in nouns, and a rod category in nouns. In verbs, however, three tendencies — Indicative, Conjunctive, and Imperative — are formed relatively strongly, with two ratios — Active, Passive, and two tenses — Präsens and Präteritum in German.

Scholars point out that the following features played an important role in the separation of German languages from Indo-European languages as a separate group.

- In a system of strong verbs, internal inflections, especially the Ablaut phenomenon, developed;
- Weak verbs began to form their past tense form with dental additions;
- As a result of the inflection of the adjectives in the words following the demonstrative pronouns, the adjectives change first to the weak and strong, and then to the mixed variable suffixes.

It is known from history that after the death of Charles the Great, the leader of the great empire founded by the Franks, the empire was divided into three parts by Charles the Great's grandsons under the Treaty of Strasbourg, which came into force in 842 and the Treaty of Verdun in 843. As a result, German, French, and Italian began to be used as the state languages in the newly formed states. It was during this period that a union of Anglo-Saxon tribes from the German tribes occupied Great Britain, and the newly formed state was named England, and English was chosen as the official language of communication in the region.

In the development of the vocabulary of these states, founded by Germanic tribes, first of all, the daily activities and unique culture of the peoples who speak these languages play an important role. Lexical units of languages such as ancient Latin, ancient Greek, ancient Slavic, Celtic, and Finnish, which serve as ancestral languages in Indo-European languages, also play an important role in the expansion of lexical units in German languages. For example, French, which was originally the language of the German Frankish tribe, was confused with Italian, Spanish, and Rhetoric in the Middle Ages, and became a language belonging to the Romance language group.

In turn, the western and northern German languages German, English, Icelandic, Danish and Scandinavian languages during the development of the three, the old, the middle and the construction of a new period of its lexical-morphological forms.

German languages are inflected languages. One of the peculiarities of inflected languages is internal inflections, that is, “the phenomenon of changing the vowel in the stem to another vowel - ablaut, the change of the vowel to another vowel with a similar pronunciation - umlaut and the formation of consonants in the word formation. New words are created based on changes that include characters such as 'change'. In analytical languages, however, suffixes play an important role in word formation. Nevertheless, under the influence of interlingual organic relations it is now possible to see the peculiarities of analytic languages in inflected languages, which in turn are characteristic of inflective, amorphous, and cohesive languages in analytic languages. For this reason, in modern German languages, such as English, German, and Swedish, affixation is used to form new words, compound words, from one word group to another. the role of new word-formation phenomena through transmission is incomparable. Due to the rapid development of cognitive linguistics in recent years, as in all languages, the cognitive properties of artificial words, as well as simple primitive words, require serious study. .

Nicolaus Janos Raag also noted that as the German language began to appear in written sources from the beginning of the Middle Ages, the need for not only derived words but also simpler, more artificial and compound words, especially compound nouns, increased in German. emphasizes. (S.89)

The first research work on the history of the development of Germanic languages, in particular the development of lexical units specific to Germanic languages, dates back to the 18th century. The work started by the Grimm brothers was continued in the 19th century by German scholars Friedrich Schlegel, German Paul, Ferdinand de Soussure, Franz Bopp, Rasmus Rask, V.G.Admoni, V.V.Shcherba. Research on German lexicology and German grammar intensified in the 20th century. In the 20th century, German scholars such as Werner Betz, Walter Hensen, and Johannes Erben began to study word formation in German from a historical perspective. By the 20th century, the emphasis on learning German was also increasing in the former Soviet Union. As a result, large scientific schools of the German language, the object of study across the country, began to emerge. For example, an F.A. from the University of St. Petersburg . Proof of our opinion is the emergence of such well-known German scientists as Brown, VM Zhirmunsky, V.P.Berkov, L.R.Zinder, T.V.Stroyeva, G.N.Eichbaum, K.A.Filippov, L.N.Grigoryeva, T.V.Ponamaryova, N.A.Golikova.

T.A.Katskova, who is currently conducting research at the University of St. Petersburg, wrote her article entitled “Pamyati Uchitelya T.V.Stroyevoy” (Sokolskoy)”. In addition to, L.R.Zinder and T.V.Stroyeva co-authored “Theoretical Phonetics and Theoretical Grammar of the German Language” and “History of the German Language”, “German Lexicology”, “Introduction to German Philology”, “German Dialectology” and “German Language”. He published textbooks and manuals and organized lectures, special courses and seminars in these areas. [<https://nemphil.spbu.ru/lib> . Page 22]

It should be noted that in languages such as Old High German, Old English, Old Scandinavian, the lexical richness of the language increased, first on the basis of suffixes and then on the basis of suffixes and prefixes formed from prefixes.

In particular, word formation and word formation based on internal inflection and affixation are common in West German languages. Some research has also been done in this regard.

In particular, the Russian scientist Ye. A. Libert in his dissertation on 10.02.20 - Comparative-historical, typological and comparative linguistics "Typology of suffixed diminutive models in the West Germanic languages" in the candidate's dissertation In order to study, he compared the reduction suffixes in the West German languages [6:24]. In this study, the scientist first compared the diminutive suffixes in English, German, Dutch, Frisian, as well as plotdich and Yiddish, which are based on Old German. In his research, Ye.A. Libert noted that the suffixes in German appeared in the form "-kin", "-lin", and later these suffixes were used to describe the palatalization, assimilation and dissimilation of sounds in West Germanic languages. He used the relevant examples to show that the first and second shifts of consonants took different forms in different German languages, that is, these suffixes "-ke (n), -je (n), -t'e, -tsje, -çe (n)" was able to show that it had arrived in appearance.

As a result of the drop of the "n" sound at the end of the word, this suffix appears in modern English as "-ie" or "-y". For example: Becky (a form of the name "Rebecca"), Cathie (a form of the name "Catherina"). Such suffixes are also found in modern English words such as "mummy - mother, daddy - father, pussy - kitten, bookie - booklet, buggie - stroller." It should be noted that a number of words in English are also formed with the suffix -ette, derived from the French. For example, Lisette - Lizahon, luncheonette - light food, and so on.

These diminutives have appeared in the Frisian and Dutch languages as "-ke, - (t) sje, -je and - (e) tsje, -pje, kje, respectively. For example, in the Frisian language: Jerske, Pieterke, kayke - key, nestke - nail; in Dutch: Annsje, Geartsje, Soukje, taffeltsje - board, stockje - stick, betje - bed and others. In Plotdich's language, the suffix is -the. For example, "bilt'e - photo". [6:13].

In modern German, on the other hand, we see that the monoftong vowels in the diminutive suffixes were replaced by diphthongs by the Middle and New Upper German period. Forexample: das Buch (book) - das Buchlein (booklet), der Tisch (table) - das Tischlein (table), der Bruder (brother) - der Bruderchen (littlebrother) and others.

Historically, lexical units of verbs and nouns were common in ancient Germans, while auxiliary word groups, especially prepositions, were less developed. It was not until the Middle Ages that most of the prefixes became prepositions as a result of the formation of nouns from artificial verbs. In Old Upper German, for example, a number of prefixes in compound verbs have become relevant words, prepositions, and forms not only in Osh form, but also in meaning. Such prefixes include: ana-, duruh-, for a-, furi-, fram-, gagen-, hinder-, in-, nidar-, ubar-, ûf-, umbi-, undar-, ûz-, wider- and others. [1:35]

Experiments show that fixed phrases in the German form "preposition + noun" are relatively rare expressions. There are many such phrases in English:

By day, at night, by car (by bus, by train, by ship, by boat, by plain ...), on foot, at home and so on.

In German, such derivations as "preposition + noun" are relatively rare:

NachJahr und Tag - within a specified period

In Bausch und Bogen - completely, completely

Von Mal zu Mal - every time

NachHause - home

ZuHause - at home

SchrittfürSchritt - step by step ...

We find that the similarity of the expressions in German and English in the form of "preposition and noun" is due to the fact that in German most of such expressions are formed by the direct connection of the preposition to the noun without the article. in which case we can come to the following conclusions.

We have witnessed that the term derivation also has its own history. In modern linguistics, Wilhelm von Humboldt distinguishes between internal and external form. After all, although each phenomenon in language is characterized by its own object of study, in fact, they require a whole organism, a system. As for the derivative nature of word-forming affixes, this functional sign is a sign that word-formers in the paradigm of affixal morphemes are functionally different from other affixes, lexical and grammatical form-formers. With the understanding of such a functional difference, two branches of linguistics called word formation and morphemics emerged.

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IMPROVING USE OF REMOTE BANKING SERVICES IN UZBEKISTAN

Sobirov Ilxom Husanovich

Samarkand State University of Architecture Construction,

Abstract: *The article shows the problems of the development of digital banking in Uzbekistan, as well as specific recommendations*

Keywords: *Digital Bank, interactive analysis, remote banking product*

Introduction

"Strategy for the reform of the banking system of the Republic of Uzbekistan for 2020-2025" approved by the Decree of the President of the Republic of Uzbekistan dated March 2, 2020 No. .

The document defines the goals, tasks and priority areas of the development of the banking system, the directions of transformation and reform of the banking system in 2020 - 2025, ways to search for possible solutions based on the experience of transforming the financial sector of foreign countries and taking into account world trends in the financial sector. Effective ways of improving the quality of banking services in commercial banks have been determined.

Mobile banking is the most advanced and popular method of providing new banking services by commercial banks. Mobile banking is a modern way of using banking services through a mobile phone (smartphone) using the Internet. This method allows users of banking services to manage their bank account remotely and provides almost all internet banking operations.

With the help of mobile banking, the client has the following options:

- monitor the status of bank cards (credit, debit, etc.);
- mobile and landline phone services,
- making payments for internet providers, utility, cable and digital television services without intermediary fees;
- conducting online conversion practices;
- making transfers from card to card;
- setting up automatic execution of transactions - payments on account or on certain dates;
- getting information about the receipt of funds in the bank account;
- get information about the transactions made through a bank card;
- making purchases in online stores;
- extinguish loans,
- adding funds to deposits;
- getting a loan;
- receiving and sending money transfer;
- Paying YHXBB fines;
- opening a bank card;
- blocking and unblocking a bank card;
- payment of interest on deposits to a bank card;
- extending the validity period of the bank card;

- making international money transfers in different currencies through the mobile application.

To use mobile banking services, the client must first open a bank account, that is, a card account (bank card). For this, it is necessary to visit the bank (or remotely), fill out the necessary documents, and download the corresponding application to the mobile phone via the Internet..

Mobile applications may be offered by banks or payment organizations. If the mobile application and the bank card belong to the same bank, it is possible to be exempted from small brokerage fees when carrying out transactions through this application. For example, there are commissions for transferring funds from one card to another or withdrawing cash. In general, the commission does not exceed 1%.

If the application is installed on the smartphone, it is enough to go through authorization in this application with a bank card, that is, it is necessary to enter the information requested about the bank card in the application..

For authorization, it is necessary to enter the information on the front of the card (name of the card holder, individual card number and card validity period). Then it is necessary to confirm the increased authorization using the code received by the bank's SMS message. After that, it is important to set a password (PIN code) to access the application and remember it, in case of theft or loss of the phone, the security of funds is provided by this password.

Multiple cards can be registered in one application. This allows you to easily switch from one card to another in the application itself and make a payment with the desired card.

Each mobile banking app has a different design and menu, but that's not a problem because they're so easy to use. The menu consists mainly of sections according to the operations desired by the user (transfers, payments, etc.). First of all, you need to enter the card information in the application, after which you can make payments.

As soon as you perform any banking operation through mobile banking, an SMS message about the operation will be sent to the phone number registered in the application.

Advantages of mobile banking:

- Using banking services remotely through a mobile application;
- Saving time;
- Works 24 hours a day.

Disadvantages of mobile banking:

- Theft or loss of a mobile phone;
- Low level of account security.

The Decree of the President of May 12, 2020 on the "Strategy for the reform of the banking system of the Republic of Uzbekistan for the years 2020-2025" also requires the careful adoption of modern technologies to establish remote banking services, the identification of new types of services and products, studying foreign best practices and increasing the popularity of banking services in our country. use, protection of consumer rights, increasing the financial literacy of the population is given special importance to be among the current issues of our time. Therefore, this decision started a new stage in the development of banking services.

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FUNDAMENTALS OF USING INNOVATION IN EDUCATIONAL TECHNOLOGIES.

Sobirov Ilxom Husanovich¹ , Komilova Mukammal²

¹Samarkand State University of Architecture Construction, Associate Professor of the "
Economics " Department

²Head of the "Economics" department of the Samarkand State University of Architecture
Construction

Abstract. *The article shows new innovative pedagogical technologies in the educational process, new information in the field of education and effectively using them in creative work.*

Keywords: *information technologies, educational technologies, student, teacher, Innovation means*

Introduction

The comprehensive development and progress of the society depends on the development and improvement of the content of education.

As the President of the Republic of Uzbekistan Sh.M. Mirziyoev stated in his Address to the Oliy Majlis, "**...it is necessary to introduce information technologies that fully meet international standards at all stages of education.**" That's why we should try to make the young generation mature in all aspects.

In contrast to the methodical lesson development of the educational process focused on the active, effective activity of the teacher, the pedagogical technology of education focuses on learners, and also creates conditions for mastering the educational material, taking into account their individual and joint activities with the teacher.

Each of the educational technologies embodies information materials about the conditions of the educational session, pedagogical goals, tasks and expected results, the plan of the educational session, methods and means of teaching. It also includes a technological map of the training session, that is, a step-by-step description of the cooperative activity of the teacher and the student according to the goal of this training session.

New innovative educational technologies, modern methods and tools of teaching are entering the educational process and are being used effectively. The teacher should not be the only source of learning, but should be the organizer of the process of students' independent work, consultant, manager of the learning process. It is these ideas that underlie the development of innovative educational technology.

The main goal of education is to increase the effectiveness of education based on the use of elements of innovative educational technologies, to create independent thinking, original specialists.

The current educational development brought a new direction - innovative pedagogy to the field. The term "innovative pedagogy" and its characteristic research appeared in Western Europe and the United States in the 60s.

Innovation (English innovation) is innovation. Innovation means purposeful changes that introduce new, relatively stable elements into a certain social unit. This is the activity of the innovator.

Lectures as effective forms of organizing training sessions (problematic lecture, lecture-seminar, virtual-technological lecture, visual lecture, binary lecture, introductory lecture,

lecture-conference, informative lecture, lecture-debate, explanatory lecture, on-Line lecture) training, video training, webinars, internet conferences, as innovative teaching methods, problem methods, interactive methods, practical games, educational projects, portfolios, graphic organizers, information and communication technologies are needed.

It is known that pedagogical technology is an educational process that is pre-planned and fully designed in the educational process, designed for a certain time, the educational process is more focused on the learner, using activated methods and modern educational tools, which guarantees the achievement of the educational goal.

Live speech of the pedagogue, games of individuals, public actions serve as a means of education. But in order to solve the relevant educational tasks, it is necessary to include them in a specific system of the work of a coach and pedagogue. The effectiveness of the educational result depends on the skillful use of the methods, tools and forms of organizing the educational process and ensuring their harmony.

When using technologies, we should pay attention to the work of monitoring the student's knowledge, comparing the important things in their growth with the main ideas that have arisen in society, determining the ways and methods of their development, choosing the appropriate one, and deeply analyzing the dialectics of the mutual transition of various tools and methods of educational influence.

Today, educational technologies are divided into the following areas:

1. Modern-traditional teaching elements are used pedagogical technologies;
2. Person-oriented pedagogical technologies in the pedagogical process;
3. Pedagogical technologies based on activation and acceleration of students' activities;
4. Pedagogical technology based on effective management of the educational process;
6. Developing educational technologies;
7. Technologies related to private educational subjects.

When innovative education technologies are used, students' interest increases, knowledge turns into skills, and the quality and efficiency of knowledge increases.

It is known that primary education creates a basis for the development of mental and spiritual qualities of students. Growing and developing the consciousness, thinking, worldview and spiritual world of the individual through the means of knowledge given to the child during this period is one of the main manifestations of organizing the child's development process in education.

In primary education, the necessary knowledge is imparted on the basis of practical activities and various interesting tools. In the process of this activity, the child, who is developed by the student's ability to achieve some result, imagine through images, create something with his own hands, begins to understand his self.

Therefore, in modern conditions, the use of project and problem-based educational technologies in educational practice develops students' ability to think independently, critically and creatively, while ensuring the effective course of the teaching process. The social requirements for improving the quality of education and increasing its effectiveness require teachers to use project and innovative educational technologies in the process of teaching purposefully and effectively.

Educational technology is based on the principles of humanity. The uniqueness of this

direction in philosophy, pedagogy and psychology is shown by paying special attention to the individuality of the student.

- Taking into account the innovative development features of modern educational methods, it is appropriate to pay special attention to the following principles in education:
- the principle of education based on structural-functional reformation;
- the principle of ensuring mutual compatibility of economic and social goals in the educational process;
- the principle of effectiveness in financing education;
- the principle of taking into account the specific features of socio-economic development between the market of labor and educational services.

Thus, based on the above-mentioned approaches, we have developed recommendations aimed at effective use of innovative educational methods in the educational process;

1. When using modern educational methods, paying special attention to the organizational and technological criteria that take into account the level of education, organizing the educational process by dividing the factors affecting various forms and methods of education into separate groups;

2. Ensuring the gradation of the use of educational methods that develop a person in obtaining a certain level of knowledge, taking into account the requirements based on the joint interests of the participants in education

3. Taking into account educational and professional flexibility in education, which reflects specific development characteristics of the learner, limits the development of the education system in some cases and reveals its role in society;

4. Showing the formation of a person's social role as a result of the educational process; use of educational methods that are not influenced by factors that lead to the stratification of psychological and professional knowledge.

5. The educational process can be viewed as an asymmetric information base, because the real assessment of the quality of the educational process is valid after a certain period has passed. Based on this situation, it is appropriate to use multi-developmental educational methods.

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DIGITALIZATION AND INNOVATIVE DEVELOPMENT STRATEGY OF THE CONSTRUCTION INDUSTRY IN UZBEKISTAN

Sobirov Ilxom Husanovich

Samarkand State University of Architecture Construction,

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Abstract. *The scope of creativity in our country is expanding every year. New houses, large factories, modern infrastructure facilities are being built. Naturally, in order to carry out such large-scale works in a high-quality and timely manner, qualified builders and designers, stable organizations are needed.*

Keywords: country, secondary, construction-contract works , implement, industry

Introduction

There are about 22,000 construction-contracting organizations in our country, and most of them do not keep information on the number of employees and professional qualifications.

The fact that all constructions in the regions have been handed over to single customer service organizations has a negative impact on the indicators. While in advanced foreign practice, construction safety, quality and results are controlled, in our country all efforts and resources are spent on process control, and quality is secondary.

It is necessary to create a healthy competitive environment among the project institutes, develop clear criteria for conducting tenders based on foreign experience, and introduce an electronic system in this regard.

Improving the norms and rules of urban planning and the activities of single customer service organizations has become a requirement of today.

The level of use of information technologies in the construction sector remains low. In particular, the processes of creating projects, carrying out expertise, issuing construction permits, and controlling construction-contract works are still being done on paper. A geoinformation system that provides information on where and what can be built for investors and entrepreneurs has not been established.

In accordance with the Decree of the President of the Republic of Uzbekistan dated March 13, 2020 No. PF-5963 "On additional measures to deepen reforms in the construction sector of the Republic of Uzbekistan", a strategy for modernization, rapid and innovative development of the construction sector of the Republic of Uzbekistan for 2020-2025 was developed.

The purpose of the strategy is to implement centralized changes in the construction industry, aimed at forming modern methods of managing the construction industry, increasing investment attractiveness in the implementation of projects, and introducing innovative solutions to the construction industry.

According to the strategy developed on the basis of the decree, the main directions of modernization, rapid and innovative development of the construction industry of the Republic of Uzbekistan were defined as follows:

- digitalization of the construction industry by creating additional subsystems and databases;
- modernization of the regulatory framework in the field of urban development and adaptation of foreign regulatory documents;
- improvement of the system of complex development of settlements;

- to improve the position of the Republic of Uzbekistan on the indicator "Obtaining construction permits" in the "Doing Business" international index by simplifying some permit procedures in the construction sector;
- development of the mortgage loan market and the construction of multi-apartment houses by attracting public funds for the construction of housing on the basis of a share in the construction of housing real estate;
- application of innovative technologies to development of building materials and construction of objects;
- step-by-step transition to enlarged indicators in mutual calculations between participants of investment projects;
- improving the principles of licensing, taking into account the liberalization of the institutional framework, strengthening the potential of project and contract works;
- improvement of the system of expert work by canceling the state expertise in the estimate part of the project;
- improvement of the construction quality control system by directing the state, authorship and technical control to the non-state sector;
- organization of modern project institutes with the participation of advanced foreign companies by selling a package of shares to advanced foreign investors;
- improvement of the system of training, retraining and improvement of qualifications of personnel in the field of construction, taking into account the use of modern and innovative methods of training;
- formation and strengthening of the institution of non-state customers, taking into account the liberalization of the state segments, when implementing projects from state and equivalent sources of financing;
- development of scientific potential in the field of construction and architecture by increasing the effectiveness of fundamental, research and applied scientific researches and developments;
- to strengthen the requirements for labor protection and safety equipment during construction and installation works at construction sites;
- development of human capital, improvement of material and technical support of construction bodies and institutions, creation of new mechanisms of social and material stimulation of workers.

It is proposed to implement the following target indicators for the implementation of the 2020-2025 strategy of modernization, rapid and innovative development of the construction industry of the Republic of Uzbekistan:

- By 2025, the operation of the electronic platforms of the national information system "Transparent Construction" will be covered up to 100% throughout the republic;
- Increase the share of Eurocode adaptation in the field of urban planning to 50% by 2025, taking into account the geological, natural-climatic, seismological and other characteristics of the Republic of Uzbekistan;
- By 2025, the share of information on the objects of the state urban planning cadastre geographic information system, taking into account the expansion of geoportal functions, will be published for open use;

- Improving the position of the Republic of Uzbekistan in the direction of "Dealing with Construction Permits" of the international index "Doing Business Index" from the current 61.7 points to 78.2 points by 2025;

- establishment of 15 modern project-research institutes by 2025 with the involvement of advanced foreign project organizations;

- By 2025, increase the share of non-governmental organizations performing customer functions to at least 25 percent of the total number of objects included in the Investment Program;

- By 2025, the share of the introduction of the "volumetric method" should be increased to 25% of the total number of construction objects in the republic..

The strategy defines the goals, tasks and priority areas of the development of the construction industry, the directions of transformation and reform of the construction industry in 2020-2025, the development of proposals and scientific ideas based on the experience of the transformation of the construction sector of foreign countries and taking into account world trends in the field of construction.

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PRINCIPLES OF SUSTAINABLE DEVELOPMENT IN ARCHITECTURAL DESIGN IN THE CONTEX OF CLIMATE CHANGE

Abdullaeva Firuza Timurovna

The department of «Interior Design» at Samarkand Architecture and
Construction University

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Abstract. *The contemporary climate crisis has become a severe threat to humanity. Over the past decade, record-high temperatures have been recorded, leading to an increased frequency of extreme weather events, such as catastrophic wildfires, devastating floods, and intense storms. Since 1992, the United Nations has recognized global warming as a significant threat to humanity and has been striving to foster international cooperation to halt the rise in global temperatures. As climate change increasingly impacts our environment, architects are turning to innovative adaptive design strategies, such as "cool roof" systems, adaptive façades that regulate temperature and natural light (e.g., Trombe walls), green roofs and façades, as well as the reuse and retrofitting of existing buildings. This paper examines these innovative methods in adaptive architecture, which enhance the resilience of buildings and communities to extreme weather events and other climate-related threats.*

Keywords: *adaptive architecture, sustainable design, eco-friendliness, energy efficiency, climate change.*

Introduction

Since the initiation of climate change negotiations in 1990, greenhouse gas emissions have risen by 62%, exacerbating the climate crisis. We are now in a critical race to adapt to a rapidly changing climate, alongside two other global threats: biodiversity loss and environmental pollution. In the coming decades, by 2050, a significant portion of the urban population is expected to face increased risks from extreme heat. Studies indicate that 1.6 billion people across more than 970 cities will regularly experience dangerously high temperatures.

Adaptive architecture, defined by its ability to adjust to changing environmental conditions, presents a promising approach to mitigating the effects of climate change. Recognizing the impact of architecture on our planet is paramount when designing and constructing buildings and cities. The construction sector is responsible for nearly 39% of global annual CO₂ emissions, underscoring the significant influence that material choices and construction methods have on our climate. [1]

Urban development often leads to an increase in temperatures in large cities, creating areas with exceptionally high temperatures known as "urban heat islands." (Table №1)

An urban heat island refers to an urban environment with higher temperatures compared to surrounding rural areas. This phenomenon occurs due to a combination of factors related to microclimate changes caused by urbanization.

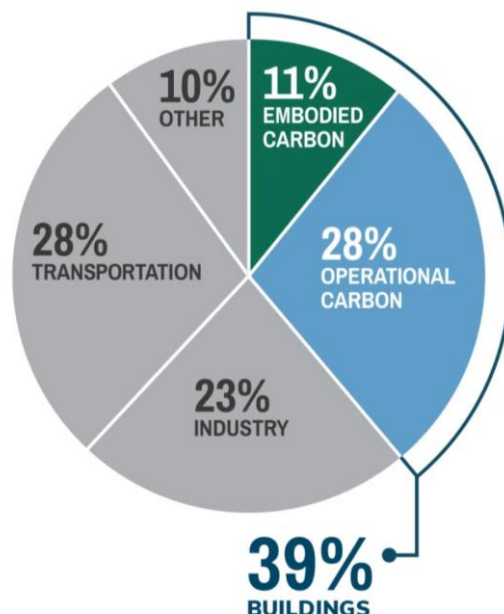


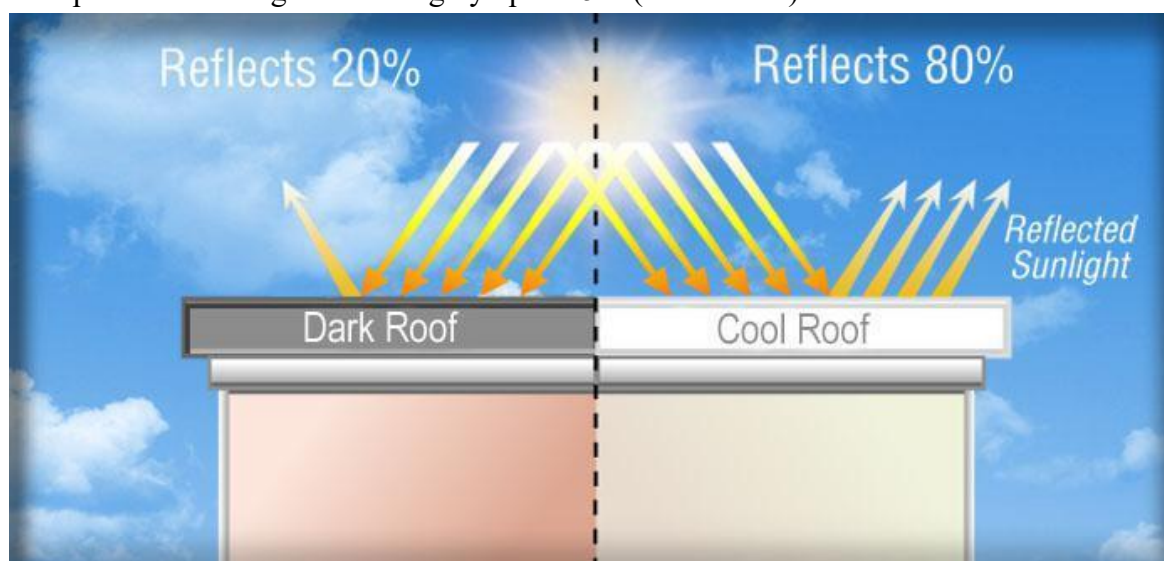
Table № 1 The global annual CO2 emissions.

Urban heat islands have detrimental effects on the urban population, including:

- Increased energy consumption for ventilation and air conditioning systems.
- Air pollution and smog formation.
- Higher greenhouse gas emissions.
- Deterioration of water quality at treatment facilities.
- Increased morbidity and mortality related to heat.

Below are steps we can take to enhance architectural resilience:

Main Body: In cities of the Mediterranean and the Middle East, residents have long recognized that white surfaces naturally reflect sunlight, making white clothing more comfortable and white houses cooler. On a typical sunny summer afternoon, a clean white roof, reflecting 80% of sunlight, will remain 30°C cooler than a grey roof that reflects only 20% of the solar radiation. According to the International Energy Agency, well-designed cities could reduce energy consumption for heating and cooling by up to 25% (Picture №1).



(Picture №1) The benefits of "cool roofs" for building owners.

This observation highlights the significant role of surface albedo—the ability of materials to reflect sunlight—in mitigating the urban heat island effect. By incorporating reflective materials and light-colored surfaces, cities can reduce heat absorption, lower temperatures, and enhance overall energy efficiency. The potential of such strategies is particularly important in light of the growing challenges posed by climate change, which exacerbate the need for sustainable urban development and energy conservation. [2]

Thanks to the advancements made by chemical engineers, modern roofing waterproofing materials and systems, predominantly in bright white colors, have been developed and are commonly referred to as "cool roofs." The principle behind their functionality lies in their ability to reflect solar radiation (solar reflectance) and exhibit high thermal emittance (the capacity to absorb, convert, and re-emit energy across a different wavelength range). An example of a material with low thermal emittance is galvanized steel, which, despite its high reflectivity, tends to heat up considerably when exposed to sunlight.

"Cool roofs" are typically applied in a liquid form. These coatings consist of cold-curing polymers, specialized additives, fillers, and reflective pigments in various hues. However, it is well-established that light-colored coatings, particularly those in white, exhibit the highest levels of solar reflectance. The surface temperature of such coatings remains close to that of the ambient air, contributing significantly to the mitigation of the urban heat island effect and reducing the demand for cooling energy.

This technology is widely used both in the construction of new roofs and in the repair and restoration of existing ones, with no need to remove the old covering. "Cool roofs" can be applied to various substrates, such as concrete, metal, rolled bitumen, single-ply polymer membranes, and are suitable for flat, pitched, and domed structures. [3]

The benefits of "cool roofs" for building owners are undeniable:

- Energy savings of up to 15% on building cooling.
- Increased comfort in spaces without air conditioning.
- Reduced heat load on machinery, electronics, and industrial equipment.
- Optimized storage temperatures for food, raw materials, and lubricants.
- Extended roof lifespan due to reduced exposure to ultraviolet rays and temperature-induced structural deformations.
- Lower maintenance and repair costs for rooftop air conditioning units.

Another significant solution for reducing air temperatures in urban environments is the greening of roofs and façades. Green roofs not only help mitigate summer heat but also absorb excess rainwater. Additionally, they can be utilized to host small-scale farms for growing vegetables and fruits.

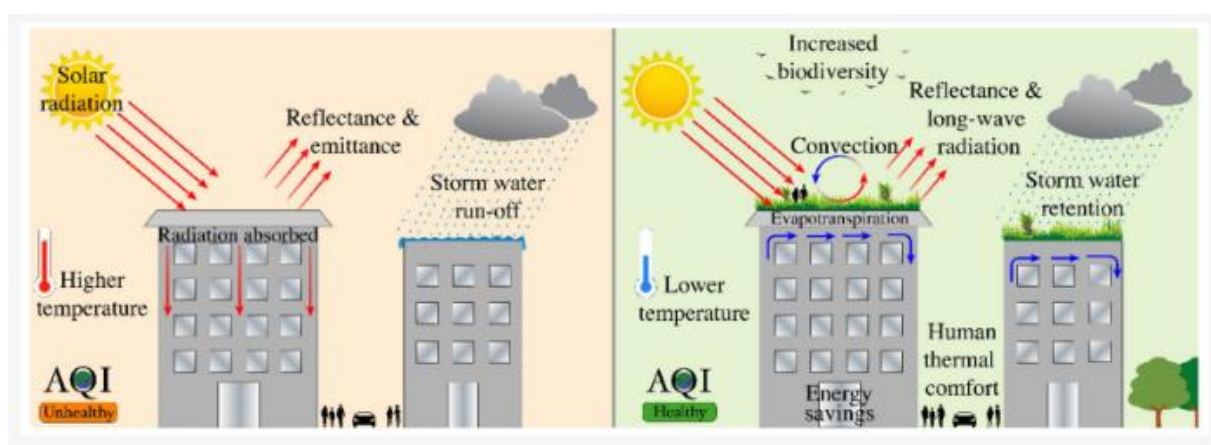
Green roofs, through the natural processes of dew formation and evaporation, have the ability to cool the urban environment during hot summer months. This is achieved by converting the solar energy absorbed by plants into thermal energy, thereby preventing surface temperature increases.

Furthermore, green roofs contribute to reducing the urban heat island effect by covering dark rooftops, which are primary sources of heat in cities. These roofs intercept solar radiation, thereby decreasing the need for air conditioning and reducing greenhouse gas emissions from power plants.

Green roofs have the potential to reduce indoor temperatures by 2-6°C in comparison to conventional roofing systems. The extent of temperature reduction depends on several factors, including the size and type of the green roof, the regional climate, as well as the building's design and insulation. (Picture №2)

Beyond their cooling effect, green roofs also enhance air quality. The vegetation on these roofs acts as a natural filter, capturing atmospheric pollutants and contaminants while absorbing harmful gases. This process reduces the emission of particulate matter and dust, thereby minimizing the formation of urban smog.

In this way, green roofs play a vital role in mitigating the impacts of climate change and fostering a healthier, more sustainable urban environment. By lowering ambient temperatures, reducing greenhouse gas emissions, and improving air quality, they contribute significantly to the adaptation of urban areas to the increasingly warmer summer conditions. [4]



(Picture №2) Schematic representing the diverse processes involved in green roofs.

To achieve energy savings and create comfortable conditions in cold climates, it is essential to incorporate elements of passive solar heating. One such element is the Trombe wall.

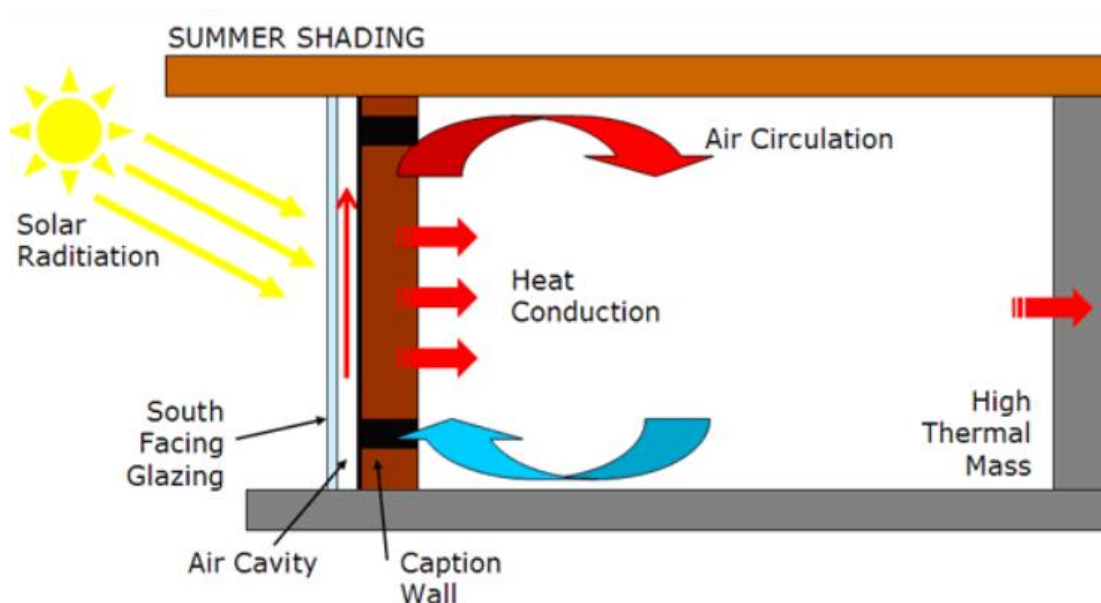
A Trombe wall is a massive, dark-colored wall with external glazing. It absorbs solar heat during the day and gradually releases it at night, thereby reducing the need for heating the building. The Trombe wall is a key component of solar architecture, which aims to minimize heating energy consumption through the use of solar energy. It helps maintain a comfortable indoor temperature without the need for additional heating. [5]

The delay in heat release to the interior depends on the thickness of the Trombe wall:

- With a wall thickness of 20 cm, the heat delay is approximately 5 hours;
- With a wall thickness of 40 cm, the delay is approximately 10-12 hours.

The Trombe wall can be made not only of concrete but also of stone or brick. To enhance heat transfer, special openings are created at the bottom and top of the wall to facilitate natural air convection, and fans are sometimes installed to promote forced circulation for more efficient heat release.

Solar rays pass through the glazing and strike the concrete wall, which is positioned approximately 100 mm away from the glazing. The ultraviolet rays from the sun heat the surface of the wall, and some of the rays are reflected from the wall as infrared radiation, which does not pass through the glass, thereby warming the air as well. (Picture №3)



(Picture №3) The Trombe wall efficiency.

The construction industry, known for its high energy intensity, accounts for up to 39% of global CO₂ emissions, with 28% of this figure attributable solely to the operation of buildings—namely, their heating, cooling, ventilation, and lighting. The remaining 11% is directly linked to the construction process itself. To minimize its ecological impact, a complete reevaluation of construction practices is necessary.

Adaptive architecture is a design approach that allows buildings to adjust to changing conditions and needs throughout their lifecycle. This is a crucial element of sustainable development, as it enhances the flexibility and longevity of buildings while reducing their environmental impact.

Adaptive reuse offers an alternative to demolition, providing an opportunity to improve the energy efficiency of existing structures. While much attention is given to the preservation of architectural heritage, most buildings lack historical significance and are demolished after 30 to 50 years. However, the global environmental crisis compels us to reconsider this approach, urging us to evaluate the "climatic value" of a building on par with its cultural and historical significance. [6]

One of the most prominent examples of adaptive reuse can be found in London, England. The Tate Modern art gallery, once the Bankside power station, was converted by Pritzker Prize-winning architects Jacques Herzog and Pierre de Meuron.

In the context of a changing climate and rapid technological progress, adaptive architecture enables buildings to transform and respond to evolving circumstances, adapting to new functions and conditions in order to maintain their relevance and avoid demolition.

Adaptive architecture, which integrates elements such as passive solar heating, cool roofs, green roofs, Trombe walls, and the adaptive reuse of existing buildings, is of paramount importance in mitigating the effects of climate change and creating a more sustainable built environment. These strategies represent a shift away from traditional construction practices, offering solutions that not only reduce energy consumption but also enhance the resilience of buildings to the challenges posed by a changing climate.

Conclusion. By incorporating passive solar design, such as Trombe walls, buildings can effectively harness the sun's energy for heating while minimizing the need for additional heating systems. Cool roofs and green roofs further reduce the urban heat island effect, improve air quality, and enhance the overall comfort of indoor spaces. Meanwhile, adaptive reuse allows us to breathe new life into existing structures, reducing the need for demolition and conserving valuable resources.

Through these innovative approaches, architects and builders can design buildings that are more adaptable to evolving climate conditions, ensuring that structures remain relevant and functional over time. This approach not only helps mitigate the environmental impact of construction but also contributes to creating healthier and more comfortable living environments.

In the broader context, by focusing on resilience, energy efficiency, and the reduction of environmental footprints, adaptive architecture plays a critical role in shaping cities and communities that are better equipped to face the challenges of climate change. By enhancing the adaptability and sustainability of our built environments, we can pave the way for a more environmentally responsible and livable future, ensuring that the needs of both current and future generations are met.

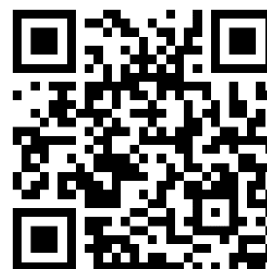
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